Lectures on Algebra III

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Preface

These are my lecture notes for the graduate course Algebra III at the University of Chinese Academy of Sciences. The topic of this course is commutative algebra. As most of the students had already followed the courses Algebra I and II, where basic algebra and homological algebra were discussed, I can concentrate on commutative algebra itself, with the help of notions and tools from previous courses.

I mainly followed the classical textbooks of Atiyah-MacDonald [2] and Matsumura [11, 12]. In some sense, this note is a mixture of some key basic parts of [2, 11, 12], with a point of view toward some current developments [1, 14, 6, 15]. For some other standard references, see [3, 7, 16, 13, 5, 10].

Chapter 1

Rings and Modules

Convention: In these notes, all rings will be commutative with unit, unless otherwise stated.

1.1 Prime ideals

The notions of rings, ideals, modules, and homomorphisms have been introduced in Algebra I & II. Here are some examples of rings.

Examples 1.1.1. 1. *The ring of integers $\mathbb{Z}$, the quotient ring $\mathbb{Z}/n\mathbb{Z}$ $n \in \mathbb{Z}$, the field of rational numbers $\mathbb{Q}$.*

2. *Let A be a ring, for any integer $n \geq 1$, we have*

$$A[X_1, \cdots, X_n] = \{\textit{Polynomials in variables}\, X_1, \ldots, X_n \,\textit{with coefficients in}\, A\}$$

and

$$A[[X_1, \cdots, X_n]] = \{\textit{Formal power series in variables}\, X_1, \ldots, X_n \,\textit{with coefficients in}\, A\}.$$

Let us recall the definition of prime ideals.

Definition 1.1.2. *Let A be a ring.*

1. *An ideal $\mathfrak{p} \subsetneq A$ is prime if $\forall\, x, y \in A$ such that $xy \in \mathfrak{p}$, then $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. Equivalently, $\mathfrak{p} \subsetneq A$ is prime if $A/\mathfrak{p} \neq 0$ is an integral domain.*

2. *An ideal $\mathfrak{m} \subsetneq A$ is maximal if there exists no ideal $\mathfrak{a}$ such that $\mathfrak{m} \subsetneq \mathfrak{a} \subsetneq A$. Equivalently, $\mathfrak{m} \subsetneq A$ is maximal if $A/\mathfrak{m} \neq 0$ is a field.*

Thus maximal ideals are prime, but not conversely in general. Also, the zero ideal (0) (simply denoted by 0) is prime if and only if A is an integral domain.

Example 1.1.3. *Let p be a prime number. Then $(p) \subset \mathbb{Z}$ is a maximal ideal. All prime ideals of $\mathbb{Z}$ are of the form 0 or (p), p prime.*

Lemma 1.1.4. *Any ring $A \neq 0$ has at least one maximal ideal.*

Proof. Apply the Zorn Lemma to $(\Sigma, \subset)$, where $\Sigma = \{\text{ideals} \subsetneq A\}, \subset$: inclusion. $\square$

Corollary 1.1.5. *Let $\mathfrak{a} \subsetneq A$ be an ideal. Then there exists a maximal ideal of A containing $\mathfrak{a}$. In particular, any non unit $x \in A$ is contained in a maximal ideal.*

Lemma 1.1.6. *Let $f : A \to B$ be a ring homomorphism, $\mathfrak{q} \subset B$ a prime ideal. Then $f^{-1}(\mathfrak{q})$ is a prime ideal of A.*

Proof. Consider the composition $A \to B \to B/\mathfrak{q}$. It induces an injection $A/f^{-1}(\mathfrak{q}) \hookrightarrow B/\mathfrak{q}$. Then $\mathfrak{q} \subset B$ prime $\Leftrightarrow$ $B/\mathfrak{q}$ integral domain $\Rightarrow$ $A/f^{-1}(\mathfrak{q})$ integral domain $\Leftrightarrow$ $f^{-1}(\mathfrak{q}) \subset A$ prime. □

Remark 1.1.7. *If $\mathfrak{q} \subset B$ maximal, then in general it is not true that $f^{-1}(\mathfrak{q}) \subset A$ maximal: e.g. consider $A = \mathbb{Z} \hookrightarrow B = \mathbb{Q}$ and $\mathfrak{q} = 0$.*

Notation 1.1.8. *For any ring A,*

$$\operatorname{Spec} A = \{\textit{prime ideals of}\, A\}.$$

By Lemma 1.1.4, if $A \neq 0$, then $\operatorname{Spec} A \neq \emptyset$. By Lemma 1.1.6, any ring homomorphism $f : A \to B$ induces a map

$$f^* : \operatorname{Spec} B \to \operatorname{Spec} A.$$

We will introduce a topology on the sets $\operatorname{Spec} A$ such that the maps f^* are continuous. In this way, we will get a functor

$$\operatorname{Spec} : \text{Rings} \longrightarrow \text{Topological Spaces}, \quad A \mapsto \operatorname{Spec} A.$$

We need some preparations.

Definition 1.1.9. *Let A be a ring.*

1. *The nilradical of A is*

$$\textit{nilrad}(A) = \{\textit{nilpotent elements in}\, A\}.$$

A is called reduced if $\textit{nilrad}(A) = 0$. In general, put $A_{red} = A/\textit{nilrad}(A)$.

2. *Let $I \subset A$ be an ideal. The radical of I is*

$$r(I) = \{x \in A \,|\, x^n \in I \,\textit{for some}\, n > 0\}.$$

In particular, $r(0) = \textit{nilrad}(A)$.

One can check nilrad(A) is an ideal of A, cf. [2] Proposition 1.7. For any ideal $I \subset A$, let $f : A \to A/I$ be the natural projection. Then we have

$$r(I) = f^{-1}(\text{nilrad}(A/I)).$$

Thus $r(I)$ is also an ideal. The ring $A_{red} = A/\text{nilrad}(A)$ is then reduced: nilrad$(A_{red}) = 0$.

Example 1.1.10. *Let $p \neq q$ be prime numbers. Then $r((p^3q^2)) = (pq)$, and $\textit{nilrad}(\mathbb{Z}/p^3q^2\mathbb{Z}) = (\overline{pq})$, where $\overline{p}$ is the image of p in $\mathbb{Z}/p^3q^2\mathbb{Z}$; similarly for $\overline{q}$.*

Proposition 1.1.11. *Let A be a ring, $I \subset A$ an ideal.*

1. *$I \subset r(I)$.*

2. *$r(I) = A \quad \Leftrightarrow \quad I = A$.*

3. *If* $\mathfrak{p} \subset A$ *is a prime ideal, then* $r(\mathfrak{p}) = r(\mathfrak{p}^n) = \mathfrak{p}$ *for any* $n \geq 1$.

4. *nilrad*$(A) = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}$.

5. $r(I) = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A, \mathfrak{p} \supset I} \mathfrak{p}$.

Proof. The proofs for 1-3 are left as exercises.

For 4, we first show $\operatorname{nilrad}(A) \subset \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}$: $\forall f \in \operatorname{nilrad}(A)$, $f^n = 0$ for some $n \geq 1$. Then $\forall \mathfrak{p} \in \operatorname{Spec} A$, $0 = f^n \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime, we have $f \in \mathfrak{p}$. Thus $f \in \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}$.

Now we show $\operatorname{nilrad}(A) \supset \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}$. We prove that if $f \in A$ is not nilpotent, then there exits a prime $\mathfrak{p} \in \operatorname{Spec} A$ such that $f \notin \mathfrak{p}$. Consider the set

$$\Sigma = \{I \subset A \,\text{ideals} \,|\, \forall\, n \geq 1, f^n \notin I\}.$$

Then $0 \in \Sigma$. Apply the Zorn Lemma to $(\Sigma, \subset)$, we find a maximal element $\mathfrak{p} \in \Sigma$. We claim that $\mathfrak{p}$ is a prime ideal. Indeed, for any elements $x, y \notin \mathfrak{p}$, $\mathfrak{p} + (x), \mathfrak{p} + (y) \supsetneq \mathfrak{p}$ thus not in Σ. Therefore, there exist m, n such that $f^m \in \mathfrak{p} + (x), f^n \in \mathfrak{p} + (y)$. Then $f^{m+n} \in \mathfrak{p} + (xy) \Rightarrow \mathfrak{p} + (xy) \notin \Sigma \Rightarrow xy \notin \mathfrak{p}$. Thus $\mathfrak{p}$ is prime and $f \notin \mathfrak{p}$.

For 5, we have $\operatorname{nilrad}(A/I) = \bigcap_{\mathfrak{q} \subset A/I \text{ prime}} \mathfrak{q}$. Then

$$r(I) = f^{-1}(\operatorname{nilrad}(A/I)) = \bigcap_{\mathfrak{q} \subset A/I \text{ prime}} f^{-1}(\mathfrak{q}) = \bigcap_{\mathfrak{p} \subset A \text{ prime}, \mathfrak{p} \supset I} \mathfrak{p}$$

□

Let I, J be ideals of A. Then $IJ \subset I \cap J$ and $I + J$ are ideals of A.

Exercise 1.1.12. *1.* $r(IJ) = r(I \cap J) = r(I) \cap r(J)$.

2. $r(I + J) = r(r(I) + r(J))$.

Let $A \neq 0$ be a ring. Recall that we want to introduce a topology on $\operatorname{Spec} A = \{\text{prime ideals of } A\}$. For any subset $M \subset A$, let

$$V(M) = \{\mathfrak{p} \in \operatorname{Spec} A \,|\, M \subset \mathfrak{p}\}.$$

One checks easily the following lemma.

Lemma 1.1.13. *1. Let* $I = (M)$ *be the ideal generated by* M*, then* $V(M) = V(I) = V(r(I))$.

2. $V(0) = \operatorname{Spec} A, \quad V(1) = \emptyset$.

3. *If* $(M_i)_{i \in \Sigma}$ *is a family of subsets of* A*, then* $V(\bigcup_{i \in \Sigma} M_i) = \bigcap_{i \in \Sigma} V(M_i)$.

4. *For any ideals* I, J*,* $V(I \cap J) = V(IJ) = V(I) \cup V(J)$.

This lemma implies that the following definition makes sense.

Definition 1.1.14. *The Zariski topology on* $\operatorname{Spec} A$ *is the toplogy such that the closed subsets are of the form* $V(M)$*,* $M \subset A$ *subsets.*

Note the following facts.

- For any ideals I, J, we have $V(I) \supset V(J) \Leftrightarrow I \subset r(J) \Leftrightarrow r(I) \subset r(J)$. In particular $V(I) = V(J) \Leftrightarrow r(I) = r(J)$.
- If $\mathfrak{p} \in \operatorname{Spec} A$, we have $V(\mathfrak{p}) = \overline{\{\mathfrak{p}\}}$. $\mathfrak{p}$ is maxmial as ideal of A if and only if it is closed as point of $\operatorname{Spec} A$, i.e. $\mathfrak{p}$ maximal $\Leftrightarrow \overline{\{\mathfrak{p}\}} = \{\mathfrak{p}\}$.

Let $X = \operatorname{Spec} A$. For any $f \in A$, set

$$X_f = D(f) = X \setminus V(f),$$

which is an open subset.

Proposition 1.1.15. *1. $(X_f)_{f\in A}$ form a basis of open sets for the Zariski toplogy on X.*

2. For any ring homomorphism $\phi : A \to B$, the induced map $\phi^* : \operatorname{Spec} B \to \operatorname{Spec} A$ is continuous.

Proof. Let $U = X \setminus V(I)$ be any open subset. Then

$$V(I) = \bigcap_{f\in I} V(f) \Rightarrow U = X \setminus \bigcap_{f\in I} V(f) = \bigcup_{f\in I} X \setminus V(f) = \bigcup_{f\in I} X_f.$$

This proves 1.

For 2, let $X = \operatorname{Spec} A, Y = \operatorname{Spec} B$, we prove $\phi^{*,-1}(X_f) = Y_{\phi(f)}$:

$$\begin{aligned} \forall\, \mathfrak{q} \in Y, \quad \mathfrak{q} \in \phi^{*,-1}(X_f) &\Leftrightarrow \phi^*(\mathfrak{q}) = \phi^{-1}(\mathfrak{q}) \in X_f \\ &\Leftrightarrow f \notin \phi^{-1}(\mathfrak{q}) \\ &\Leftrightarrow \phi(f) \notin \mathfrak{q} \\ &\Leftrightarrow \mathfrak{q} \in Y_{\phi(f)}. \end{aligned}$$

□

Remark 1.1.16. *One can characterize the essential image of the functor*

$$\operatorname{Spec} : \textit{Rings} \longrightarrow \textit{Topological Spaces}.$$

This is the class of the so called "spectral spaces", cf. [8].

Examples 1.1.17. *1.* $\operatorname{Spec} \mathbb{Z} = \{(0), (2), (3), (5), \cdots\}$. *For any prime p, (p) is a maximal ideal $\Rightarrow \overline{\{(p)\}} = (p)$, i.e. $(p) \in \operatorname{Spec} \mathbb{Z}$ is a closed point. On the other hand, $\overline{\{(0)\}} = \operatorname{Spec} \mathbb{Z}$.*

2. Let k be a field. Then $\operatorname{Spec} k[X] = \{(0), (f(X)) | \, f(X)$ irreducible polynomials$\}$. For any irreducible polynomial $f(X)$, the ideal $(f(X)) \subset k[X]$ is maxmial, $\overline{\{(f(X))\}} = \{(f(X))\}$, and $\overline{\{(0)\}} = \operatorname{Spec} k[X]$. (Recall that $k[X]$ is a PID.)

If k is algebraically closed, then any irreducible polynomial is of the form $f(X) = X - \alpha, \alpha \in k$. So we have a bijection $\operatorname{Spec} k[X] \simeq k \amalg \{0\}$ (disjoint union).

3. What is $\operatorname{Spec} \mathbb{Z}[X]$? Consider the natural inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[X]$, which induces

$$\phi : \operatorname{Spec} \mathbb{Z}[X] \to \operatorname{Spec} \mathbb{Z} = \{(0), (2), (3), (5), \cdots\}, \quad \mathfrak{p} \mapsto \mathfrak{p} \cap \mathbb{Z}.$$

There are two cases:

- *If* $\mathfrak{p} \cap \mathbb{Z} = 0$, *then* $\mathfrak{p} = 0$ *or* $\mathfrak{p} = (f(X))$ *with* $f(X) \in \mathbb{Z}[X]$ *irreducible polynomial.*
- *If* $\mathfrak{p} \cap \mathbb{Z} = (p)$, *consider the map* $\pi : \mathbb{Z}[X] \twoheadrightarrow \mathbb{F}_p[X]$, $\sum a_i X^i \mapsto \sum_i \overline{a}_i X^i$. *Then* $\pi(\mathfrak{p})$ *is a prime ideal of* $\mathbb{F}_p[X]$. *By the above 2, we have* $\pi(\mathfrak{p}) = (g(X))$ *with* $g(X) \in \mathbb{F}_p[X]$ *an irreducible polynomial. Let* $f(X) \in \mathbb{Z}[X]$ *be such that* $\pi(f(X)) = g(X)$. *Then* $\mathfrak{p} = (p, f(X))$.

We write $\mathbb{A}^1_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[X]$, *called the affine line over* $\mathbb{Z}$. *Similarly, let* $\mathbb{A}^1_{\mathbb{Q}} = \operatorname{Spec} \mathbb{Q}[X]$ *and* $\mathbb{A}^1_{\mathbb{F}_p} = \operatorname{Spec} \mathbb{F}_p[X]$ *for each* p. *Then the above consideration implies that*

$$\mathbb{A}^1_{\mathbb{Z}} \simeq \mathbb{A}^1_{\mathbb{Q}} \amalg \coprod_p \mathbb{A}^1_{\mathbb{F}_p}, \quad \phi^{-1}(0) \simeq \mathbb{A}^1_{\mathbb{Q}}, \quad \phi^{-1}((p)) \simeq \mathbb{A}^1_{\mathbb{F}_p}, \quad \forall\, p \textit{ prime}.$$

4. *Consider the ring* $\mathbb{Z} \times \mathbb{Z} = \mathbb{Z} \oplus \mathbb{Z}$. *Its prime ideals are of the form* $\mathfrak{p} \oplus \mathbb{Z}, \mathbb{Z} \oplus \mathfrak{q}$ $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec} \mathbb{Z}$. *Thus*
$$\operatorname{Spec} \mathbb{Z} \times \mathbb{Z} \simeq \operatorname{Spec} \mathbb{Z} \amalg \operatorname{Spec} \mathbb{Z}.$$
In general, for any rings A *and* B, *we have*
$$\operatorname{Spec} A \times B \simeq \operatorname{Spec} A \amalg \operatorname{Spec} B.$$

Exercise 1.1.18. *Let* $X = \operatorname{Spec} A$. *For* $f \in A$, *set* $X_f = X \setminus V(f)$. *Then*

1. $X_f \cap X_g = X_{fg}$.
2. $X_f = \emptyset \Leftrightarrow f$ *nilpotent.*
3. $X_f = X \Leftrightarrow f$ *is a unit.*
4. $X_f = X_g \Leftrightarrow r((f)) = r((g))$.
5. X *is quasi-compact for the Zariski topology.*

Exercise 1.1.19. *Let* $\phi : A \to B$ *be a ring homomorphism, and* $\phi^* : \operatorname{Spec} B \to \operatorname{Spec} A$ *the induced continuous map. If* ϕ *is surjective, then* ϕ^* *induces a homeomorphism of topological spaces*
$$\phi^* : \operatorname{Spec} B \xrightarrow{\sim} V(\ker \phi) \subset \operatorname{Spec} A.$$
In particular, $\operatorname{Spec} A \simeq \operatorname{Spec} A_{red}$, *and if* $I \subset A$ *is an ideal, we have*
$$\operatorname{Spec} A/I \simeq \operatorname{Spec} A/r(I) \simeq V(r(I)) = V(I) \subset \operatorname{Spec} A.$$

1.2 Nakayama lemma

Before we discuss Nakayama lemma, let us briefly review some basics of module theory.

Let A be a ring. Ideals of A are examples of A-modules (submodules of the A-module A). Let Mod_A be the category of A-modules. This is an abelian category.

For any $M \in \mathrm{Mod}_A$, we have three covariant functors:

$$\begin{aligned}
\operatorname{Hom}_A(M, \cdot) : &\quad \mathrm{Mod}_A \to \mathrm{Mod}_A, \quad N \mapsto \operatorname{Hom}_A(M, N),\\
M \otimes_A - : &\quad \mathrm{Mod}_A \to \mathrm{Mod}_A, \quad N \mapsto M \otimes_A N,\\
- \otimes_A M : &\quad \mathrm{Mod}_A \to \mathrm{Mod}_A, \quad N \mapsto N \otimes_A M,
\end{aligned}$$

and the functors $M \otimes_A -$ and $- \otimes_A M$ are naturally isomorphic to each other; on the other hand, we have also a contravariant functor:

$$\mathrm{Hom}_A(\cdot, M): \quad \mathrm{Mod}_A \to \mathrm{Mod}_A, \quad N \mapsto \mathrm{Hom}_A(N, M).$$

We will often omit the subscript A in the notations $\mathrm{Hom}_A(M, N)$ and $M \otimes_A N$. Recall the following basic facts:

1. For any $M, N, P \in \mathrm{Mod}_A$, there exists a canonical isomorphism $\mathrm{Hom}(M \otimes N, P) \simeq \mathrm{Hom}(M, \mathrm{Hom}(N, P))$.

2. Let $0 \to N' \to N \to N''$ be a sequence of homomorphisms of A-modules. Then it is exact if and only if for any $M \in \mathrm{Mod}_A$, the sequence $0 \to \mathrm{Hom}(M, N') \to \mathrm{Hom}(M, N) \to \mathrm{Hom}(M, N'')$ is exact.

3. Let $M' \to M \to M'' \to 0$ be a sequence of homomorphisms of A-modules. Then it is exact if and only if for any $N \in \mathrm{Mod}_A$, the sequence $0 \to \mathrm{Hom}(M'', N) \to \mathrm{Hom}(M, N) \to \mathrm{Hom}(M', N)$ is exact.

4. Let $M' \to M \to M'' \to 0$ be an exact sequence of homomorphisms of A-modules. Then for any $N \in \mathrm{Mod}_A$, the sequence $M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ is exact.

Example 1.2.1. *$0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$ exact, consider $N = \mathbb{Z}/2\mathbb{Z}$, then $0 \to \mathbb{Z} \otimes N \xrightarrow{\times 2 \otimes id} \mathbb{Z} \otimes N$ is not exact, since it is isomorphic to $0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z}$.*

The above example shows that $- \otimes N$ is not left exact in general.

Definition 1.2.2. *An A-module N is flat if $- \otimes N$ is an exact functor.*

Example 1.2.3. *The free module $A^n (n \geq 1)$ is flat. More generally, any projective module is flat.*

Let $f : A \to B$ be a ring homomorphism. Then B can be viewed as an A-module via f. It is then an A-algebra. For any module $N \in \mathrm{Mod}_B$, we can view it as an A-module: $\forall\, a \in A, x \in N, \quad ax := f(a)x$. For any $M \in \mathrm{Mod}_A$, the A-module $B \otimes_A M$ admits a B-module structure: for any $b, b' \in B, x \in M, \quad b(b' \otimes x) := (bb') \otimes x$. In this way, we get functors

$$\mathrm{Mod}_B \to \mathrm{Mod}_A, \quad N \mapsto N, \quad \text{and} \quad \mathrm{Mod}_A \to \mathrm{Mod}_B, \quad M \mapsto B \otimes_A M,$$

which are adjoint: for any $M \in \mathrm{Mod}_A, N \in \mathrm{Mod}_B$, we have natural isomorphisms

$$\mathrm{Hom}_A(M, N) \simeq \mathrm{Hom}_B(B \otimes_A M, N).$$

Let B, C be two A-algebras given by ring homomorphisms $f : A \to B, g : A \to C$ respectively. Then the A-module $D = B \otimes_A C$ admits a multiplication such that it is an A-algebra. In fact, D can be defined as the puschout of B and C over A in the category of rings. We have natural ring homomorphisms $p_1 : B \to D, \quad p_2 : C \to D$. The triple (D, p_1, p_2) satisfies the universal property: for any A-algebra R and homomorphisms $q_1 : B \to R, q_2 : C \to R$ of A-algebras, there exists a unique homomorphism $\phi : D \to R$ of A-algebras, such that $q_1 = \phi \circ p_1, \quad q_2 = \phi \circ p_2$. The morphism $A \to D$ is given by

$$a \mapsto f(a) \otimes 1 = 1 \otimes g(a) = a(1 \otimes 1).$$

Example 1.2.4. *Let $B = A[X_1, \cdots, X_n]$, then for any A-algebra C, we have*

$$B \otimes_A C \simeq C[X_1, \cdots, X_n].$$

If $C = A[Y_1, \cdots, Y_m]$, then

$$B \otimes_A C \simeq A[X_1, \cdots, X_n, Y_1, \cdots, Y_m].$$

Now we move toward Nakayama lemma. Let A be a ring. Recall that we have the notion of finitely generated A-modules.

Proposition 1.2.5. *Let M be a finitely generated A-module, $\mathfrak{a} \subset A$ an ideal, $\phi \in \mathrm{Hom}_A(M, M)$ such that $\phi(M) \subset \mathfrak{a}M$. Then ϕ satisfies an equation of the form*

$$\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0, \quad a_1, \ldots, a_n \in \mathfrak{a}, n \geq 1.$$

Proof. Let $x_1, \ldots, x_n$ be a set of generators of M. Then $\phi(x_i) \in \mathfrak{a}M, 1 \leq i \leq n$ by assumption. For any i, write

$$\phi(x_i) = \sum_{j=1}^{n} a_{ij}x_j, \quad a_{ij} \in \mathfrak{a},$$

that is

$$\phi \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (a_{ij}) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Then

$$(\delta_{ij}\phi - a_{ij}) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0, \quad \text{where } \delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

and we view $(\delta_{ij}\phi - a_{ij}) \in M_{n\times n}(R)$ with $R \subset \mathrm{End}_A(M)$ the commutative subring generated by ϕ and the scalars (i.e. the image of the natural morphism $A \to \mathrm{End}_A(M)$). Then we have

$$\det(\delta_{ij}\phi - a_{ij}) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0$$

and thus $\det(\delta_{ij}\phi - a_{ij}) = 0 \in R \subset \mathrm{End}_A(M)$. Expanding out the determinant, we get

$$\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0, \quad a_1, \ldots, a_n \in \mathfrak{a}.$$

□

Remark 1.2.6. *If $\mathfrak{a} = A$ and $M \simeq A^n$ finite free, the above is the classcial Cayley-Hamilton theorem.*

Corollary 1.2.7. *Let M be a finitely generated A-module, $\mathfrak{a} \subset A$ an ideal such that $\mathfrak{a}M = M$. Then there exists an element $x \in A$ such that $x - 1 \in \mathfrak{a}$ and $xM = 0$.*

Proof. Take $\phi = id \in \mathrm{End}_A(M)$ and $x = 1 + a_1 + \cdots + a_n$ where a_i are as above for $1 \leq i \leq n$. The image of x under $A \to \mathrm{End}_A(M)$ is $id^n + a_1 id^{n-1} + \cdots + a_n = 0$. This means $xM = 0$. □

We will take some special ideal $\mathfrak{a}$. Here come the Jacobson radicals.

Definition 1.2.8. *Let A be a ring. Its Jacobson radical $\mathfrak{R} = J(A)$ (or $rad(A)$) is the intersection of all maximal ideals of A.*

Lemma 1.2.9. $x \in \mathfrak{R} \quad \Leftrightarrow \quad xy - 1 \in A^{\times}, \quad \forall\, y \in A.$

Proof. $\Rightarrow$: if $\exists\, y$ such that $xy-1$ is not unit, then there is a maximal ideal $\mathfrak{m}$ such that $xy-1 \in \mathfrak{m}$. But $x \in \mathfrak{m} \Rightarrow 1 \in \mathfrak{m}$, contradracation!

$\Leftarrow$: We prove if $x \notin \mathfrak{R}$, then $\exists\, y \in A$ such that $xy-1 \notin A^{\times}$. Indeed, there exists a maximal ideal $\mathfrak{m}$ such that $x \notin \mathfrak{m}$. Then $\mathfrak{m} + (x) = (1)$. So there exists $y \in A$ such that $xy - 1 \in \mathfrak{m}$. □

A special class of rings with simple Jacobson radicals is as follows.

Definition 1.2.10.

1. *The ring A is called a local ring, if there exists a unique maximal ideal $\mathfrak{m}$. In this case, $k = A/\mathfrak{m}$ is called the residue field of A.*
2. *The ring A is called semi-local, if there exist only finitely many maximal ideals.*

Examples 1.2.11.

1. *Any field is a local ring.*
2. *Let p be a prime number, then $\mathbb{Z}_{(p)} = \{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \notin p\mathbb{Z}\} \subset \mathbb{Q}$ is a local ring with maximal ideal $p\mathbb{Z}_{(p)}$ and residue field $\mathbb{F}_p$.*
3. *Let k be a field, $n \geq 1$, then the formal power series ring $k[[X_1, \cdots, X_n]]$ is local with maximal ideal $\mathfrak{m} = (X_1, \cdots, X_n)$ and residue field k.*
4. *For any $n \geq 2$, the ring $\mathbb{Z}/n\mathbb{Z}$ is semi-local. It is local $\Leftrightarrow n = p^r$, where p is a prime number and $r \geq 1$.*

Lemma 1.2.12.

1. *Let A be a ring, $\mathfrak{m} \subset A$ an ideal such that any $x \in A \setminus \mathfrak{m}$ is a unit. Then A is local and $\mathfrak{m}$ is its maximal ideal.*
2. *Let A be a ring, $\mathfrak{m} \subset A$ a maximal ideal such that for any $x \in \mathfrak{m}, 1+x$ is a unit. Then A is local.*

Proof. Exercise. □

If A is local, its Jacobson radical $\mathfrak{R} = \mathfrak{m}$. For a general ring, we have

$$\mathfrak{R} \supset nilrad(A).$$

Now we come back to finitely generated A-modules.

Proposition 1.2.13 (Nakayama Lemma)**.** *Let A be a ring, M a finitely generated A-module, $\mathfrak{a} \subset \mathfrak{R}$ an ideal. If $\mathfrak{a}M = M$, then $M = 0$.*

Proof. By Corollary 1.2.7, there exists $x \in A$ such that $x - 1 \in \mathfrak{a}$ and $xM = 0$. Then $1 - x \in \mathfrak{R}$. By Lemma 1.2.9, $1 - (1-x) = x$ is a unit. Thus $M = x^{-1}xM = 0$. □

Remark 1.2.14. *The finiteness condition is important, without it the proposition is not true: consider $A = \mathbb{Z}_{(p)} \subset M = \mathbb{Q}$, then $\mathfrak{m}\mathbb{Q} = \mathbb{Q}$ but $\mathbb{Q} \neq 0$.*

Corollary 1.2.15. *Let M a finitely generated A-module, $N \subset M$ a submodule, $\mathfrak{a} \subset \mathfrak{R}$ an ideal. If $M = \mathfrak{a}M + N$, then $M = N$.*

Proof. Apply Proposition 1.2.13 to M/N. □

Corollary 1.2.16. *Let A be a local ring with maximal ideal $\mathfrak{m}$ and residue field k. Let M be a finitely generated A-module. Suppose $x_i, 1 \leq i \leq n$ are elements of M such that $\overline{x_i}, 1 \leq i \leq n$ form a basis of the k-vector space M/mM. Then $x_i, 1 \leq i \leq n$ generates M.*

Proof. Let $N = (x_1, \cdots, x_n) \subset M$ be the submodule genetated by $x_i, 1 \leq i \leq n$. Then the projection $M \to M/\mathfrak{m}M$ induces a surjection $N \twoheadrightarrow M/\mathfrak{m}M$ by assumption. Thus $N + \mathfrak{m}M = M$. By Corollary 1.2.15 we have $N = M$. □

Proposition 1.2.17. *Let A be a ring, M a finitely generated A-module. If $f : M \to M$ is a surjective A-module homomorphism. Then f is bijective.*

Proof. Let $R = A[X]$ be the one variable polynormial ring over A. Then M can be viewed as an R-module via f: $\forall\, P(X) \in R, m \in M$, we define

$$P(X)m = P(f)(m).$$

In particular $Xm = f(m)$. Since f is surjective, we have $f(M) = M$. Let $I = (X) \subset R$, then

$$IM = M.$$

Now M is finitely generated over R. By Corollary 1.2.7, there exists $Q(X) \in R$ such that $Q(X) - 1 \in I = (X)$ and $Q(X)M = 0$. Write $Q(X) = 1 - XG(X)$ with $G(X) \in R = A[X]$. It annihilates M. So $m = XG(X)m, \forall\, m \in M$. This is

$$m = fG(f)(m).$$

It follows that f is injective. □

Exercise 1.2.18. *[2] Chapter 2, exercises 5, 8, 10, 12, 13.*

Chapter 2

Localization

In basic algebra, we construct the rational numbers from integers by

$$\mathbb{Q} = \{\frac{a}{b} \,|\, a, b \in \mathbb{Z}, b \neq 0\}, \quad \text{and} \quad \frac{a}{b} = \frac{r}{s} \quad \Leftrightarrow \quad as - br = 0.$$

We will generalize this construction to arbitrary rings and modules.

2.1 Localization of rings

Let A be a ring. Recall a multiplicative subset of A is a subset $S \subset A$ such that $1 \in S$ and for any $s_1, s_2 \in S$ we have $s_1 s_2 \in S$.

Examples 2.1.1. *The following sets are multiplicative subsets of A.*

1. *For any $a \in A$, $S = \{1, a, a^2, \cdots\}$.*
2. *For any $\mathfrak{p} \in \operatorname{Spec} A$, $S = A \setminus \mathfrak{p}$.*
3. *$S = \{s \in A \,|\, s \text{ non zero-divisor}\}$. If A is an integral domain, $S = A \setminus \{0\}$.*
4. *For any ideal $\mathfrak{a} \subset A$, $S = 1 + \mathfrak{a}$.*

Definition 2.1.2. *Let S be a multiplicative subset of A. Consider the equivalence relation $\sim$ on $A \times S$: $\forall\, (a, s), (b, t) \in A \times S$,*

$$(a, s) \sim (b, t) \quad \Leftrightarrow \quad (at - bs)u = 0, \quad \text{for some}\, u \in S.$$

Write $\frac{a}{s}$ for the equivalence class of (a, s). Then the quotient set

$$S^{-1}A := A \times S / \sim = \{\frac{a}{s} \,|\, a \in A, s \in S\}$$

is a commutative ring:

$$\frac{a}{s} + \frac{b}{t} = \frac{at + bs}{st}, \quad \frac{a}{s} \cdot \frac{b}{t} = \frac{ab}{st}, \quad 1 = \frac{1}{1}.$$

$S^{-1}A$ is called the localization of A with respective to S.

Exercise 2.1.3. *Check the above $+, \cdot$ are well defined.*

We have a natural ring homomorphism $f : A \to S^{-1}A, \quad x \mapsto \frac{x}{1}$.

Proposition 2.1.4. *1. $f : A \to S^{-1}A$ has the following properties:*

(a) $\forall\, s \in S$, $f(s)$ is a unit in $S^{-1}A$.

(b) if $a \in \mathrm{Ker} f$, then $\exists\, s \in S$ such that $as = 0$.

(c) any $x \in S^{-1}A$ can be written as $f(a)f(s)^{-1}$ for some $a \in A, s \in S$.

2. Let $g : A \to B$ be a ring homomorphism. If $\forall\, s \in S$ $g(s)$ is a unit, then there exists unique $h : S^{-1}A \to B$ such that $g = h \circ f$. If moreover for any $a \in \mathrm{Ker} g, \exists\, s \in S$ such that $as = 0$ and any $x \in B$ can be written as $g(a)g(s)^{-1}$ for some $a \in A, s \in S$, then the homomorphism h is an isomorphism.

Proof. Exercise. □

Examples 2.1.5. *1. $S^{-1}A = 0 \Leftrightarrow 0 \in S$.*

2. For $f \in A, S = \{f^n\}_{n \geq 0}$, write $A_f := S^{-1}A = \{\frac{a}{f^n} \,|\, a \in A, n \geq 0\}$.

3. For $\mathfrak{p} \in \mathrm{Spec}\, A, S = A \backslash \mathfrak{p}$, $A_\mathfrak{p} := S^{-1}A$ is a local ring, with maximal ideal $\mathfrak{m} = \{\frac{a}{s} \,|\, a \in \mathfrak{p}, s \in S\} = f(\mathfrak{p})A_\mathfrak{p}$. Indeed, $\forall\, \frac{b}{t} \notin \mathfrak{m} \Rightarrow b \notin \mathfrak{p}$, i.e. $b \in S \Rightarrow \frac{b}{t}$ is a unit. In the case $A = \mathbb{Z} \supset \mathfrak{p} = (p)$, p prime, $\mathbb{Z}_{(p)} = S^{-1}\mathbb{Z}$.

4. Let $S_0 = \{$non zero-divisors$\}$. Then $S_0^{-1}A$ is called the total quotient ring of A. Note that $f : A \to S_0^{-1}A$ is injective. In case A is an integral domain, $S_0 = A \setminus \{0\}$, $S_0^{-1}A$ is a field, called the fraction field of A, which we denote also $\mathrm{Frac}\, A$, *e.g.* $\mathbb{Q} = \mathrm{Frac}\, \mathbb{Z}$.

We want to describe the ideals of $S^{-1}A$ in terms of ideals of A.

Notation 2.1.6. *Let $f : A \to B$ be a ring homomorphism. For any ideal $\mathfrak{a} \subset A$, we write $\mathfrak{a}^e := f(\mathfrak{a})B$ the ideal of B generated by $f(\mathfrak{a})$. For any ideal $\mathfrak{b} \subset B$, we write $\mathfrak{b}^c := f^{-1}(\mathfrak{b})$ the ideal of A given by the inverse image of $\mathfrak{b}$. Sometimes we also write $\mathfrak{b}^c = \mathfrak{b} \cap A$.*

For $\mathfrak{a}, \mathfrak{b}$ as above, note

$$\mathfrak{a} \subset \mathfrak{a}^{ec}, \quad \mathfrak{b} \supset \mathfrak{b}^{ce}, \quad \mathfrak{a}^e = \mathfrak{a}^{ece}, \quad \mathfrak{b}^c = \mathfrak{b}^{cec}.$$

In the following we consider the natural homomorphism $f : A \to S^{-1}A$. For any ideal $\mathfrak{a} \subset A$,

$$\mathfrak{a}^e = f(\mathfrak{a})S^{-1}A = S^{-1}\mathfrak{a} = \{\frac{a}{s} \,|\, a \in \mathfrak{a}, s \in S\}.$$

Proposition 2.1.7. *1. Any ideal of $S^{-1}A$ is of the form $S^{-1}\mathfrak{a}$, for some ideal $\mathfrak{a} \subset A$.*

2. If $\mathfrak{a} \subset A$ is an ideal, then $\mathfrak{a}^{ec} = (S^{-1}\mathfrak{a})^c = \bigcup_{s \in S}(\mathfrak{a} : s)$, where $(\mathfrak{a} : s) = \{x \in A \,|\, xs \in \mathfrak{a}\}$.

3. $\mathfrak{a} = \mathfrak{b}^c$ for some $\mathfrak{b} \subset S^{-1}A$ ideal $\Leftrightarrow \forall\, s \in S, \overline{s} \in A/\mathfrak{a}$ is a non zero-divisor.

4. $f^* : \mathrm{Spec}\, S^{-1}A \to \mathrm{Spec}\, A$ is injectie, with image $\{\mathfrak{p} \in \mathrm{Spec}\, A \,|\, \mathfrak{p} \cap S = \emptyset\}$.

5. The operation $I \to S^{-1}I$ commutes with finite sums, products, interesections, radicals.

Proof. For 1, 2, 3, 5, see [2]. We only prove 4.

For any $\mathfrak{q} \in \operatorname{Spec} S^{-1}A$, $\mathfrak{q}^c \in \operatorname{Spec} A$. If $\mathfrak{q}^c \cap S \neq \emptyset$, then $1 \in S^{-1}\mathfrak{q}^c \subset \mathfrak{q}$ absurd. Conversely, for any $\mathfrak{p} \in \operatorname{Spec} A$, $A/\mathfrak{p}$ is an integral domain. Let $\overline{S}$ be the image of S in $A/\mathfrak{p}$, which is a multiplicative subset of $A/\mathfrak{p}$ and

$$S^{-1}A/S^{-1}\mathfrak{p} \simeq \overline{S}^{-1}(A/\mathfrak{p}).$$

Then $\overline{S}^{-1}(A/\mathfrak{p})$ is either 0 or a subring of the fraction field of $A/\mathfrak{p}$. So $S^{-1}\mathfrak{p}$ is either $S^{-1}A$ or a prime ideal. Moreover we have $S^{-1}\mathfrak{p} = S^{-1}A \Leftrightarrow \mathfrak{p} \cap S \neq \emptyset$. This proves that $\{\mathfrak{p} \in \operatorname{Spec} A \,|\, \mathfrak{p} \cap S = \emptyset\} \to \operatorname{Spec} S^{-1}A, \mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ is an inverse to $f^* : \operatorname{Spec} S^{-1}A \to \{\mathfrak{p} \in \operatorname{Spec} A \,|\, \mathfrak{p} \cap S = \emptyset\}, \mathfrak{q} \mapsto \mathfrak{q}^c$.

□

Corollary 2.1.8. *1. For any prime ideal $\mathfrak{p} \subset A$, we have a homeomorphism*

$$\operatorname{Spec} A_{\mathfrak{p}} \simeq \{\mathfrak{q} \in \operatorname{Spec} A \,|\, \mathfrak{q} \subset \mathfrak{p}\}.$$

2. For any $f \in A$, we have a homeomorphism

$$\operatorname{Spec} A_f \simeq D(f) = \operatorname{Spec} A \setminus V(f).$$

Recall we have also

$$\operatorname{Spec} A/\mathfrak{p} \simeq V(\mathfrak{p}) = \{\mathfrak{q} \in \operatorname{Spec} A \,|\, \mathfrak{q} \supset \mathfrak{p}\}.$$

$A_{\mathfrak{p}}$ is a local ring with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. Let $S = A \setminus \mathfrak{p}$ and $\overline{S}$ be its image in $A/\mathfrak{p}$. Then $\overline{S} = \{\text{non zero-divisors of } A/\mathfrak{p}\} = A/\mathfrak{p} \setminus \{0\}$, and we have

$$A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} = S^{-1}A/S^{-1}\mathfrak{p} = \overline{S}^{-1}(A/\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p}).$$

2.2 Localization of modules

Let A be a ring, $S \subset A$ a multiplicative subset, $M \in \operatorname{Mod}_A$.

Definition 2.2.1. *Consider the equivalence relation $\sim$ on $M \times S$:*

$$(m, s) \sim (m', s') \quad \Leftrightarrow \quad \exists\, t \in S, \quad t(ms' - sm') = 0.$$

The equivalence class of (m, s) is written as $\frac{m}{s}$. Then the quotient set

$$S^{-1}M := M \times S/\sim = \{\frac{m}{s} \,|\, m \in M, s \in S\}$$

is a $S^{-1}A$-module:

$$\frac{a}{s} \cdot \frac{m}{t} = \frac{am}{st}, \quad \frac{a}{s} \in S^{-1}A, \quad \frac{m}{t} \in S^{-1}M.$$

One can check that the above is well defined.

Notation 2.2.2. *For any $\mathfrak{p} \in \operatorname{Spec} A$, write $M_{\mathfrak{p}} = S^{-1}M$ for $S = A \setminus \mathfrak{p}$. For any $f \in A$, write $M_f = S^{-1}M$ for $S = \{f^n\}_{n \geq 0}$.*

If $u : M \to N$ is a homomorphism of A-modules, then

$$S^{-1}u : S^{-1}M \to S^{-1}N, \quad \frac{m}{s} \mapsto \frac{u(m)}{s}$$

is a homomorphism of $S^{-1}A$-modules. In this way we get a functor

$$S^{-1} : \mathrm{Mod}_A \to \mathrm{Mod}_{S^{-1}A}.$$

Proposition 2.2.3. *The functor S^{-1} is exact.*

Proof. Exercise. □

Corollary 2.2.4. *Let $N, P \subset M$ be submodules. Then we have*

1. $S^{-1}(N + P) = S^{-1}N + S^{-1}P$,
2. $S^{-1}(N \cap P) = S^{-1}N \cap S^{-1}P$,
3. $S^{-1}(M/N) \simeq S^{-1}M/S^{-1}N$.

Proof. Exercise. □

Proposition 2.2.5. *For any $M \in \mathrm{Mod}_A$, there exists a unique isomorphism of $S^{-1}A$-modules*

$$\phi : S^{-1}A \otimes_A M \to S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s}.$$

Proof. Consider the map

$$S^{-1}A \times M \to S^{-1}M, \quad (\frac{a}{s}, m) \mapsto \frac{am}{s}.$$

This is A-bilinear, thus there exists an A-linear homomorphism

$$\phi : S^{-1}A \otimes_A M \to S^{-1}M, \quad \frac{a}{s} \otimes m \mapsto \frac{am}{s}.$$

Then ϕ is $S^{-1}A$-linear and surjective. One then checks by definition $\mathrm{Ker}\phi = 0$. □

Corollary 2.2.6. *$S^{-1}A$ is flat over A.*

Corollary 2.2.7. *For any $M, N \in \mathrm{Mod}_A$, there exists a unique isomorphism of $S^{-1}A$-modules*

$$\phi : S^{-1}M \otimes_{S^{-1}A} S^{-1}N \simeq S^{-1}(M \otimes_A N), \quad \frac{m}{s} \otimes \frac{n}{t} \mapsto \frac{m \otimes n}{st}.$$

In particular, for any $\mathfrak{p} \in \mathrm{Spec}\, A$, $M_\mathfrak{p} \otimes_{A_\mathfrak{p}} N_\mathfrak{p} \simeq (M \otimes_A N)_\mathfrak{p}$ as $A_\mathfrak{p}$-modules.

Proof. Since $S^{-1}M \simeq S^{-1}A \otimes_A M \simeq M \otimes_A S^{-1}A$, we have

$$\begin{aligned} S^{-1}M \otimes_{S^{-1}A} S^{-1}N &\simeq M \otimes_A S^{-1}A \otimes_{S^{-1}A} S^{-1}N \\ &\simeq M \otimes_A S^{-1}A \otimes_{S^{-1}A} S^{-1}A \otimes_A N \\ &\simeq S^{-1}A \otimes_A (M \otimes_A N) \\ &\simeq S^{-1}(M \otimes_A N). \end{aligned}$$

□

We give some applications of the technique of localization.

Proposition 2.2.8. *Let $M \in \mathrm{Mod}_A$. TFAE:*

1. $M = 0$,
2. $M_{\mathfrak{p}} = 0$ *for any prime ideal* $\mathfrak{p} \subset A$,
3. $M_{\mathfrak{m}} = 0$ *for any maximal ideal* $\mathfrak{m} \subset A$.

Proof. $1 \Rightarrow 2 \Rightarrow 3$ are trivial. We show $3 \Rightarrow 1$. Suppose $M \neq 0$. Take $0 \neq x \in M$ and consider $\mathfrak{a} = ann(x) = \{y \in A \,|\, yx = 0\}$. Then $\mathfrak{a} \subsetneq A$ is a proper ideal. There exists a maximal ideal $\mathfrak{m} \supset \mathfrak{a}$. Since $\frac{x}{1} \in M_{\mathfrak{m}} = 0$, there exists $s \in A \setminus \mathfrak{m}$ such that $xs = 0$. Then $s \in ann(x) = \mathfrak{a} \subset \mathfrak{m}$. Contradication! $\square$

Corollary 2.2.9. *Let $\phi : M \to N$ be a homomorphism of A-modules. TFAE:*

1. ϕ *is injective,*
2. $\phi_{\mathfrak{p}} : M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ *is injective for any prime ideal* $\mathfrak{p} \subset A$,
3. $\phi_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ *is injective for any maximal ideal* $\mathfrak{m} \subset A$.

Similar statements hold with "injective" replaced by "surjective".

Proof. $1 \Rightarrow 2$: since S^{-1} is exact.
$2 \Rightarrow 3$: trivial.
$3 \Rightarrow 1$: apply Proposition 2.2.8 to $M' = \mathrm{Ker}\phi$.
Similarly for surjectivity. $\square$

Proposition 2.2.10. *Let $M \in \mathrm{Mod}_A$. TFAE:*

1. M *is flat over* A,
2. $M_{\mathfrak{p}}$ *is flat over* $A_{\mathfrak{p}}$ *for any prime ideal* $\mathfrak{p} \subset A$,
3. $M_{\mathfrak{m}}$ *is flat over* $A_{\mathfrak{m}}$ *for any maximal ideal* $\mathfrak{m} \subset A$.

Proof. $1 \Rightarrow 2$: $M_{\mathfrak{p}} \simeq M \otimes_A A_{\mathfrak{p}}$, then apply the base change property of flatness.
$2 \Rightarrow 3$: trivial.
$3 \Rightarrow 1$: Let $0 \to N_1 \to N_2$ be any exact sequence of A-modules. Then $0 \to N_{1,\mathfrak{m}} \to N_{2,\mathfrak{m}}$ is exact for any maximal ideal $\mathfrak{m} \subset A$ as S^{-1} is exact. Since $M_{\mathfrak{m}}$ flat over $A_{\mathfrak{m}}$, the sequence $0 \to N_{1,\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}} \to N_{2,\mathfrak{m}} \otimes_{A_{\mathfrak{m}}} M_{\mathfrak{m}}$ is exact. This is isomorphic to $0 \to (N_1 \otimes_A M)_{\mathfrak{m}} \to (N_2 \otimes_A M)_{\mathfrak{m}}$. Apply Corollary 2.2.9, we see $0 \to N_1 \otimes_A M \to N_2 \otimes_A M$ is exact, i.e. M is flat over A. $\square$

Remark 2.2.11. *Let A be a ring and $X = \mathrm{Spec}\, A$. The constructions of this chapter can be easily upgraded to construct a sheaf of rings $\mathcal{O}_X$ on X, and more generally for any $M \in \mathrm{Mod}_A$, one can construct a sheaf $\widetilde{M}$ of $\mathcal{O}_X$-modules. The pair $(X, \mathcal{O}_X)$ is the affine scheme associated to A. In this way, the category Mod_A is equivalent to the category of quasi-coherent $\mathcal{O}_X$-modules on X. For more details, see [8] sections II.2 and II.5.*

Exercise 2.2.12. *[2] Chapter 3, exercises 2, 3, 15, 19, 21, 22.*

Chapter 3

Flatness

We study some further properties of flat modules and algebras.

3.1 Flat modules

Let A be a ring, $M \in \mathrm{Mod}_A$, recall that M is flat over A if $M \otimes - : \mathrm{Mod}_A \to \mathrm{Mod}_A$ is an exact functor. TFAE:

1. M is flat over A.
2. for any exact sequence of A-modules $0 \to N' \to N$, the sequence $0 \to N' \otimes M \to N \otimes M$ is exact.
3. for any $N \in \mathrm{Mod}_A$, $\mathrm{Tor}_n^A(M, N) = 0, \forall\, n \geq 1$.
4. for any $N \in \mathrm{Mod}_A$, $\mathrm{Tor}_1^A(M, N) = 0$.

We will give some further equivalent conditions.

Lemma 3.1.1. *Let $M, N \in \mathrm{Mod}_A$, and $x_i \in M, y_i \in N$ be such that $\sum x_i \otimes y_i = 0$ in $M \otimes N$. Then there exist finitely generated submodules $M_0 \subset M, N_0 \subset N$ such that $x_i \in M_0, y_i \in N_0$ and $\sum x_i \otimes y_i = 0$ in $M_0 \otimes N_0$.*

Proof. Exercise (see [2] Corollary 2.13). ☐

Corollary 3.1.2. *M flat over $A \Leftrightarrow$ for any exact sequence $0 \to N' \to N$ of finitely generated A-modules, the sequence $0 \to M \otimes N' \to M \otimes N$ exact.*

Proof. The direction $\Rightarrow$ is OK. For any exact sequence $0 \to N' \xrightarrow{f} N$ of A-modules, we want to show the sequence $0 \to M \otimes N' \xrightarrow{id \otimes f} M \otimes N$ exact, i.e. $id \otimes f$ is injective. Let $u = \sum x_i \otimes y_i \in \mathrm{Ker}(id \otimes f) \subset M \otimes N'$, with $x_i \in M, y_i \in N'$. Then $\sum x_i \otimes f(y_i) = 0$ in $M \otimes N$. Let $N_0' \subset N'$ be the submodule generated by y_i and $u_0 = \sum x_i \otimes y_i \in M \otimes N_0'$, which is mapped to 0 under

$$M \otimes N_0' \to M \otimes N' \to M \otimes N, u_0 \mapsto u \mapsto 0.$$

Let $N_0 \subset N$ be a finitely generated submodule such that $N_0 \supset f(N_0')$ and $\sum x_j \otimes f(y_i) = 0 \in M \otimes N_0$. Then f induces an injection $N_0' \to N_0$. We have $id \otimes f_0 : M \otimes N_0' \to M \otimes N_0$ and $id \otimes f_0(u_0) = 0$. By assumption $u_0 = 0$ in $M \otimes N_0'$. Thus $u = 0$ and $\mathrm{Ker}(id \otimes f) = 0$. ☐

Corollary 3.1.3. *M flat over $A \Leftrightarrow$ for any finitely generated $N \in \mathrm{Mod}_A, \mathrm{Tor}_1^A(M,N) = 0$.*

Remark 3.1.4. *One can also prove the above corollary from the following two basic facts:*

- *Let $\mathcal{I}$ be a directed poset, then the direct limit functor $\varinjlim : \mathrm{Mod}_A^{\mathcal{I}} \to \mathrm{Mod}_A$ is exact.*
- *The direct limit functor commutes with Tor: $\mathrm{Tor}_*^A(M, \varinjlim_i N_i) \cong \varinjlim_i \mathrm{Tor}_*^A(M, N_i)$.*

Corollary 3.1.5. *TFAE:*

1. *M is flat over A.*
2. *For any finitely generated ideal $I \subset A$, the sequence $0 \to I \otimes M \to M = A \otimes M$ is exact.*
3. *For any finitely generated ideal $I \subset A$, $\mathrm{Tor}_1^A(M, A/I) = 0$.*

Proof. $1 \Rightarrow 2 \Leftrightarrow 3$ are OK. We show $3 \Rightarrow 1$: By Corollary 3.1.3, it suffices to show $\mathrm{Tor}_1^A(M,N) = 0$ for any finitely generated $N \in \mathrm{Mod}_A$. Take such a N. Let $x_1, \ldots, x_n$ be a set of generators of N and set $N_i = (x_1, \ldots, x_i)$. Then we get

$$0 \subset N_1 \subset N_2 \subset \cdots \subset N_{i-1} \subset N_i \subset \cdots \subset N.$$

Consider the exact sequence $0 \to N_{i-1} \to N_i \to N_i/N_{i-1} \to 0$. We get a long exact sequence

$$\cdots \to \mathrm{Tor}_1^A(M, N_{i-1}) \to \mathrm{Tor}_1^A(M, N_i) \to \mathrm{Tor}_1^A(M, N_i/N_{i-1}) \to \cdots$$

Let $I_i = ann(x_i)$, then $N_i/N_{i-1} \simeq A/I_i$. By induction, it suffices to show $\mathrm{Tor}_1^A(A/I) = 0$, where $I \subset A$ is an ideal. The short exact sequence $0 \to I \to A \to A/I$ induces

$$0 = \mathrm{Tor}_1^A(M, A) \to \mathrm{Tor}_1^A(M, A/I) \to I \otimes M \to M \to \cdots$$

Now $3 \Leftrightarrow 2 \Rightarrow$ for any ideal I, $0 \to I \otimes M \to M$ is exact $\Rightarrow$ for any ideal I, $\mathrm{Tor}_1^A(M, A/I) = 0$. □

Proposition 3.1.6. *M flat over $A \Leftrightarrow$ If $a_i \in A, x_i \in M, 1 \le i \le r$ and $\sum_{i=1}^r a_i x_i = 0$, then there exists an integer s and elements $b_{ij} \in A$, $y_j \in M$, $1 \le j \le s$, such that $\sum_{i=1}^r a_i b_{ij} = 0, \forall\, j$ and $x_i = \sum_{j=1}^s b_{ij} y_j \ \forall\, i$.*

Proof. $\Rightarrow$: If M flat and $\sum_{i=1}^r a_i x_i = 0$, consider the exact sequence

$$0 \to K \to A^r \xrightarrow{f} A,$$

where $f(b_1, \cdots, b_r) = \sum_{i=1}^r a_i b_i$ and $K = \mathrm{Ker} f$, we get an exact sequence

$$0 \to K \otimes M \to M^r \xrightarrow{f_M} M$$

with $f_M(t_1, \cdots, t_r) = \sum_{i=1}^r a_i t_i$. Then $(x_1, \cdots, x_r) \in \mathrm{Ker} f_M = K \otimes M$, so there exist an integer s and for each $1 \le j \le s$ elements $\beta_j \in K \subset A^r$ and $y_j \in M$ such that

$$(x_1, \cdots, x_r) = \sum_{j=1}^s \beta_j \otimes y_j.$$

Write $\beta_j = (b_{1j}, \cdots, b_{rj})$, $\quad b_{ij} \in A$, then $x_i = \sum_{j=1}^s b_{ij} y_j$ for each i. As $\beta_j \in K$, we have $f(\beta_j) = 0$, i.e. $\sum_{i=1}^r a_i b_{ij} = 0, \forall\, j$.

$\Leftarrow$: for any finitely generated ideal $I \subset A$, we show $f : I \otimes M \to M$ is injective. Let $a_1, \ldots, a_r$ be a set of generators of I. For any $\xi \in I \otimes M$, write

$$\xi = \sum_{i=1}^r a_i \otimes x_i, \quad \text{then} \quad f(\xi) = \sum_{i=1}^r a_i x_i.$$

If $\xi \in \mathrm{Ker} f$, i.e. $\sum_{i=1}^r a_i x_i = 0$, by assumption, $\exists\, b_{ij} \in A, y_j \in M$ such that $\sum_i a_i b_{ij} = 0$ for any j and $x_i = \sum_{j=1}^s b_{ij} y_j$ for any i. Then

$$\xi = \sum_{i=1}^r a_i \otimes x_i = \sum_{i=1}^r \sum_{j=1}^s a_i b_{ij} \otimes y_j = 0, \quad \Rightarrow \mathrm{Ker} f = 0.$$

$\square$

Corollary 3.1.7. *Let A be a local ring with $\mathfrak{m} \subset A$ maximal ideal and $k = A/\mathfrak{m}$ residue field. Let M be an A-module. Suppose (either $\mathfrak{m}$ is nilpotent, or) M is finitely generated over A. Then*

$$M \quad \textit{free} \quad \Leftrightarrow \quad M \quad \textit{projective} \quad \Leftrightarrow \quad M \quad \textit{flat}.$$

Proof. The directions $\Rightarrow \cdots \Rightarrow$ are OK.

We show if M flat, then it is free. It suffices to prove: if $x_1, \ldots, x_n \in M$ such that $\overline{x_1}, \ldots, \overline{x_n} \in M/\mathfrak{m}M = M \otimes_A k$ are linearly independt over k, then $x_1, \ldots, x_n$ are linearly independent over A. We prove this by induction on n. For $n = 1, x = x_1$, if $\exists\, a \in A$ such that $ax = 0$. Since M flat, by Proposition 3.1.6 there exists $y_1, \ldots, y_r \in M$ and $b_1, \ldots, b_r \in A$ such that $x = \sum b_i y_j$ and $ab_i = 0$ for any i. Since $\overline{x} \neq 0$ in $M/\mathfrak{m}M$, there exists an i such that $b_i \notin \mathfrak{m}$. Then b_i is a unit, and thus $a = 0$. Suppose $n > 1$ and $\sum_{i=1}^n a_i x_i = 0$. By Proposition 3.1.6, there exist $y_1, \ldots, y_r \in M, b_{ij} \in A, 1 \leq j \leq r$ such that $x_i = \sum_{j=1}^r b_{ij} y_j$ and $\sum_{i=1}^n a_i b_{ij} = 0$. By assumption, $x_n \notin \mathfrak{m}M$. Then $b_{nj} \notin \mathfrak{m}$ for some j, i.e. it is a unit. Then

$$a_1 b_{1j} + \cdots + a_n b_{nj} = 0 \quad \Rightarrow a_n = \sum_{i=1}^{n-1} c_i a_i, c_i = -b_{ij} b_{nj}^{-1}.$$

Now we have

$$\sum_{i=1}^n a_i x_i = a_1(x_1 + c_1 x_n) + \cdots + a_{n-1}(x_{n-1} + c_{n-1} x_n) = 0,$$

modulo $\mathfrak{m}$ and since $\overline{x_1} + \overline{c_1 x_n}, \ldots, \overline{x_{n-1}} + \overline{c_{n-1} x_n}$ are linearly independent over k, by induction hypothesis, $a_1 = \cdots = a_{n-1} = 0$, and then $a_n = \sum_{i=1}^{n-1} c_i a_i = 0$. $\square$

Remark 3.1.8. *Let A be a local ring, $M \in \mathrm{Mod}_A$ any A-module. Then*

- *M projective $\Leftrightarrow$ M free. The direction $\Leftarrow$ is clear. The direction $\Rightarrow$ is due to Kaplansky.*
- *We know M projective $\Rightarrow$ M flat. Without the finiteness condition, the direction $\Leftarrow$ is not true in general: consider $A = \mathbb{Z}_{(p)}, M = \mathbb{Q}$.*

We can combine Proposition 2.2.10 and Corollary 3.1.7 together.

Corollary 3.1.9. *Let A be a ring, M a finitely generated A-module. Then M is flat over $A \Leftrightarrow$ for any prime ideal $\mathfrak{p} \subset A$, $M_\mathfrak{p}$ is free over $A_\mathfrak{p} \Leftrightarrow$ for any maximal ideal $\mathfrak{m} \subset A$, $M_\mathfrak{m}$ is free over $A_\mathfrak{m}$.*

3.2 Faithful flatness

Let A be a ring, $M \in \mathrm{Mod}_A$.

Definition 3.2.1. *M is faithfully flat (f.f.) over A if for any sequence of A-modules of homomorphism $\mathcal{S}$, $\mathcal{S} \otimes M$ is exact $\Leftrightarrow \mathcal{S}$ is exact.*

By definition, faithfully flat is flat, but the converse is not ture in general: let $B \neq 0, C \neq 0$ be rings and consider $A = B \times C$. Then B is a projective A-module, thus flat over A, but it is not faithfully flat over A (since $\operatorname{Spec} B \to \operatorname{Spec} A$ is not surjective, see the following Proposition 3.3.1).

Theorem 3.2.2. *TFAE:*

1. *M is faithfully flat over A.*
2. *M is flat over A and for any $0 \neq N \in \mathrm{Mod}_A, N \otimes M \neq 0$.*
3. *M is flat over A and for any maximal ideal $\mathfrak{m} \subset A$, $\mathfrak{m}M \neq M$.*

Proof. $1 \Rightarrow 2$: If $N \in \mathrm{Mod}_A$ such that $N \otimes M = 0$ then $0 \to N \to 0$ is exact since $0 \to N \otimes M \to 0$ is exact and M faithfully flat over A, thus $N = 0$.

$2 \Rightarrow 3$: Since $A/\mathfrak{m} \neq 0$, $A/\mathfrak{m}M \otimes M = M/\mathfrak{m}M \neq 0$.

$3 \Rightarrow 2$: $\forall\, 0 \neq x \in N$, $Ax \simeq A/I$ for some ideal $I \subsetneq A$ ($I = ann(x)$). Take a maximal ideal $\mathfrak{m}$ such that $I \subset \mathfrak{m}$. Then

$$M \supset \mathfrak{m}M \supset IM \Rightarrow A/I \otimes M = M/IM \neq 0.$$

Consider the exact sequence $0 \to Ax \to N$. Since M is flat, the sequence $0 \to Ax \otimes M \to N \otimes M$ is exact. Now

$$Ax \otimes I \simeq M/IM \neq 0 \Rightarrow N \otimes M \neq 0.$$

$2 \Rightarrow 1$: Let $\mathcal{S} : N' \xrightarrow{f} N \xrightarrow{g} N''$ sequence of A-module such that $N' \otimes M \xrightarrow{f_M} N \otimes M \xrightarrow{g_M} N' \otimes M$ is exact. Since M is flat, $\mathrm{Im}(g \circ f) \otimes = \mathrm{Im}(g_M \circ f_M) = 0$. By assumption $\mathrm{Im}(g \circ f) = 0$, i.e. $g \circ f = 0$. Let $H(\mathcal{S}) = \mathrm{Ker} g / \mathrm{Im} f$. Then $H(\mathcal{S}) \otimes M = H(\mathcal{S} \otimes M) = 0$ since $\mathcal{S} \otimes M$ is exact. By assumption, $H(\mathcal{S}) = 0$, i.e. $\mathcal{S}$ is exact.

□

Corollary 3.2.3. *Let A, B be local rings with maximal ideals $\mathfrak{m}_A, \mathfrak{m}_B$ repsectively, $f : A \to B$ a local homomorphism, i.e. $f(\mathfrak{m}_A) \subset \mathfrak{m}_B$. Let $M \neq 0$ be a finitely generated B-module. Then M is flat over $A \Leftrightarrow M$ is faithfully flat over A. In particular, B is flat over $A \Leftrightarrow B$ is faithfully flat over A.*

Proof. By assumption, $\mathfrak{m}_A M \subset \mathfrak{m}_B M$. By Nakayama lemma, $M \neq 0 \Rightarrow \mathfrak{m}_B M \neq M \Rightarrow \mathfrak{m}_A M \neq M$. Then apply Theorem 3.2.2. □

Exercise 3.2.4. 1. *Let M be a faithfully flat A-module, B any A-algebra, then $M \otimes_A B$ is faithfully flat over B.*

2. *Let M be a faithfully flat B-module, B a faithfully flat A-algebra, then M is faithfully flat over A.*

3. *Let M be a faithfully flat B-module, B an A-algebra. If M is faithfully flat over A, then B is faithfully flat over A.*

3.3 Flat ring homomorphisms

A ring homomorphism $f : A \to B$ is called flat (resp. faithfully flat) if B is flat (resp. faithfully flat) over A as an A-module via f.

Proposition 3.3.1. *Let $f : A \to B$ be a faithfully flat ring homomorphism. Then*

1. *For any $N \in \mathrm{Mod}_A$, the map $N \to N \otimes_A B, \quad x \mapsto x \otimes 1$ is injective. In particular, f is injecture.*

2. *For any ideal $I \subset A$, we have $IB \cap A = I$.*

3. *The map $f^* : \operatorname{Spec} B \to \operatorname{Spec} A$ is surjective.*

Proof. 1: For any $0 \neq x \in N$, we show $x \otimes 1 \neq 0$. Consider $0 \neq Ax \subset N$, then $Ax \otimes B \subset N \otimes B$ since B is flat over A. As B is faithfully flat over A, we have

$$Ax \otimes B = (x \otimes 1)B \neq 0 \Rightarrow x \otimes 1 \neq 0.$$

2: $A \to B$ is faithfully flat, then $A/I \to B \otimes A/I = B/IB$ is faithfully flat, thus injective by 1. This implies $IB \cap A = I$.

3: For any $\mathfrak{p} \in \operatorname{Spec} A$, the induced map $A_\mathfrak{p} \to B_\mathfrak{p}$ is faithfully flat. Theorem 3.2.2 $\Rightarrow \mathfrak{p}B_\mathfrak{p} \neq B_\mathfrak{p}$. Take any maximal ideal $\mathfrak{m}$ of $B_\mathfrak{p}$ such that $\mathfrak{m} \supset \mathfrak{p}B_\mathfrak{p}$. Then

$$\mathfrak{m} \cap A_\mathfrak{p} \supset \mathfrak{p}A_\mathfrak{p} \Rightarrow \mathfrak{m} \cap A_\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}.$$

Consider the following commutative diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ A_\mathfrak{p} \arrow[r] & B_\mathfrak{p}. \end{tikzcd}$$

Let $P = \mathfrak{m} \cap B \in \operatorname{Spec} B$, then $P \cap A = \mathfrak{p}$. □

Theorem 3.3.2. *Let $f : A \to B$ be a ring homomorphism. TFAE:*

1. *f is faithfully flat.*

2. *f is flat and $f^* : \operatorname{Spec} B \to \operatorname{Spec} A$ is surjective.*

3. *f is flat and for any maximal ideal $\mathfrak{m} \subset A$, there exists a maximal ideal $\mathfrak{m}' \subset B$ such that $\mathfrak{m} = f^{-1}(\mathfrak{m}')$.*

Proof. $1 \Rightarrow 2$ is OK.

$2 \Rightarrow 3$: There exists $\mathfrak{p}' \in \operatorname{Spec} B$ such that $\mathfrak{m} = f^{-1}(\mathfrak{p}')$. Take any $\mathfrak{m}' \subset B$ maximal ideal such that $\mathfrak{m}' \supset \mathfrak{p}'$. Then $\mathfrak{m} = f^{-1}(\mathfrak{m}')$.

$3 \Rightarrow 1$: For any $\mathfrak{m} \subset A$ maximal ideal, let $\mathfrak{m}' \subset B$ be a maximal ideal such that $\mathfrak{m} = f^{-1}(\mathfrak{m}')$. Then $\mathfrak{m}B \subset \mathfrak{m}' \subsetneq B$. By Theorem 3.2.2, B is faithfully flat over A. $\square$

Theorem 3.3.3. *Let $f : A \to B$ be a flat ring homomorphism. Then the going-down theorem holds:*

for any prime ideals $\mathfrak{p}, \mathfrak{p}' \in \operatorname{Spec} A$ such that $\mathfrak{p} \subset \mathfrak{p}'$ and any prime ideal $\mathfrak{q}' \in \operatorname{Spec} B$ such that $\mathfrak{p}' = f^{-1}(\mathfrak{q}')$, there exists a prime ideal $\mathfrak{q} \in \operatorname{Spec} B$ such that $\mathfrak{q} \subset \mathfrak{q}'$ and $\mathfrak{p} = f^{-1}(\mathfrak{q})$

Proof. Let $\mathfrak{p}, \mathfrak{p}', \mathfrak{q}'$ be as above. Then we get the induced homomorphism

$$f_{\mathfrak{p}'} : A_{\mathfrak{p}'} \to B_{\mathfrak{q}'},$$

which is flat (since it is the composition of $A_{\mathfrak{p}'} \to B_{\mathfrak{p}'}$ and $B_{\mathfrak{p}'} \to B_{\mathfrak{q}'}$), thus faithfully flat by Corollary 3.2.3 (this map is local). By Proposition 3.3.1, the map $f^*_{\mathfrak{p}'} : \operatorname{Spec} B_{\mathfrak{q}'} \to \operatorname{Spec} A_{\mathfrak{p}'}$ is surjective. Thus there exists $\mathfrak{q}^* \in \operatorname{Spec} B_{\mathfrak{q}'}$ such that $f_{\mathfrak{p}'}^{-1}(\mathfrak{q}^*) = \mathfrak{p}A_{\mathfrak{p}'}$. Consider the following commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A_{\mathfrak{p}'} & \longrightarrow & B_{\mathfrak{q}'}. \end{array}$$

Take $\mathfrak{q} = \mathfrak{q}^* \cap B \in \operatorname{Spec} B$. Then $\mathfrak{q} \subset P'$ and $f^{-1}(\mathfrak{q}) = \mathfrak{p} \in \operatorname{Spec} A$. $\square$

Exercise 3.3.4. *Let $A \subset B$ be integral doamins which have the same field of fractions. Prove that if B is faithfully flat over A, then $A = B$.*

Remark 3.3.5. *Let $f : A \to B$ be a faithfully flat ring homomorphism. Then one can describe the category Mod_A as the category of B-modules together with descent data with respect to the homomorphism $f : A \to B$. This is Grothendieck's faithfully flat descent theorem for modules, see [7].*

Chapter 4

Integral extensions

4.1 Integral dependence, going-up and going-down theorems

Let B be a ring, $A \subset B$ a subring, then B is also called an extension ring of A.

Definition 4.1.1. *An element $x \in B$ is said to be integral over A if x satisfies an equation of the form*

$$x^n + a_1 x^{n-1} + \cdots + a_n = 0, \quad a_1, \ldots, a_n \in A.$$

Example 4.1.2. *Consider $\mathbb{Z} \subset \mathbb{Q}$. If $x = \frac{r}{s} \in \mathbb{Q}, (r, s) = 1$ integral over $\mathbb{Z}$, then $s = \pm 1, x \in \mathbb{Z}$.*

Proposition 4.1.3. *Let $A \subset B$ be as above. TFAE:*

1. *$x \in B$ integral over A.*
2. *$A[x]$ is a finitely generated A-module.*
3. *$A[x] \subset C \subset B$, where C is a subring of B, which is finitely generated as an A-module.*
4. *There exists a faithful $A[x]$-module M, which is finitely generated as an A-module.*

Proof. $1 \Rightarrow 2$: Since x satisfies an equation of the form $x^n + a_1 x^{n-1} + \cdots + a_n = 0, \quad a_1, \ldots, a_n \in A$, for any $r \geq 0, x^{n+r} = -(a_1 x^{n+r-1} + \cdots + a_n x^r)$. Thus for any $m \geq 1$, $x^m \in A + Ax + \cdots + Ax^{n-1}$, i.e. $A[x]$ is generated over A by $1, x, \ldots, x^{n-1}$.

$2 \Rightarrow 3$: Take $C = A[x]$.

$3 \Rightarrow 4$: Take $M = C$, which is faithful as an $A[x]$-module.

$4 \Rightarrow 1$: Let $\phi \in \mathrm{End}_A(M)$ be the multiplication by x, then $\phi(M) = xM \subset AM = M$. Since M is finitely generated over A, there exist $a_1, \ldots, a_n \in A$ such that $\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$. The condition that M is faithful over $A[x]$ implies that $x^n + a_1 x^{n-1} + \cdots + a_n = 0$. □

Corollary 4.1.4. *Let $x_i \in B, 1 \leq i \leq n$ be integral over A. Then the subring $A[x_1, \ldots, x_n]$ of B is a finitely generated A-module.*

Remark 4.1.5. *A finitely generated A-algebra B is also called of finite type over A. It is called of finite over A if it is finitely generated as an A-module. The above corollary says that finite type + integral = finite.*

Corollary 4.1.6. *Let $C = \{x \in B \mid x \text{ integral over } A\}$. Then $C \subset B$ is a subring containing A*

Proof. For any $x, y \in C$, $A[x, y]$ is a finitely generated A-module and a subring of B. Since $A[x \pm y] \subset A[x, y]$ and $A[xy] \subset A[x, y]$, by Proposition 4.1.3 3, $x \pm y, xy$ integral over A, thus $x \pm y, xy \in C$. So C is a subring of B. $\square$

Definition 4.1.7. 1. *Let $A \subset B$ be an extension of rings, $C = \{x \in B \mid x \text{ integral over } A\}$ is called the integral closure of A in B. If $C = A$ then A is said to be integrally closed in B. If $C = B$ then B is said to be integral over A.*

2. *A ring homomorphism $f : A \to B$ is said to be integral (or B an integral A-algebra) if B is integral over $f(A)$.*

Corollary 4.1.8. *If $A \subset B \subset C$ are rings such that B is integral over A and C is integral over B, then C is integral over A.*

Proof. For any $x \in C$, we have an equation $x^n + b_1x^{n-1} + \cdots + b_n = 0, b_i \in B$. Consider $B' = A[b_1, \cdots, b_n] \subset B$. By Proposition 4.1.3 It is a finitely generated A-module since b_i are integral over A. Then x is integral over B'. By Proposition 4.1.3 again, $B'[x]$ is a finitely generated B'-module. Thus $B'[x]$ is finitely generated as an A-module. This implies that x is integral over A by Proposition 4.1.3. $\square$

Corollary 4.1.9. *Let $A \subset B$ be an extention of rings and C the integral closure of A in B. Then C is integrally closed in B.*

One checks directly by definition the following proposition.

Proposition 4.1.10. *Let $A \subset B$ be rings such that B is integral over A.*

1. *If $\mathfrak{b} \subset B$ is an ideal and $\mathfrak{a} = \mathfrak{b} \cap A$, then $B/\mathfrak{b}$ is integral over $A/\mathfrak{a}$.*

2. *If $S \subset A$ is a multiplicative subset, then $S^{-1}B$ is integral over $S^{-1}A$.*

We discuss some properties of integral extensions.

Proposition 4.1.11. *Let $A \subset B$ be integral domains with B integral over A. Then B is a field $\Leftrightarrow$ A is a field.*

Proof. Suppose that A is a field. For any $0 \neq y \in B$, let $y^n + a_1y^{n-1} + \cdots + a_n = 0, a_i \in A$ be an equation such that n is the smallest possible degree. Since B is an integral domain, we have $a_n \neq 0$, Then

$$y^{-1} := -a_n^{-1}(y^{n-1} + a_1y^{n-2} + \cdots + a_{n-1}) \in B$$

is the inverse of y. This implies that B is a field.

Suppose B is a field. For any $0 \neq x \in A$, $x^{-1} \in B$ integral over A, thus we have an equation $x^{-m} + a'_1x^{-m+1} + \cdots + a'_m = 0, a'_i \in A$. Then

$$x^{-1} := -(a'_1 + \cdots + a'_m x^{m-1}) \in A$$

is the inverse of x. So A is a field.

$\square$

Corollary 4.1.12. *Let $A \subset B$ be an integral extension. Let $\mathfrak{q} \in \operatorname{Spec} B, \mathfrak{p} = \mathfrak{q} \cap A$. Then $\mathfrak{q}$ is maximal $\Leftrightarrow$ $\mathfrak{p}$ is maximal.*

Proof. By Proposition 4.1.10, $B/\mathfrak{q}$ is integral over $A/\mathfrak{p}$. Proposition 4.1.11 implies that $B/\mathfrak{q}$ is a field $\Leftrightarrow$ $A/\mathfrak{p}$ is a field. □

Corollary 4.1.13. *Let $A \subset B$ be an integral extension. Let $\mathfrak{q}, \mathfrak{q}' \in \operatorname{Spec} B$ be such that $\mathfrak{q} \subset \mathfrak{q}'$ and $\mathfrak{q} \cap A = \mathfrak{q}' \cap A$. Then $\mathfrak{q} = \mathfrak{q}'$.*

Proof. Let $\mathfrak{p} = \mathfrak{q} \cap A = \mathfrak{q}' \cap A$. By Proposition 4.1.10, $B_{\mathfrak{p}}$ is integral over $A_{\mathfrak{p}}$. Let

$$\mathfrak{m} = \mathfrak{p}A_{\mathfrak{p}}, \quad \mathfrak{n} = \mathfrak{q}B_{\mathfrak{p}} \subset \mathfrak{n}' = \mathfrak{q}'B_{\mathfrak{p}}.$$

Then $\mathfrak{n} \cap A_{\mathfrak{p}} = \mathfrak{n}' \cap A_{\mathfrak{p}} = \mathfrak{m}$. Since $\mathfrak{m}$ is a maximal ideal, by Corollary 4.1.12, $\mathfrak{n}'$ and $\mathfrak{n}$ are maximal ideals. Then $\mathfrak{n}' = \mathfrak{n}$ and thus $\mathfrak{q}' = \mathfrak{q}$. □

Theorem 4.1.14. *Let $A \subset B$ be an integral extension of rings. Then $\operatorname{Spec} B \to \operatorname{Spec} A$ is surjective.*

Proof. For any $\mathfrak{p} \in \operatorname{Spec} A$, by Proposition 4.1.10 $B_{\mathfrak{p}}$ is integral over $A_{\mathfrak{p}}$. Let $\mathfrak{m} \subset B_{\mathfrak{p}}$ be a maximal ideal, then Corollary 4.1.12 implies that $\mathfrak{m} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$ is maximal. Consider the following commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A_{\mathfrak{p}} & \longrightarrow & B_{\mathfrak{p}}. \end{array}$$

Let $\mathfrak{q} = \mathfrak{m} \cap B$, then $\mathfrak{q} \cap A = \mathfrak{p}$. □

Theorem 4.1.15. *Let $A \subset B$ be an integral extension of rings. Then the going-up theorem holds:*

for any prime ideals $\mathfrak{p}, \mathfrak{p}' \in \operatorname{Spec} A$ such that $\mathfrak{p} \subset \mathfrak{p}'$ and any prime ideal $\mathfrak{q} \in \operatorname{Spec} B$ such that $\mathfrak{p} = \mathfrak{q} \cap A$, there exists a prime ideal $\mathfrak{q}' \in \operatorname{Spec} B$ such that $\mathfrak{q} \subset \mathfrak{q}'$ and $\mathfrak{p}' = \mathfrak{q}' \cap A$.

Proof. Let $\mathfrak{p}, \mathfrak{p}', \mathfrak{q}$ be as above. Then by Proposition 4.1.10 $B/\mathfrak{q}$ is integral over $A/\mathfrak{p}$. Let $\overline{\mathfrak{p}'} \subset A/\mathfrak{p}$ be the image of $\mathfrak{p}'$. By Theorem 4.1.14, there exists $\overline{\mathfrak{q}'} \in \operatorname{Spec} B/\mathfrak{q}$ such that $\overline{\mathfrak{q}'} \cap A/\mathfrak{p} = \overline{\mathfrak{p}'}$. Let $\mathfrak{q}' \subset B$ be the inverse image of $\overline{\mathfrak{q}'}$ under $B \to B/\mathfrak{q}$. Then $\mathfrak{q}' \in \operatorname{Spec} B$ and $\mathfrak{q}' \cap A = \mathfrak{p}', \mathfrak{q}' \supset \mathfrak{q}$. □

Proposition 4.1.16. *Let $A \subset B$ be rings and C the integral closure of A in B. Then for any multiplicative subset $S \subset A$, $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.*

We will study integral extensions of integral doamins.

Definition 4.1.17. *An integral domain is siad to be integrally closed (or normal) if it is integrally closed in its field of fractions.*

Example 4.1.18. *The integral domains $\mathbb{Z}, k[X_1, \cdots, X_n]$ (k is a field) are integrally closed. More generally, any unique fractorization domain (UFD) is integrally closed.*

Proposition 4.1.19. *Let A be an integral domain. TFAE:*

1. *A is integrally closed.*

2. *$A_{\mathfrak{p}}$ is integrally closed for any prime ideal $\mathfrak{p} \subset A$.*

3. *$A_{\mathfrak{m}}$ is integrally closed for any maximal ideal $\mathfrak{m} \subset A$.*

Proof. Let C be the integral closure of A in $K = \mathrm{Frac}A$, then $A \subset C \subset K$. By Proposition 4.1.16 $C_{\mathfrak{p}}$ is the integral closure of $A_{\mathfrak{p}}$ in K for any $\mathfrak{p} \in \mathrm{Spec}\, A$. Let $f : A \hookrightarrow C$ be the inclusion. Then A is integrally closed $\Leftrightarrow$ f is surjective $\Leftrightarrow$ $f_{\mathfrak{p}}$ is surjective for any $\mathfrak{p} \in \mathrm{Spec}\, A$ $\Leftrightarrow$ $f_{\mathfrak{m}}$ is surjective for any maximal ideal $\mathfrak{m}$, by Corollary 2.2.9. The proposition follows. □

Next, we will follow [2] to prove a going-down theorm for integral extensions of integral doamins. We need some preparations.

Definition 4.1.20. *Let $A \subset B$ be rings with $\mathfrak{a} \subset A$ an ideal. An element $x \in B$ is said to be integral over $\mathfrak{a}$ if x satisfies $x^n + a_1x^{n-1} + \cdots + a_n = 0, a_1, \ldots, a_n \in \mathfrak{a}$. Then integral closure of $\mathfrak{a}$ in B is the set of elements of B which are integral over $\mathfrak{a}$.*

Lemma 4.1.21. *Let C be the integral closure of A in B, $\mathfrak{a}^e$ the extension of $\mathfrak{a}$ in C. Then the integral closure of $\mathfrak{a}$ in B is $r(\mathfrak{a}^e) \subset C$.*

Proof. If $x \in B$ integral over $\mathfrak{a}$, we have $x^n + a_1x^{n-1} + \cdots + a_n = 0, a_1, \ldots, a_n \in \mathfrak{a}$. In particular $x \in C$ and $x^n \in \mathfrak{a}^e$. Thus $x \in r(\mathfrak{a}^e)$. Conversely, for any $x \in r(\mathfrak{a}^e)$, we have $x^n \in \mathfrak{a}^e = \mathfrak{a}C$ for some $n \geq 1$. Write $x^n = \sum_{i=1}^m a_ix_i, a_i \in \mathfrak{a}, x_i \in C$. Since x_i are integral over A, $M := A[x_1, \ldots, x_m]$ is a finitely generated A-module. Now $x^nM \subset \mathfrak{a}M$ implies that x^n is integral over $\mathfrak{a}$, and thus x integral over $\mathfrak{a}$. □

Lemma 4.1.22. *Let $A \subset B$ be integral domains with A integrally closed. Let $x \in B$ be integral over an ideal $\mathfrak{a} \subset A$. Then x is algebraic over $K = \mathrm{Frac}A$ and if its minimal polynomial over K is $t^n + a_1t^{n-1} + \cdots + a_n$, then $a_1, \ldots, a_n \in r(\mathfrak{a})$.*

Proof. The statement that x is algebraic over K is OK. Let $L|K$ be an extension of fields which contains all conjugates $x_1, \ldots, x_n$ of x. If $x^m + b_1x^{m-1} + \cdots + b_m = 0$ with $b_i \in \mathfrak{a} \subset A \subset K$, then $x_i^m + b_1x_i^{m-1} + \cdots + b_m = 0$. Thus x_i are integral over $\mathfrak{a}$. Since the coefficients a_i are polynomials of x_i, they are also integral over $\mathfrak{a}$. Now apply Lemma 4.1.21 to the extension $A \subset B = K$ and $C = A$. We get $a_1, \ldots, a_n \in r(\mathfrak{a})$. □

Lemma 4.1.23. *Let $f : A \to B$ be a ring hommomorphism and $\mathfrak{p} \in \mathrm{Spec}\, A$. Then $\mathfrak{p} \in \mathrm{Im}f^*$ (i.e. $\mathfrak{p} = f^{-1}(\mathfrak{q}) = \mathfrak{q}^c$ for some $\mathfrak{q} \in \mathrm{Spec}\, B$) $\Leftrightarrow$ $\mathfrak{p}^{ec} = \mathfrak{p}$.*

Proof. $\Rightarrow$: If $\mathfrak{p} = \mathfrak{q}^c$ for some $\mathfrak{q} \in \mathrm{Spec}\, B$, then $\mathfrak{p}^{ec} = \mathfrak{q}^{cec} = \mathfrak{q}^c = \mathfrak{p}$.

$\Leftarrow$: If $\mathfrak{p}^{ec} = \mathfrak{p}$, then $f^{-1}(f(\mathfrak{p})) = \mathfrak{p}$. Let $S = f(A \setminus \mathfrak{p}) \subset B$, then $\mathfrak{p}^e \cap S = \emptyset$. Consider the following commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A_{\mathfrak{p}} & \longrightarrow & B_{\mathfrak{p}} = S^{-1}B. \end{array}$$

There exists a maximal ideal $\mathfrak{m} \subset B_{\mathfrak{p}}$ such that $\mathfrak{p}B_{\mathfrak{p}} = \mathfrak{p}^eB_{\mathfrak{p}} \subset \mathfrak{m}$. Let $\mathfrak{q} = \mathfrak{m} \cap B \in \mathrm{Spec}\, B$, then $\mathfrak{q}^c = \mathfrak{p}$.

□

Theorem 4.1.24. *Let $A \subset B$ be an integral extension with A, B integral domains and A integrally closed. Then the going-down theorem holds:*

for any prime ideals $\mathfrak{p}, \mathfrak{p}' \in \operatorname{Spec} A$ such that $\mathfrak{p} \subset \mathfrak{p}'$ and any prime ideal $\mathfrak{q}' \in \operatorname{Spec} B$ such that $\mathfrak{p}' = f^{-1}(\mathfrak{q}')$, there exists a prime ideal $\mathfrak{q} \in \operatorname{Spec} B$ such that $\mathfrak{q} \subset \mathfrak{q}'$ and $\mathfrak{p} = f^{-1}(\mathfrak{q})$

Proof. Let $\mathfrak{p}, \mathfrak{p}', \mathfrak{q}'$ be as above. Consider the following commutative diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ A_{\mathfrak{p}'} \arrow[r] & B_{\mathfrak{q}'}. \end{tikzcd}$$

Applying Lemma 4.1.23 to $A \to B_{\mathfrak{q}'}$, it suffices to show $\mathfrak{p}B_{\mathfrak{q}'} \cap A = \mathfrak{p}$. The inclusion $\mathfrak{p}B_{\mathfrak{q}'} \cap A \supset \mathfrak{p}$ always holds. So we need to show $\mathfrak{p}B_{\mathfrak{q}'} \cap A \subset \mathfrak{p}$.

For any $x \in \mathfrak{p}B_{\mathfrak{q}'}$, write $x = \frac{y}{s}$ with $y \in \mathfrak{p}B, s \in B \setminus \mathfrak{q}'$. By Lemma 4.1.21, y is integral over $\mathfrak{p}$. By Lemma 4.1.22, the minimal equation of y over $K = \operatorname{Frac} A$ is of the form

$$y^r + u_1 y^{r-1} + \cdots + u_r = 0, \quad u_1, \cdots, u_r \in \mathfrak{p}.$$

Now suppose $x \in \mathfrak{p}B_{\mathfrak{q}'} \cap A$ and $x = \frac{y}{s}$. Then $s = yx^{-1}$ with $x^{-1} \in K$. The minimal equation of s over K is of the form

$$s^r + v_1 s^{r-1} + \cdots + v_r = 0, \quad v_i = \frac{u_i}{x^i}, \quad 1 \leq i \leq r.$$

Then $x^i v_i = u_i \in \mathfrak{p}$. Since $s \in B$ is integral over A, by Lemma 4.1.22 we have $v_i \in A$. If $x \notin \mathfrak{p}$, we have $v_i \in \mathfrak{p}, 1 \leq i \leq r$. Then $s^r \in \mathfrak{p}B \subset \mathfrak{q}' \Rightarrow s \in \mathfrak{q}'$, a contradication. Thus we have proved $\mathfrak{p}B_{\mathfrak{q}'} \cap A \subset \mathfrak{p}$. □

Remark 4.1.25. *In the above we have followed [2] to prove Theorem 4.1.24. See Matsumura's book [11] (Theorem 5, vi), v)) for a different proof, using Galois theory and the going-up Theorem 4.1.15.*

4.2 Integral closures of integral domains and valuation rings

Let A be an integral domain with fraction field $K = \operatorname{Frac} A$. We get an inclusion $A \subset K$. Recall that A is called integrally closed if its integral closure in K is itself. Can we describe the integral closure of a general integral domain A? The answer is yes, and it will involve valuation rings.

Definition 4.2.1. *Let B be an integral domain with fraction field K. Then B is called a valuation ring of K if for any $0 \neq x \in K$, either $x \in B$ or $x^{-1} \in B$ (or both).*

Proposition 4.2.2. *Let B be a valuation ring of K. Then we have:*

1. *B is a local ring.*

2. *If B' is a ring such that $B \subset B' \subset K$, then B' is also a valuation ring of K.*

3. *B is integrally closed.*

Proof. 1. Let $\mathfrak{m} = \{x \in B \mid x \text{ non-unit}\}$. Then $x \in \mathfrak{m} \Leftrightarrow$ either $x = 0$ or $x^{-1} \notin B$. For any $a \in B, x \in \mathfrak{m}$, we have $ax \in \mathfrak{m}$: otherwise $(ax)^{-1} \in B \Rightarrow x^{-1} = a(ax)^{-1} \in B$, a contradication since $x \in \mathfrak{m}$. For any $0 \neq x, 0 \neq y \in \mathfrak{m}$, we have either $xy^{-1} \in B$ or $x^{-1}y \in B$. If $xy^{-1} \in B$, $x + y = (1 + xy^{-1})y \in \mathfrak{m}$; if $x^{-1}y \in B$, $x + y = (1 + x^{-1}y)x \in \mathfrak{m}$. Therefore, $\mathfrak{m}$ is an ideal of B and it is the unique maximal ideal. It follows that B is a local ring.

2. This follows by definition.

3. Let $x \in K$ be any element integral over B. We have $x^n + b_1x^{n-1} + \cdots + b_n = 0, b_1, \ldots, b_n \in B$. If $x \notin B$, then $x^{-1} \in B \Rightarrow x = -(b_1 + b_2x^{-1} + \cdots + b_nx^{1-n}) \in B$, contradication! Thus $x \in B$. It follows that B is integrally closed.

□

Theorem 4.2.3. *Let A be an integral domain with fraction field $K = FracA$ and $\mathfrak{p} \in \operatorname{Spec} A$ a prime ideal. Then there exists a valuation ring B of K such that $A \subset B$ and $\mathfrak{m}_B \cap A = \mathfrak{p}$.*

Proof. Considering $A \subset A_{\mathfrak{p}} \subset K$, we may assume A is local with $\mathfrak{p} = \mathfrak{m}_A$. Let $\Sigma = \{B \subset K \text{ subring} \mid B \supset A, B \neq \mathfrak{p}B\}$. Then $A \in \Sigma$. By Zorn Lemma, there exists a maximal element $R \in \Sigma$. Since $\mathfrak{p}R \neq R$, there exists a maximal ideal $\mathfrak{m} \subset R$ such that $\mathfrak{m} \supset \mathfrak{p}R$. Consider the local ring $R_{\mathfrak{m}}$. Then $R \subset R_{\mathfrak{m}} \in \Sigma \Rightarrow R = R_{\mathfrak{m}}$, i.e. R is a local ring. We have $\mathfrak{p} \subset \mathfrak{m} \Rightarrow \mathfrak{m} \cap A = \mathfrak{p}$ since $\mathfrak{p}$ is maximal. We claim R is a valuation ring of K, then the theorem will be proved.

Indeed, for any $x \in K$ and $x \notin R$, $R[x] \notin \Sigma \Rightarrow 1 \in \mathfrak{p}R[x] = R[x]$. Then we have

$$1 = a_0 + a_1x + \cdots + a_nx^n, \quad a_i \in \mathfrak{p}R \subset \mathfrak{m}.$$

Since $1 - a_0$ is a unit of R, we get

$$1 = b_1x + \cdots + b_nx^n, \quad b_i = (1 - a_0)^{-1}a_i \in \mathfrak{m}. \tag{4.1}$$

Let n be the smallest possible degree of such an equation. If $x^{-1} \notin R$, replacing x by x^{-1} we get similarly

$$1 = c_1x^{-1} + \cdots + c_nx^{-n}, \quad c_i \in \mathfrak{m}. \tag{4.2}$$

Let m be the smallest possible degree of such an equation. If $n \geq m$, multiplying b_nx^n on both sides of equation 4.2 we get

$$b_nx^n = c_1b_nx^{n-1} + \cdots + c_mb_nx^{n-m}. \tag{4.3}$$

Puting it into equation 4.1, we get a new equation with degree $n-1$ for x, a contradication to the choice of n! If $n \leq m$, we get a similar contradication with the role of x replaced by x^{-1}. Thus, $x^{-1} \in R$. We have proved by definition that R is a valuation ring of K. □

Corollary 4.2.4. *Let A be an integral domain with fraction field $K = FracA$. Then the integral closure $\overline{A}$ of A in K is given by*

$$\overline{A} = \bigcap_{A \subset B \subset K, B \text{ val. ring of } K} B.$$

Proof. If $A \subset B \subset K$ for a valuation ring B of K, then $\overline{A} \subset B$ since B is integrally closed. Conversely, if $x \in K$ such that $x \notin \overline{A}$, we will find a valuation ring B of K such that $A \subset B \subset K$ and $x \notin B$.

Let $y = x^{-1}$ and $A' = A[y] \subset K$, then $1 \notin yA'$: otherwise $1 = a_1 y + \cdots + a_n y^n \Rightarrow x$ is integral over A, thus $x \in \overline{A}$, a contradication. Since y is not a unit in A', there exists a maximal ideal $\mathfrak{m}' \subset A'$ such that $y \in \mathfrak{m}'$. Applying Theorem 4.2.3 to A' and $\mathfrak{m}'$, we find a valuation ring B of K such that $B \supset A'$ and $\mathfrak{m}_B \cap A' = \mathfrak{m}'$. Then $y = x^{-1} \in \mathfrak{m}_B \Rightarrow x \notin B$. □

Let $A \subset K$ be a valuation ring, then $A^\times \subset K^\times$ is a subgroup. Set $\Gamma = K^\times / A^\times$. For any $\xi = [x], \eta = [y] \in \Gamma$ with $x, y \in K^\times$, we define

$$\xi \geq \eta \Leftrightarrow xy^{-1} \in A.$$

One can check that this is a well defined total ordering. We write the additive law on Γ as $+$ (althourgh it is induced from the multiplication on $K^\times$) and get a totally ordered abelian group $(\Gamma, \leq, +)$. Let $v : K^\times \to \Gamma = K^\times / A^\times$ be the canonical homomorphism. Then for any $x, y \in K^\times$, we have

- $v(xy) = v(x) + v(y)$,
- $v(x + y) \geq \min(v(x), v(y))$ $(x \neq -y)$.

This means that v is a valuation of K in the following sense:

Definition 4.2.5. *Let $(\Gamma, \leq, +)$ be a totally ordered abelian group and K be a field. A valuation of K with values in Γ is a map $v : K^\times \to \Gamma$ such that for any $x, y \in K^\times$*

1. *$v(xy) = v(x) + v(y)$,*
2. *$v(x + y) \geq \min(v(x), v(y))$ $(x \neq -y)$.*

Let $v(0) = \infty$ and $\forall\, a \in \Gamma, \infty \geq a$. In this way we can extend v to a map $v : K \to \Gamma \cup \{\infty\}$ satisfying similar properties as above.

Lemma 4.2.6. *Let $v : K \to \Gamma \cup \{\infty\}$ be a valuation. Then $R_v = \{x \in K \mid v(x) \geq 0\}$ is a valuation ring of K with maximal ideal $\mathfrak{m}_v = \{x \in K \mid v(x) > 0\}$.*

Proof. Exercise. □

One can define the notion of isomorphism (or equivalence) for valuations. The above constructions show that the set of isomorphisim classes of valuations of K is in bijection with the set of isomorphism classes of valuation rings of K.

4.3 Noether's normalization lemma and Hilbert Nullstellensatz

In this section, we study integral extensions for finitely generated algebras over a field. Then we give some applications.

Recall that an A-algebra B is called a finitely generated A-algebra if $B \simeq A[X_1, \cdots, X_n]/I$ as A-algebras, where $I \subset A[X_1, \cdots, X_n]$ is an ideal. In the following we will consider the case $A = k$ is a field.

Theorem 4.3.1. *Let k be a field and $A \neq 0$ be a finitely generated k-algebra. Then there exist $t_1, \cdots, t_n \in A$ which are algebraically independent over k such that A is finite and integral over $k[t_1, \cdots, t_n]$.*

Proof. Let $y_1, \cdots, y_m \in A$ be such that $A = k[y_1, \cdots, y_m]$. If $y_1, \cdots, y_m$ are algebraically independent over k, then we are done. Otherwise, there exists a polynomial $f(X_1, \cdots, X_m) \in k[X_1, \cdots, X_m]$ such that

$$f(y_1, \cdots, y_m) = 0.$$

Set $d = \deg(f) + 1$. For any monomial $\alpha = aX_1^{d_1} \cdots X_m^{d_m}$, set

$$\ell(\alpha) = dd_1 + d^2 d_2 + \cdots + d^{m-1} d_{m-1} + d_m.$$

Then $\forall\, \alpha \neq \beta$ monomials of f, we have $\ell(\alpha) \neq \ell(\beta)$. For $1 \leq i \leq m-1$, set $x_i = y_i - y_m^{d^i}$, then

$$f(x_1 + y_m^d, x_2 + y_m^{d^2}, \cdots, x_{m-1} + y_m^{d^{m-1}}, y_m) = 0.$$

For any $\alpha = aX_1^{d_1} \cdots X_m^{d_m}$ monomial of f, we get $a y_m^{\ell(\alpha)}$ appearing in the equation. Thus y_m is integral over $k[x_1, \cdots, x_{m-1}]$. Since $A = k[x_1, \cdots, x_{m-1}, y_m]$, A is integral over $k[x_1, \cdots, x_{m-1}]$. By induction on m, there exists a k-subalgebra $B \subset k[x_1, \cdots, x_{m-1}]$ isomorphic to the polynomial algebra and $k[x_1, \cdots, x_{m-1}]$ is integral over B. Then A is integral over B and it is finitely generated as a B-module.

□

Corollary 4.3.2 (Weak Nullstellensatz)**.** *Let A be a finitely generated k-algebra. If A is a field, then it is a finite extension of k.*

Proof. By Theorem 4.3.1, there exists a k-subalgebra $R \subset A$ isomorphic to a polynomial algebra such that A is integral over R. We prove $R = k$. If not, let $R = k[x_1, \cdots, x_n], n \geq 1$. Then $x_1^{-1} \in A \setminus R$ integral over R. But this is impossible since R is integrally closed. □

Corollary 4.3.3. *Assume k is algebraically closed. Then any maximal ideal of $k[X_1, \cdots, X_n]$ is of the form $(X_1 - a_1, \cdots, X_n - a_n), a_i \in k$.*

Proof. Let $\mathfrak{m} \subset k[X_1, \cdots, X_n]$ be any maximal ideal. Then $k[X_1, \cdots, X_n]/\mathfrak{m}$ is a finite extension of k by Corollary 4.3.2. Thus $k[X_1, \cdots, X_n]/\mathfrak{m} \simeq k$ as k is algebraically closed. Let $a_i = X_i \bmod \mathfrak{m} \in k$, then $(X_1 - a_i, \cdots, X_n - a_n) \subset \mathfrak{m}$. But $(X_1 - a_i, \cdots, X_n - a_n)$ is already a maximal ideal, thus $(X_1 - a_i, \cdots, X_n - a_n) = \mathfrak{m}$. □

Corollary 4.3.4. *Assume k is algebraically closed. Let $I \subsetneq k[X_1, \cdots, X_n]$ be a proper ideal. Then there exists a point $(a_1, \cdots, a_n) \in k^n$ such that $f(a_1, \cdots, a_n) = 0, \quad \forall\, f \in I$.*

Proof. Take a maximal ideal $\mathfrak{m}$ such that $I \subset \mathfrak{m}$. By Corollary 4.3.3, $\mathfrak{m} = (X_1 - a_i, \cdots, X_n - a_n), a_i \in k$. Then $\forall\, f \in I$, $f \in \mathfrak{m} \Leftrightarrow f \bmod \mathfrak{m} = 0 \Rightarrow f(a_1, \cdots, a_n) = 0$. □

Theorem 4.3.5 (Hilbert Nullstellensatz)**.** *Let A be a finitely generated k-algebra. Then the Jacobson radical equals to the nil radical:*

$$J(A) = nilrad(A).$$

Proof. We may assume $A \neq 0$. If $J(A) \supsetneq nilrad(A)$, take $a \in J(A) \setminus nilrad(A)$, then $S = \{1, a, a^2, \cdots\} \subset A$ is a multiplicative subset. Consider the ring $A_a \neq 0$. Take a maximal ideal $\mathfrak{p} \subset A_a$. Let $\mathfrak{q} = \mathfrak{p} \cap A$ be the inverse image of $\mathfrak{p}$ under the natural map $A \to A_a$. By Corollary 4.3.2, $A_a/\mathfrak{p}$ is a finite extension of k. Since $A/\mathfrak{q} \subset A_a/\mathfrak{p}$ is a k-subalgebra, $A/\mathfrak{q}$ is integral over k, thus by Proposition 4.1.11 $A/\mathfrak{q}$ is a field, i.e. $\mathfrak{q}$ is a maximal ideal. By construction, $a \notin \mathfrak{q}$, contradication to $a \in J(A)$. □

The following corollary is clear (apply Theorem 4.3.5 to $k[X_1, \cdots, X_n]/I$).

Corollary 4.3.6. *Let $I \subsetneq k[X_1, \cdots, X_n]$ be a proper ideal. Then*

$$r(I) = \bigcap_{I \subset \mathfrak{m}, \mathfrak{m} \text{ max. ideal}} \mathfrak{m}.$$

Exercise 4.3.7. *[2] Chapter 5, exercises 1, 4, 5, 7, 12, 13, 14, 15.*

Chapter 5

Noetherian and Artinian rings

We define and study Noetherian and Artinian rings (resp. modules) in this chapter.

5.1 Chain conditions

Let $(\Sigma, \leq)$ be a partially ordered set. One checks easily TFAE:

1. Every increasing sequence $x_1 \leq x_2 \leq \cdots$ in Σ is stationary.

2. Every non-empty subset of Σ has a maximal element.

Let A be a ring, $M \in \mathrm{Mod}_A$, and $\Sigma = \{\text{submodules of}\, M\}$. For the partially ordered set $(\Sigma, \subset)$, we call the above 1 the ascending chain condition (a.c.c. for short) and the above 2 the maximal condition. For the partially ordered set $(\Sigma, \supset)$, we call the above 1 the descending chain condition (d.c.c. for short) and the above 2 the minimal condition.

Definition 5.1.1. *1. M is said to be Noetherian if it satifies a.c.c.*

2. M is said to be Artinian if it satifies d.c.c.

Examples 5.1.2. *1. A finite abelian group satifies both a.c.c. and d.c.c. as a $\mathbb{Z}$-module.*

2. $\mathbb{Z}$ satifies a.c.c. but not d.c.c.: for any $0 \neq a \in \mathbb{Z}$, consider the chain $(a) \supsetneq (a^2) \supsetneq \cdots \supsetneq (a^n) \supsetneq \cdots$. Similarly for $k[X]$.

3. Let p be a prime number. Consider the abelian group $G = \{x \in \mathbb{Q}/\mathbb{Z} \,|\, p^n x = 0, n \geq 0\}$. Then it satisfies d.c.c. but not a.c.c.

4. Let p be a prime number. Consider the subgroup $H \subset \mathbb{Q}$ defined by $H = \{\frac{m}{p^n} \,|\, m, n \in \mathbb{Z}, n \geq 0\}$. We have an exact sequence $0 \to \mathbb{Z} \to H \to G \to 0$, with G the group defined in 3. Then H satisfies neither d.c.c. nor a.c.c.

Proposition 5.1.3. *Let A be a ring, $M \in \mathrm{Mod}_A$. Then M is Noetherian $\Leftrightarrow$ every submodule of M is finitely generated.*

Proof. $\Rightarrow$: for any submodule $N \subset M$, let $\Sigma = \{N' \subset N \,|\, N' \,\text{finitely generated}\}$. Then $0 \in \Sigma$ and since M Noetherian, Σ has a maximal element, say N_0. If $N_0 \neq N$, consider $N_0 + Ax$ for some $x \in N \setminus N_0$, then $N_0 \subsetneq N_0 + Ax \in \Sigma$, contradication! So $N_0 = N$ which is finitely generated.

$\Leftarrow$: let $M_1 \subset M_2 \subset \cdots$ be an ascending chain of submodules. Then $N := \bigcup_{n=1}^{\infty} M_n \subset M$ is a submodule, thus finitely generated by hypothesis, say by $x_1, \cdots, x_r$. Let $x_i \in M_{n_i}$ and $n = \max_{1 \le i \le r} n_i$. Then $N = M_n$, i.e. the chain is stationary.

□

Proposition 5.1.4. *Let $0 \to M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \to 0$ be an exact sequence of A-modules. Then:*

1. *M Noetherian $\Leftrightarrow M', M''$ Noetherian.*
2. *M Artinian $\Leftrightarrow M', M''$ Artinian.*

Proof. We only prove 1; the proof for 2 is similar.

$\Rightarrow$: An ascending chain of submodules of M' or M'' gives rise to a chain in M, thus stationary.

$\Leftarrow$: Let $(L_n)_{n\ge1}$ be an ascending chain of submodules of M. Then $(\alpha^{-1}(L_n))_{n\ge1}$ and $(\beta(L_n))_{n\ge1}$ are ascending chains in M' and M'' repsectively. For $n >> 1$, both chains are stationary. Thus $(L_n)_{n\ge1}$ is stationary. □

Corollary 5.1.5. *If $M_i (1 \le i \le n)$ are Noetherian (resp. Artinian) modules, then so is $\oplus_{i=1}^n M_i$.*

In the following we wil consider chains in which the inclusions are strict. Let

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0$$

be a chain. We call the length of this chain is n.

Definition 5.1.6. *A composition series of M is a finite chain as above, which is maximal, i.e. each $M_{i-1}/M_i (1 \le i \le n)$ is simple.*

Proposition 5.1.7. *Suppose that M has a composition series of length n. Then every composition series of M has length n, and every chain in M can be extended to a composition series.*

Proof. Let $\ell(M) =$ the least length of composition series of M. Write $\ell(M) = +\infty$ if M has no composition sereis. We will prove:

Claim 1. If $\ell(M) < +\infty$, then $N \subsetneq M \Rightarrow \ell(N) < \ell(M)$.

Let (M_i) be a composition series of M of minimal length. Consider $N_i = N \cap M_i \subset N$. Then $N_{i-1}/N_i \subset M_{i-1}/M_i$ and thus $N_{i-1}/N_i = M_{i-1}/M_i$ or 0. By definition, we have $\ell(N) \le \ell(M)$. If $\ell(N) = \ell(M) = n$, then $N_{i-1}/N_i = M_{i-1}/M_i$ for $i = 1, \cdots, n$. This implies $N = M$.

Claim 2. Any chain in M has length $\le \ell(M)$.

We can assume $\ell(M) < +\infty$. If $M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_k = 0$ is a chain of length k. By Claim 1, $\ell(M) > \ell(M_1) > \cdots > \ell(M_k) = 0 \Rightarrow \ell(M) \ge k$.

By Claim 2, we get that all composition series have the same length. Consider any chain, if its length is $\ell(N)$, it must be a composition series; if its length $< \ell(M)$, new terms can be inserted until the length is $\ell(M)$.

□

Proposition 5.1.8. *M has a composition series (i.e. $\ell(M) < +\infty$) $\Leftrightarrow$ M satisfies both chain conditions.*

Proof. The direction $\Rightarrow$ is OK. For the direction $\Leftarrow$, since M satisfies the maximal condition, if $M \neq 0$, there exists a maximal submodule $M_1 \subsetneq M_0 = M$; if $M_1 \neq 0$, it satisfies the maximal condition, so there exists a maximal submodule $M_2 \subsetneq M_1$, $\cdots$ We get $M_0 \supsetneq M_1 \supsetneq \cdots$, which must be finite since M satisfies d.c.c. This is a composition series of M by construction. $\square$

Definition 5.1.9. *M is called of finite length if it has a composition series (equivalently if it satisfies both a.c.c. and d.c.c. by the above proposition). In this case, $\ell(M)$ = length of a composition series.*

One proves by the same method as in group theory that the **Jordan-Hölder Theorem** holds: if $(M_i)_{0\leq i\leq n}, (M'_i)_{0\leq i\leq n}$ are any two composition series of M, then $M_{i-1}/M_i \simeq M'_{i-1}/M'_i, 1 \leq i \leq n$.

Lemma 5.1.10. *The function $\ell(-)$ on the class of A-modules of finite length is additive, i.e. if $0 \to M' \to M \to M'' \to 0$ is an exact sequence of finite length A-modules, then $\ell(M) = \ell(M') + \ell(M'')$.*

Proof. Exercise. $\square$

Proposition 5.1.11. *Let k be a field. For k-vector spaces V, TFAE:*

1. *finite dimension*
2. *finite length*
3. *a.c.c.*
4. *d.c.c.*

Moreover, if these are satisfied, length = dimension.

Proof. Exercise. $\square$

5.2 Noetherian rings

Definition 5.2.1. *A ring A is called Noetherian (resp. Artinian) if it is so as an A-module, i.e. if it satisfies a.c.c. (resp. d.c.c.) for ideals.*

By Proposition 5.1.3, A is Noetherian $\Leftrightarrow$ every ideal of A is finitely genereated. In this case, one checks easily that for any ideal $\mathfrak{a} \subset A$, there exists an integer n such that $r(\mathfrak{a})^n \subset \mathfrak{a}$. In particular, the nil radical $nilrad(A)$ is nilpotent.

Examples 5.2.2.

1. *Any field is both Noetherian and Artinian. For any integer $n \geq 1$, the quotient ring $\mathbb{Z}/n\mathbb{Z}$ is both Noetherian and Artinian. (Later we will see that in fact any Artinian ring is also Noetherian.)*
2. *The rings $\mathbb{Z}, k[X]$ are Noetherian but not Artinian.*
3. *A subring of a Noetherian ring need not be Noetherian: consider the integral domain $k[X_1, X_2, \cdots]$, the polynomial ring with variables $X_1, X_2, \cdots$ over a field k. It is not Noetherian, but its fraction field is.*

4. *In classical algebraic geometry, almost all the rings invovled are Noetherian rings (ensured by Corollary 5.2.9). They also occupy a central position in modern algebraic geometry ([7]). However, in recent years, there have been a large class of non Neotherian rings which are extremely important in number theory and algebraic geometry. These are the perfectoid rings (other than fields) introduced by Scholze in [15] (see also [6]).*

The proofs for the following propositions and corollaries are easy, so we omit.

Proposition 5.2.3. 1. *Let A be a Noetherian (resp. Artinian) ring, M a finitely generated A-module, then M is a Noetherian (resp. Artinian) A-module.*

2. *Let A be a Noetherian (resp. Artinian) ring, $\mathfrak{a} \subset A$ an ideal, then $A/\mathfrak{a}$ is a Noetherian (resp. Artinian) ring.*

Corollary 5.2.4. *Let A be a Noetherian ring, $M \in \mathrm{Mod}_A$. Then M Noetherian $\Leftrightarrow$ M is finitely generated over A.*

Corollary 5.2.5. *Let $A \subset B$ be rings. Suppose that A is a Noetherian ring and B is finitely generated as an A-module, then B is a Noetherian ring.*

Remark 5.2.6. *Let $A \subset B$ be rings. Suppose that B is finitely generated as an A-module. Then A Noetherian $\Leftrightarrow$ B Noetherian.*

The direction $\Leftarrow$ is due to Eakin and Nagata independently.

Proposition 5.2.7. *Let A be a Noetherian ring, $S \subset A$ a multiplicative subset, then $S^{-1}A$ is Noetherian. In particular, for any $\mathfrak{p} \in \operatorname{Spec} A$, $A_{\mathfrak{p}}$ is Noetherian.*

Theorem 5.2.8 (Hilbert Basis Theorem)**.** *If A is a Noetherian ring, then the polynomial ring $A[X]$ is also a Noetherian ring.*

Proof. Let $\mathfrak{a} \subset A[X]$ be any ideal, we need to prove that it is finitely generated. Set $I = \{\text{leading coefficients of } f \mid f \in \mathfrak{a}\}$. Then $I \subset A$ is an ideal, thus finitely generated since A is Noetherian. Let $a_1, \cdots, a_n$ be a set of generators of I. Then for any $1 \leq i \leq n$, we have $f_i = a_i X^{r_i} + \text{lower terms} \in \mathfrak{a} \subset A[X]$. Set $\mathfrak{a}' = (f_1, \cdots, f_n) \subset \mathfrak{a}$ and $r = \max_{1 \leq i \leq n} r_i$. For any $f = aX^m + \text{lower terms} \in \mathfrak{a}$, $a \in I$, if $m \geq r$, write $a = \sum_{i=1}^n u_i a_i, u_i \in A$, then

$$f - \sum_i u_i f_i X^{m-r_i} \in \mathfrak{a}$$

and it has degree $< m$. Proceeding in this way, we get $f = g + h, \deg g < r, h \in \mathfrak{a}'$. Let $M = A + AX + \cdots + AX^{r-1} \subset A[X]$, then we have

$$\mathfrak{a} = \mathfrak{a} \cap M + \mathfrak{a}'.$$

Sincec M is finitely generated over A, it is Noetherian, thus its submodule $\mathfrak{a} \cap M$ is finitely generated as an A-module, say by $g_1, \cdots, g_m$. Then $f_1, \cdots, f_n, g_1, \cdots, g_m$ generate $\mathfrak{a}$. Therefore, $A[X]$ is a Noetherian ring. $\square$

Corollary 5.2.9. 1. *A Noetherian $\Rightarrow$ $A[X_1, \cdots, X_n]$ Noetherian.*

2. *Let B be a finitely generated A-algebra. If A is Noetherian, then so is B. In particular, any finitely generated k-algebra is Noetherian, where k is a field.*

5.3 Artinian rings

We discuss the properties of Artinian rings.

Proposition 5.3.1. *Let A be a ring such that $\mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$ for some maximal ideals $\mathfrak{m}_1, \cdots, \mathfrak{m}_n$. Then A Noetherian $\Leftrightarrow$ A Artinian.*

Proof. Consider the chain

$$A \supset \mathfrak{m}_1 \supset \mathfrak{m}_1\mathfrak{m}_2 \supset \cdots \supset \mathfrak{m}_1 \cdots \mathfrak{m}_n = 0.$$

Each factor $\mathfrak{m}_1 \cdots \mathfrak{m}_{i-1}/\mathfrak{m}_1 \cdots \mathfrak{m}_i$ is a vector space over $A/\mathfrak{m}_i$. Thus a.c.c. $\Leftrightarrow$ d.c.c. for each factor by Proposition 5.1.11. By Proposition 5.1.4, a.c.c. (resp. d.c.c.) for each factor $\Leftrightarrow$ a.c.c. (resp. d.c.c.) for A. Thus a.c.c. $\Leftrightarrow$ d.c.c. for A.

□

Proposition 5.3.2. *Let A be an Artinian ring. Then every prime ideal is maximal.*

Proof. For any $\mathfrak{p} \in \operatorname{Spec} A$, by Proposition 5.2.3, $B := A/\mathfrak{p}$ is an Artinian integral domain. For any $0 \neq x \in B$, consider $(x) \supset (x^2) \supset \cdots$. We have $(x^n) = (x^{n+1})$ for some n. So $x^n = x^{n+1}y$ for some $y \in B$. Since B is an integral domain and $x \neq 0$, we get $1 = xy$, i.e. x is a unit. This implies that B is a field, i.e. $\mathfrak{p}$ is a maximal ideal. □

Corollary 5.3.3. *Let A be an Artinian ring. Then $J(A) = nilrad(A)$.*

Lemma 5.3.4. *Let A be a ring.*

1. *If $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$ are prime ideals and $\mathfrak{a} \subset A$ an ideal such that $\mathfrak{a} \subset \bigcup_{i=1}^n \mathfrak{p}_i$, then there exists an i such that $\mathfrak{a} \subset \mathfrak{p}_i$.*

2. *If $\mathfrak{a}_1, \cdots, \mathfrak{a}_n$ are ideals and $\mathfrak{p} \subset A$ a prime ideal such that $\mathfrak{p} \supset \bigcap_{i=1}^n \mathfrak{a}_i$, then there exists an i such that $\mathfrak{p} \supset \mathfrak{a}_i$. If $\mathfrak{p} = \bigcap_{i=1}^n \mathfrak{a}_i$ then $\mathfrak{p} = \mathfrak{a}_i$ for some i.*

Proof. See [2] Proposition 1.11. □

Proposition 5.3.5. *Let A be an Artinian ring. Then it has only finitely many maximal ideals.*

Proof. We may assume $A \neq 0$. Consider $\Sigma = \{\mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_r \,|\, \mathfrak{m}_i \subset A \,\text{maximal ideals}\}$. Then there exists a minimal element, say $\mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n$. For any maximal ideal $\mathfrak{m} \subset A$, $\mathfrak{m} \cap \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n \in \Sigma$ thus

$$\mathfrak{m} \cap \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n = \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n,$$

that is $\mathfrak{m} \supset \mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n$. By Lemma 5.3.4, $\mathfrak{m} \supset \mathfrak{m}_i$ for some i. Then $\mathfrak{m} = \mathfrak{m}_i$. □

Proposition 5.3.6. *Let A be an Artinian ring. Then $\mathfrak{R} := J(A) = nilrad(A)$ is nilpotent.*

Proof. Consider the chain

$$\mathfrak{R} \supset \cdots \supset \mathfrak{R}^n \supset \cdots .$$

By d.c.c. we have $\mathfrak{R}^k = \mathfrak{R}^{k+1} = \cdots =: \mathfrak{a}$ for some $k >> 0$. If $\mathfrak{a} \neq 0$, let $\Sigma = \{\mathfrak{b} \subset A \,\text{ideal} \,|\, \mathfrak{a}\mathfrak{b} \neq 0\}$. Then $\mathfrak{a} \in \Sigma$. There exists a minimal element $\mathfrak{c} \in \Sigma$. Then we have an element $x \in \mathfrak{c}$ such that $x\mathfrak{a} \neq 0$. Since $(x) \subset \mathfrak{c}$ and $(x) \in \Sigma$, $\mathfrak{c} = (x)$. Since

$$(x\mathfrak{a})\mathfrak{a} = x\mathfrak{a}^2 = x\mathfrak{a} \neq 0$$

and $x\mathfrak{a} \subset (x)$, we have $x\mathfrak{a} = (x) = \mathfrak{c} \Rightarrow x = xy$ for some $y \in \mathfrak{a}$. Then

$$x = xy = xy^2 = \cdots = xy^n = \cdots .$$

By $y^n = 0$ for $n >> 0$, since $y \in \mathfrak{a} = \mathfrak{R}^k \subset \mathfrak{R} = nilrad(A)$. So $x = 0$, a contradication. Thus $\mathfrak{a} = 0$. □

A chain of prime ideals of a ring A is a finite strictly increasing sequence

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n.$$

We call its length is n.

Definition 5.3.7. *Let A be a ring. Then* $\dim A$ = *supremum of the lengths of all chains of prime ideals.*

By definition, we have

$$\dim A = \sup_{\mathfrak{p} \in \operatorname{Spec} A} \dim A_{\mathfrak{p}} \geq 0, \quad \text{or} = +\infty.$$

Theorem 5.3.8. *Let A be a ring. Then A is Artinian $\Leftrightarrow$ A is Noetherian and* $\dim A = 0$*.*

Proof. $\Rightarrow$: By Proposition 5.3.2, $\dim A = 0$. By Proposition 5.3.5, A has finitely many maximal ideals, say $\mathfrak{m}_1, \cdots, \mathfrak{m}_n$. By Proposition 5.3.6, there exists an integer $k >> 0$ such that

$$\prod_{i=1}^{n} \mathfrak{m}_i^k \subset (\bigcap_{i=1}^{n} \mathfrak{m}_i)^k = \mathfrak{R}^k = 0,$$

where $\mathfrak{R} = J(A) = nilrad(A)$. The condition of Proposition 5.3.1 is satisfied, thus A is Noetherian.

$\Leftarrow$: We use the following fact, which we will prove in the next chapter (see Theorem 6.1.10):

Let A be a Noetherian ring, then it has finitely many minimal prime ideals.

This implies that

$$nilrad(A) = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r$$

with $\mathfrak{p}_i$ all the minimal prime ideals of A. On the other hand, $nilrad(A)$ is nilpotent since A is Noetherian. So

$$\prod_{i=1}^{r} \mathfrak{p}_i^k \subset (\bigcap_{i=1}^{r} \mathfrak{p}_i)^k = 0, \quad k >> 0.$$

As $\dim A = 0$, all $\mathfrak{p}_i$ are maximal ideals. By Proposition 5.3.1 again, A is Artinian.

□

Corollary 5.3.9. *Let A be a ring. Then A is Artinian $\Leftrightarrow$ $\ell(A) < +\infty$.*

From the above corollary, we obtain that for $M \in \mathrm{Mod}_A$ with A an Artinian ring, then M is finitely generated if and only if it is of finite length.

Let A be an Artinian ring, then $\operatorname{Spec} A = \{\mathfrak{m}_1, \cdots, \mathfrak{m}_n\}$ with $\mathfrak{m}_i$ all the maximal ideals of A. If A is moreover local, $\operatorname{Spec} A = \{\mathfrak{m}\}$ and $\mathfrak{m} = nilrad(A)$ is nilpotent. Thus for an Artinian local ring A, $\forall\, x \in A$, either x is a unit (i.e. $x \notin \mathfrak{m}$) or x is nilpotent (i.e. $x \in \mathfrak{m}$).

Example 5.3.10. *For any prime p and integer $n \geq 1$, $\mathbb{Z}/p^n\mathbb{Z}$ is an Artinian local ring.*

Proposition 5.3.11. *Let A be a Noetherian local ring with maximal ideal $\mathfrak{m}$. Then exactly one of the following is true:*

1. $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}, \forall\, n \geq 1$.

2. $\mathfrak{m}^n = 0$ *for some n, in which case A is Artinian.*

Proof. Suppose $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for some n. Write $M = \mathfrak{m}^n$, which is a finitely generated A-module since A is Noetherian. Then $M = \mathfrak{m}M$, by Nakayama lemma, $M = \mathfrak{m}^n = 0$.

For any prime ideal $\mathfrak{p} \subset A$, we have $0 = \mathfrak{m}^n \subset \mathfrak{p} \Rightarrow \mathfrak{m} \subset \mathfrak{p}$ and thus $\mathfrak{p} = \mathfrak{m}$. That is, $\dim A = 0$. By Theorem 5.3.8, A is Artinian. Alternatively, one can deduce directly that A is Artinian from $\mathfrak{m}^n = 0$ by applying Proposition 5.3.1. □

Let A be a local ring with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$, we get a k-vector space $\mathfrak{m}/\mathfrak{m}^2$. If $\mathfrak{m}$ is finitely generated, e.g. A is Noetherian, then $\dim_k(\mathfrak{m}/\mathfrak{m}^2) < +\infty$.

Proposition 5.3.12. *Let A be an Artinian local ring. TFAE:*

1. *Every ideal in A is principal.*

2. $\mathfrak{m}$ *is principal.*

3. $\dim_k(\mathfrak{m}/\mathfrak{m}^2) \leq 1$.

Proof. $1 \Rightarrow 2 \Rightarrow 3$ are trivial. We show $3 \Rightarrow 1$:

If $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 0$, then $\mathfrak{m} = \mathfrak{m}^2$. By Nakayama lemma, $\mathfrak{m} = 0$, i.e. A is a field. If $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 1$, by Nakayama lemma again, $\mathfrak{m} = (x)$ for some $x \in A$. For any $0 \neq \mathfrak{a} \subsetneq A$ ideal, we show $\mathfrak{a}$ is principal. Since $\mathfrak{a} \subset \mathfrak{m} = nilrad(A)$ nilpotent, there exists $r >> 0$ such that $\mathfrak{a} \subset \mathfrak{m}^r = (x^r)$ and $\mathfrak{a} \not\subseteq \mathfrak{m}^{r+1} = (x^{r+1})$. This means that $\exists\, y \in \mathfrak{a}$ and $a \in A$ such that $y = ax^r \notin (x^{r+1}) \Rightarrow a \notin (x) = \mathfrak{m} \Rightarrow a$ is a unit in $A \Rightarrow x^r = a^{-1}y \in \mathfrak{a} \Rightarrow (x^r) = \mathfrak{m}^r \subset \mathfrak{a}$. Thus $\mathfrak{a} = \mathfrak{m}^r = (x^r)$.

□

Example 5.3.13. $\mathbb{Z}/p^n\mathbb{Z}, k[X]/(f^n)$ *with $f \in k[X]$ irreducible satisfy the conditions in Proposition 5.3.12. But $k[X^2, X^3]/(X^4)$ does not:* $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 2$.

Before passing to the structure theorem for Artinian rings, we need to discuss the Chinese reminder theorem. Let A be a ring, $\mathfrak{a}, \mathfrak{b} \subset A$ ideals. Recall that $\mathfrak{a}$ and $\mathfrak{b}$ are coprime if $\mathfrak{a}+\mathfrak{b} = A$, in which case we have $\mathfrak{a}\cap\mathfrak{b} = \mathfrak{a}\mathfrak{b}$. Now let $\mathfrak{a}_1, \cdots, \mathfrak{a}_n$ be ideals of A, $\phi_i : A \to A/\mathfrak{a}_i$ the natural map. We get

$$\phi : A \to \prod_{i=1}^{n} A/\mathfrak{a}_i, \quad \mathrm{Ker}\phi = \bigcap_{i=1}^{n} \mathfrak{a}_i.$$

Thus ϕ is injective $\Leftrightarrow \bigcap_{i=1}^n \mathfrak{a}_i = 0$.

Lemma 5.3.14. 1. *If $\mathfrak{a}_i, \mathfrak{a}_j$ are coprime for any $i \neq j$, then $\bigcap_{i=1}^n \mathfrak{a}_i = \prod_{i=1}^n \mathfrak{a}_i$.*

2. *ϕ is surjective $\Leftrightarrow \mathfrak{a}_i, \mathfrak{a}_j$ are coprime for any $i \neq j$.*

Proof. See [2] Proposition 1.10. □

Corollary 5.3.15 (Chinese reminder theorem)**.** *If $\mathfrak{a}_i, \mathfrak{a}_j$ are coprime for any $i \neq j$, then*

$$A/\prod_{i=1}^{n} \mathfrak{a}_i \xrightarrow{\sim} \prod_{i=1}^{n} A/\mathfrak{a}_i.$$

Originally, the Chinese reminder theorem says that for $n_1, \cdots, n_r \in \mathbb{Z}$ such that n_i and n_j are coprime for any $i \neq j$, then $\mathbb{Z}/\prod_{i=1}^{r} n_i\mathbb{Z} \simeq \prod_{i=1}^{r} \mathbb{Z}/n_i\mathbb{Z}$.

Theorem 5.3.16. *Let A be an Artinian ring. Then A is uniquely isomorphic to a finite direct product of Artinian local rings.*

Proof. Let $\mathfrak{m}_i (1 \leq i \leq n)$ be the distinct maximal ideals of A. Then by Lemma 5.3.14 we have $\prod_{i=1}^{n} \mathfrak{m}_i^k = \bigcap_{i=1}^{n} \mathfrak{m}_i^k$ for any $k \geq 1$. By Proposition 5.3.6, for $k >> 0$,

$$\prod_{i=1}^{n} \mathfrak{m}_i^k \subset (\bigcap_{i=1}^{n} \mathfrak{m}_i)^k = \mathfrak{R}^k = 0.$$

By Corollary 5.3.15 we get $A \simeq \prod_{i=1}^{n} A/\mathfrak{m}_i^k$, where $A/\mathfrak{m}_i^k$ are Artinian local rings for $1 \leq i \leq n$.

To prove the uniqueness, if $A \simeq \prod_{i=1}^{m} A_i$ and A_i are Artinian local rings, let $\phi_i : A \to A_i$ be the projection and $\mathfrak{a}_i = \mathrm{Ker}\phi_i$. By Lemma 5.3.14 $\mathfrak{a}_i$ are coprime in pairs and $\bigcap_{i=1}^{m} \mathfrak{a}_i = 0$. By the theory of primary decomposition (which we will discuss in the next chapter), $\mathfrak{a}_i$ are uniquely determined by A, and thus $A_i \simeq A/\mathfrak{a}_i$ are uniquely determined by A. (In fact $m = n$ and $\mathfrak{a}_i = \mathfrak{m}_i^k, A_i \simeq A/\mathfrak{m}_i^k$ for some k).

□

Exercise 5.3.17. *[2] Chapter 6, exercises 1, 2; Chapter 7, exercises 1, 2, 12; Chapter 8, exercises 2, 3, 4.*

Chapter 6

Associated primes and Primary decomposition

Convention: in this chapter, all rings will be Noetherian.

We study the theory of associated primes and primary decomposition that are used to determine the structure of certian Noether rings, see Theorems 5.3.8, 5.3.16 and the next chapter.

6.1 Associated primes

Let A be a Noetherian ring, $M \in \mathrm{Mod}_A$. For any $x \in M$, we have the ideal

$$ann(x) = \{a \in A \,|\, ax = 0\} \subset A.$$

The submodule $Ax \subset M$ is isomorphic to $A/ann(x)$.

Definition 6.1.1. *The set of associated primes of M is*

$$\mathrm{Ass}(M) = \mathrm{Ass}_A(M) := \{\mathfrak{p} \in \mathrm{Spec}\, A \,|\, \mathfrak{p} = ann(x)\, \textit{for some } x \in M\}.$$

Note that $\mathfrak{p} \in \mathrm{Ass}(M) \Leftrightarrow$ there exists a submodule of M which is isomorphic to $A/\mathfrak{p}$.

Proposition 6.1.2. *Let $\mathfrak{p}$ be a maximal element of $\{ann(x) \,|\, x \in M, x \neq 0\}$. Then $\mathfrak{p} \in \mathrm{Ass}(M)$. In particular $M \neq 0 \Leftrightarrow \mathrm{Ass}(M) \neq \emptyset$.*

Proof. We need to show $\mathfrak{p}$ is prime. Let $\mathfrak{p} = ann(x)$ with $0 \neq x \in M$. If $a, b \in A$ such that $ab \in \mathfrak{p}$ and $b \notin \mathfrak{p}$, we claim $a \in \mathfrak{p}$. Indeed, we have $bx \neq 0$ and $abx = 0$. Since $\mathfrak{p} = ann(x) \subset ann(bx)$, by the choice of $\mathfrak{p}$ we get $\mathfrak{p} = ann(bx)$. Then $a \in ann(bx) = \mathfrak{p}$. The proposition is proved. □

Corollary 6.1.3. *We have*

$$\{a \in A \,|\, ax = 0\, \textit{for some}\, 0 \neq x \in M\} = \bigcup_{0 \neq x \in M} ann(x) = \bigcup_{\mathfrak{p} \in \mathrm{Ass}(M)} \mathfrak{p}.$$

Lemma 6.1.4. *Let $S \subset A$ be a multiplicative subset, $M \in \mathrm{Mod}_A$, $S^{-1}M$ the induced $S^{-1}A$-module, $f : A \to S^{-1}A$ the natural map with induced $f^* : \mathrm{Spec}\, S^{-1}A \hookrightarrow \mathrm{Spec}\, A$. Then*

$$\mathrm{Ass}_A(S^{-1}M) = f^*\big(\mathrm{Ass}_{S^{-1}A}(S^{-1}M)\big) = \mathrm{Ass}_A(M) \cap \{\mathfrak{p} \in \mathrm{Spec}\, A \,|\, \mathfrak{p} \cap S = \emptyset\}.$$

Proof. The first equality follows from defintion. We will show the second equality, which is $f^*\big(\mathrm{Ass}_{S^{-1}A}(S^{-1}M)\big) = \mathrm{Ass}_A(M) \cap \mathrm{Im} f^*$. Let $\mathfrak{p} \in \mathrm{Ass}_A(M) \cap \mathrm{Im} f^*$, then $\mathfrak{p} \cap S = \emptyset$ and $\mathfrak{p} = ann_A(x), 0 \neq x \in M$. Consider $\frac{x}{1} \in S^{-1}M$. If

$$\frac{a}{s} \in ann_{S^{-1}A}(\frac{x}{1}), \quad \text{i.e.} \quad \frac{a}{s}\frac{x}{1} = \frac{ax}{s} = 0 \in S^{-1}M,$$

there exists $t \in S$ such that $tax = 0 \in M$. Then $t \notin \mathfrak{p}$ and

$$ta \in ann(x) = \mathfrak{p} \Rightarrow a \in \mathfrak{p} \Rightarrow \frac{a}{s} \in \mathfrak{p}S^{-1}A, \quad \text{thus} \quad ann_{S^{-1}A}(\frac{x}{1}) \subset \mathfrak{p}S^{-1}A.$$

Conversely for any $\frac{a}{s} \in \mathfrak{p}S^{-1}A = S^{-1}\mathfrak{p}$ with $a \in \mathfrak{p}$, we have

$$ax = 0 \Rightarrow \frac{a}{s}\frac{x}{1} = 0 \Rightarrow \frac{a}{s} \in ann_{S^{-1}A}(\frac{x}{1}),$$

so

$$ann_{S^{-1}A}(\frac{x}{1}) = \mathfrak{p}S^{-1}A$$

and $\mathfrak{p} \in f^*\big(\mathrm{Ass}_{S^{-1}A}(S^{-1}M)\big)$. Now for any $P \in \mathrm{Ass}_{S^{-1}A}(S^{-1}M)$, we can assume $P = ann_{S^{-1}A}(\frac{x}{1}), \quad x \in M$. Write $\mathfrak{p} = P \cap S^{-1}A \in \mathrm{Im} f^*$, which is finitely generated since A is Noetherian. Then there exists $t \in S$ such that $\mathfrak{p} = ann_A(tx) \in \mathrm{Ass}_A(M)$. □

Corollary 6.1.5. *We have*

$$\mathfrak{p} \in \mathrm{Ass}_A(M) \quad \Longleftrightarrow \quad \mathfrak{p}A_{\mathfrak{p}} \in \mathrm{Ass}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}).$$

For $M \in \mathrm{Mod}_A$, recall the set

$$\mathrm{Supp}(M) = \mathrm{Supp}_A(M) = \{\mathfrak{p} \in \mathrm{Spec}\, A \mid M_{\mathfrak{p}} \neq 0\}.$$

We have

- $M \neq 0 \Leftrightarrow \mathrm{Supp}(M) \neq \emptyset$.
- If M is finitely generated, then $\mathrm{Supp}(M) = V(Ann(M))$, where $Ann(M) = \{a \in A \mid aM = 0\}$.

Theorem 6.1.6. *Let A be a Noetherian ring, $M \in \mathrm{Mod}_A$, then $\mathrm{Ass}(M) \subset \mathrm{Supp}(M)$, and any minimal element of $\mathrm{Supp}(M)$ is in $\mathrm{Ass}(M)$.*

Proof. The inclusion $\mathrm{Ass}(M) \subset \mathrm{Supp}(M)$ follows from Corollary 6.1.5. Alternatively, if $\mathfrak{p} \in \mathrm{Ass}(M)$, we have an exact sequence $0 \to A/\mathfrak{p} \to M$, which induces an exact sequence $0 \to A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \to M_{\mathfrak{p}}$ by tensoring with the flat A-algebra $A_{\mathfrak{p}}$. Then $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \neq 0 \Rightarrow M_{\mathfrak{p}} \neq 0 \Rightarrow \mathfrak{p} \in \mathrm{Supp}(M)$.

Now let $\mathfrak{p} \in \mathrm{Supp}(M)$ be a minimal element. By Corollary 6.1.5, $\mathfrak{p} \in \mathrm{Ass}_A(M) \quad \Leftrightarrow \mathfrak{p}A_{\mathfrak{p}} \in \mathrm{Ass}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}})$. So we may assume A is local with $\mathfrak{p} = \mathfrak{m} \subset A$ its maximal ideal and $M \neq 0$. Since $\mathfrak{p} \in \mathrm{Supp}(M)$ is minimal, for any prime ideal $\mathfrak{q} \subsetneq \mathfrak{p}$, we have $M_{\mathfrak{q}} = 0$. Thus $\mathrm{Supp}(M) = \{\mathfrak{p}\} \supset \mathrm{Ass}(M) \neq \emptyset$. Then we have $\mathrm{Ass}(M) = \{\mathfrak{p}\}$. □

Corollary 6.1.7. *Let $I \subset A$ be an ideal. Then the minimal prime ideals containining I are precisely the minimal associated prime ideals of the A-module A/I.*

Proof. $Ann(A/I) = I$ and by Theorem 6.1.6 $\mathrm{Ass}(A/I) \subset V(I) = \mathrm{Supp}(A/I)$. □

Proposition 6.1.8. *Let A be a Noetherian ring and $M \neq 0$ a finitely generated A-module, then there exists a chain of submodules*

$$0 = M_0 \subset \cdots \subset M_{n-1} \subset M_n = M$$

such that $M_i/M_{i-1} \simeq A/\mathfrak{p}_i$ for some $\mathfrak{p}_i \in \mathrm{Spec}\, A$, $1 \leq i \leq n$.

Proof. $M \neq 0 \Rightarrow \exists\, M_1 \subset M$ such that $M_1 \simeq A/\mathfrak{p}_1$ for some $\mathfrak{p}_1 \in \mathrm{Ass}(M)$. If $M_1 \neq M$, we consider M/M_1 and find M_2 such that $M_1 \subset M_2$ and $M_2/M_1 \simeq A/\mathfrak{p}_2$ with $\mathfrak{p}_2 \in \mathrm{Ass}(M/M_1)$, ... Since M is finitely generated thus a Noetherian module, the a.c.c. holds. Thus the process must stop in finite steps. □

Lemma 6.1.9. *Let $0 \to M' \to M \to M''$ be an exact sequence of A-modules, then*

$$\mathrm{Ass}(M) \subset \mathrm{Ass}(M') \cup \mathrm{Ass}(M'').$$

Proof. For any $\mathfrak{p} \in \mathrm{Ass}(M)$, take $N \subset M$ such that $N \simeq A/\mathfrak{p}$. If $N \cap M' = 0$, then N isomorphic to a submodule of M'' and thus $\mathfrak{p} \in \mathrm{Ass}(M'')$. If $N \cap M' \neq 0$, take $0 \neq x \in N \cap M'$. Since $N \simeq A/\mathfrak{p}$ and $A/\mathfrak{p}$ is an integral domain, we have $ann(x) = \mathfrak{p} \in \mathrm{Ass}(M')$. □

Theorem 6.1.10. *Let A be a Noetherian ring and M a finitely generated A-module, then $\mathrm{Ass}(M)$ is a finite set. In particular, for any ideal $I \subset A$, the set of minimal prime ideals containing I is finite.*

Proof. By Proposition 6.1.8, there exists a chain of submodules

$$0 = M_0 \subset \cdots \subset M_{n-1} \subset M_n = M$$

such that $M_i/M_{i-1} \simeq A/\mathfrak{p}_i$ for some $\mathfrak{p}_i \in \mathrm{Spec}\, A$, $1 \leq i \leq n$. By Lemma 6.1.9,

$$\mathrm{Ass}(M) \subset \mathrm{Ass}(M_1) \cup \mathrm{Ass}(M_2/M_1) \cup \cdots \cup \mathrm{Ass}(M_n/M_{n-1}).$$

Now

$$\mathrm{Ass}(M_i/M_{i-1}) = \mathrm{Ass}(A/\mathfrak{p}_i) = \{\mathfrak{p}_i\} \quad \Rightarrow \quad \mathrm{Ass}(M) \subset \{\mathfrak{p}_1, \cdots, \mathfrak{p}_n\}.$$

□

6.2 Primary decomposition

Definition 6.2.1.

1. *$M \in \mathrm{Mod}_A$ is said to be co-primary if $\#\mathrm{Ass}(M) = 1$.*
2. *Let $M \in \mathrm{Mod}_A$, a submodule $N \subset M$ is said to be a primary submodule if M/N is co-primary. If $\mathrm{Ass}(M/N) = \{\mathfrak{p}\}$, we say N is $\mathfrak{p}$-primary or N belongs to $\mathfrak{p}$.*

Proposition 6.2.2. *TFAE:*

1. *M is co-primary.*
2. *$M \neq 0$ and if $a \in A$ is a zero divisor of M, then a is locally nilpotent on M, i.e. $\forall\, x \in M$, $\exists\, n > 0$ such that $a^n x = 0$.*

Proof. $1 \Rightarrow 2$: Suppose $\mathrm{Ass}(M) = \{\mathfrak{p}\}$, then $M \neq 0$. For any $0 \neq x \in M$, we have $\mathrm{Ass}(Ax) = \{\mathfrak{p}\}$. Since

$$\mathrm{Supp}(Ax) = V(Ann(Ax)) = V(ann(x)),$$

by Theorem 6.1.6 $\mathfrak{p}$ is the unique minimal element of $\mathrm{Supp}(Ax) = V(ann(x))$. Thus $\mathfrak{p} = r(ann(x))$. This means that for any $a \in \mathfrak{p}$, $a^n \in ann(x)$ for some n, i.e. $a^n x = 0$.

$2 \Rightarrow 1$: Let $\mathfrak{p} = \{a \in A \,|\, a \,\text{is locally nilpotent on}\, M\}$, which is an ideal of A. For any $\mathfrak{q} \in \mathrm{Ass}(M)$, there exists $x \in M$ such that $\mathfrak{q} = ann(x)$. By definition $\mathfrak{p} \subset \mathfrak{q}$. By assumption $\bigcup_{\mathfrak{q} \in \mathrm{Ass}(M)} \mathfrak{q} \subset \mathfrak{p}$. Thus $\mathrm{Ass}(M) = \{\mathfrak{p}\}$. □

Definition 6.2.3. *An ideal $\mathfrak{q} \subset A$ is called primary if $\mathfrak{q} \neq A$ and the only zero divisors of $A/\mathfrak{q}$ are nilpotent element, i.e. $\forall\, x, y \in A$ such that $xy \in \mathfrak{q}$ and $x \notin \mathfrak{q}$, then $y^n \in \mathfrak{q}$ for some n.*

By definition, if $\mathfrak{q}$ is a primary ideal then $r(\mathfrak{q})$ is a prime ideal. Any prime ideal of A is a primary ideal.

Proposition 6.2.2 implies that this definition for ideals coincides with our previous definition for submodules.

Corollary 6.2.4. *We have*

$$\begin{aligned} \mathfrak{q} \subset A \quad \textit{primary ideal} \quad &\Longleftrightarrow \quad \mathfrak{q} \subset A \quad \textit{primary submodule} \\ &\Longleftrightarrow \quad A/\mathfrak{q} \quad \textit{co-primary} \quad A\textit{-module.} \end{aligned}$$

If $\mathfrak{q}$ is a primary ideal, we have then

$$\mathrm{Ass}(A/\mathfrak{q}) = \{r(\mathfrak{q})\}.$$

If $r(\mathfrak{q}) = \mathfrak{p}$, we say that $\mathfrak{q}$ is $\mathfrak{p}$-primary or $\mathfrak{q}$ belongs to $\mathfrak{p}$. For any $x \in A$ and ideal $\mathfrak{q} \subset A$, recall the ideal

$$(\mathfrak{q} : x) = \{y \in A \,|\, yx \in \mathfrak{q}\}.$$

We have always $\mathfrak{q} \subset (\mathfrak{q} : x)$. If $\mathfrak{q}$ is $\mathfrak{p}$-primary, then one can compute (recall $0 \subset \mathfrak{q} \subset \mathfrak{p} \subset A$)

$$(\mathfrak{q} : x) = \begin{cases} A, & \text{if}\, x \in \mathfrak{q}, \\ \mathfrak{q}, & \text{if}\, x \notin \mathfrak{p}, \\ \text{also } \mathfrak{p}\text{-primary}, & \text{if}\, x \notin \mathfrak{q}. \end{cases}$$

Let $\mathfrak{m} \subset A$ be a maximal ideal and $\mathfrak{q} \subset A$ an ideal. Since A is Noetherian, TFAE:

1. $\mathfrak{q}$ is $\mathfrak{m}$-primary.

2. $r(\mathfrak{q}) = \mathfrak{m}$.

3. $\mathfrak{m}^k \subset \mathfrak{q} \subset \mathfrak{m}$, for some $k \geq 1$.

Indeed, $1 \Rightarrow 2$ is OK. $2 \Rightarrow 1$: $r(\mathfrak{q}) = \mathfrak{m}$ implies that $\mathfrak{m}$ is the only prime ideal containing $\mathfrak{q}$, i.e. $\mathrm{Spec}\, A/\mathfrak{q} = \{\mathfrak{m}/\mathfrak{q}\}$. Then it follows by definition that $\mathfrak{q}$ is $\mathfrak{m}$-primary. $3 \Rightarrow 2$: take radicals. $2 \Rightarrow 3$: $\mathfrak{m} = r(\mathfrak{q})$ is finitely generated.

We remark that in general a prime power is not necessarily primary.

Let $M \in \mathrm{Mod}_A$, $N \subset M$ a submodule. Recall that it is primary if M/N is co-primary, i.e. $\mathrm{Ass}(M/N) = \{\mathfrak{p}\}$.

Definition 6.2.5. *Let $N \subset M$ be a submodule. A primary decomposition of N is an equation*

$$N = Q_1 \cap \cdots \cap Q_r$$

with each Q_i a primary submodule of M.

Lemma 6.2.6. *For $\mathfrak{p} \in \operatorname{Spec} A$ and Q_1, Q_2 $\mathfrak{p}$-primary submodules of M. Then $Q_1 \cap Q_2$ is also $\mathfrak{p}$-primary.*

Proof. We have the injection

$$M/Q_1 \cap Q_2 \hookrightarrow M/Q_1 \oplus M/Q_2,$$

thus $\operatorname{Ass}(M/Q_1 \cap Q_2) \subset \operatorname{Ass}(M/Q_1) \cup \operatorname{Ass}(M/Q_2) = \{\mathfrak{p}\}$. Since $\operatorname{Ass}(M/Q_1 \cap Q_2) \neq \emptyset$, we have $\operatorname{Ass}(M/Q_1 \cap Q_2) = \{\mathfrak{p}\}$. $\square$

This lemma implies that any primary decomposition can be simplified to an irredundant (or a minimal, or reduced) one: $N = Q_1 \cap \cdots \cap Q_r$ with no Q_i can be omitted and the associated primes of $M/Q_i (1 \leq i \leq r)$ are all distinct.

Lemma 6.2.7. *Let $N = Q_1 \cap \cdots \cap Q_r$ be an irredundant primary decomposition and Q_i belong to $\mathfrak{p}_i$ (i.e. $\operatorname{Ass}(M/Q_i) = \{\mathfrak{p}_i\}$), then $\operatorname{Ass}(M/N) = \{\mathfrak{p}_1, \cdots, \mathfrak{p}_r\}$.*

Proof. We have the injection

$$M/N \hookrightarrow M/Q_1 \oplus \cdots \oplus M/Q_r,$$

thus $\operatorname{Ass}(M/N) \subset \bigcup_{i=1}^r \operatorname{Ass}(M/Q_i) = \{\mathfrak{p}_1, \cdots, \mathfrak{p}_r\}$. Conversely, since $(Q_2 \cap \cdots \cap Q_r)/N$ is isomorphic to a non-zero submodule of M/Q_1, we get $\operatorname{Ass}(Q_2 \cap \cdots \cap Q_r/N) = \{\mathfrak{p}_1\}$. Since $(Q_2 \cap \cdots \cap Q_r)/N \subset M/N$, by definition $\mathfrak{p}_1 \in \operatorname{Ass}(M/N)$. Similarly $\mathfrak{p}_2, \cdots, \mathfrak{p}_r \in \operatorname{Ass}(M/N)$. $\square$

Proposition 6.2.8. *Let $N \subset M$ be a $\mathfrak{p}$-primary submodule. Let $\mathfrak{q} \in \operatorname{Spec} A$ and $f : M \to M_\mathfrak{q}$ the canonical map. Then*

1. *If $\mathfrak{p} \not\subseteq \mathfrak{q}$, $N_\mathfrak{q} = M_\mathfrak{q}$ ($\Rightarrow M = f^{-1}(N_q)$).*
2. *If $\mathfrak{p} \subset \mathfrak{q}$, $N = f^{-1}(N_\mathfrak{q})$.*

Proof. 1: Note $M_\mathfrak{q}/N_\mathfrak{q} = (M/N)_\mathfrak{q}$ and by Lemma 6.1.4 $\operatorname{Ass}_A(M_\mathfrak{q}/N_\mathfrak{q}) = \operatorname{Ass}_A(M/N) \cap \{\mathfrak{p}' \mid \mathfrak{p}' \subset \mathfrak{q}\}$. As N is $\mathfrak{p}$-primary, $\operatorname{Ass}_A(M/N) = \{\mathfrak{p}\}$. If $\mathfrak{p} \not\subseteq \mathfrak{q}$, $\operatorname{Ass}_A(M_\mathfrak{q}/N_\mathfrak{q}) = \emptyset \Rightarrow M_\mathfrak{q} = N_\mathfrak{q}$.

2: $\operatorname{Ass}_A(M/N) = \{\mathfrak{p}\} \Rightarrow \mathfrak{p}$ is the set of zero-divisors of M/N. If $\mathfrak{p} \subset \mathfrak{q}$, $A \setminus \mathfrak{q}$ does not contain zero-divisors of M/N, thus $M/N \to (M/N)_\mathfrak{q} = M_\mathfrak{q}/N_\mathfrak{q}$ is injective. This implies $N = f^{-1}(N_\mathfrak{q})$. $\square$

Corollary 6.2.9. *Let $N = Q_1 \cap \cdots \cap Q_r$ be an irredundant primary decomposition and Q_i belong to $\mathfrak{p}_i$. Let $\mathfrak{p}_j \in \operatorname{Ass}(M/N) = \{\mathfrak{p}_1, \cdots, \mathfrak{p}_r\}$ be a minimal element. Then $Q_j = f_{\mathfrak{p}_j}^{-1}(N_{\mathfrak{p}_j})$, where $f_{\mathfrak{p}_j} : M \to M_{\mathfrak{p}_j}$ is the canonical map. In particular Q_j is uniquely determined by N.*

Proof.
$$N = Q_1 \cap \cdots \cap Q_r \quad \Rightarrow \quad N_{\mathfrak{p}_j} = Q_{1\mathfrak{p}_j} \cap \cdots \cap Q_{r\mathfrak{p}_j}.$$
For $\mathfrak{p}_i, 1 \leq i \leq r$, apply Proposition 6.2.8: if $i \neq j$, as $\mathfrak{p}_i \not\subseteq \mathfrak{p}_j$, we have $Q_{i\mathfrak{p}_j} = M_{\mathfrak{p}_j}$ thus $N_{\mathfrak{p}_j} = Q_{j\mathfrak{p}_j}$, and
$$Q_j = f_{\mathfrak{p}_j}^{-1}(Q_{j\mathfrak{p}_j}) = f_{\mathfrak{p}_j}^{-1}(N_{\mathfrak{p}_j}).$$
□

Theorem 6.2.10. *Let A be a Noetherian ring and $M \neq 0 \in \mathrm{Mod}_A$. Then for any $\mathfrak{p} \in \mathrm{Ass}(M)$, we can choose a $\mathfrak{p}$-primary submodule $Q(\mathfrak{p})$ such that*
$$0 = \bigcap_{\mathfrak{p} \in \mathrm{Ass}(M)} Q(\mathfrak{p}).$$
In particular, if $\mathrm{Ass}(M)$ is finite (e.g. M is finitely generated), we get an irredundant primary decomposition $0 = \bigcap_{\mathfrak{p} \in \mathrm{Ass}(M)} Q(\mathfrak{p})$.

Proof. For any $\mathfrak{p} \in \mathrm{Ass}(M)$, consider
$$\Sigma = \{N \subset M \,|\, \mathfrak{p} \notin \mathrm{Ass}(N)\}.$$
Then $0 \in \Sigma$. If $\{N_\lambda\}_\lambda$ is a chain with $N_\lambda \in \Sigma$, then $\bigcup_\lambda N_\lambda \in \Sigma$ since $\mathrm{Ass}(\bigcup_\lambda N_\lambda) = \bigcup_\lambda \mathrm{Ass}(N_\lambda)$. By Zorn Lemma, there exists a maximal element; choose one $Q = Q(\mathfrak{p})$. As $\mathfrak{p} \in \mathrm{Ass}(M), \mathfrak{p} \notin \mathrm{Ass}(Q)$ we have $M \neq Q$. If there exists $\mathfrak{q} \in \mathrm{Ass}(M/Q)$ and $\mathfrak{q} \neq \mathfrak{p}$, by definition there exists a sumodule $Q'/Q \subset M/Q$ such that $Q'/Q \simeq A/\mathfrak{q}$. Then $\mathrm{Ass}(Q') \subset \mathrm{Ass}(Q) \cup \{\mathfrak{q}\}$ and thus $\mathfrak{p} \notin \mathrm{Ass}(Q') \Rightarrow Q' \in \Sigma$, a contradication. Thus $\mathrm{Ass}(M/Q) = \{\mathfrak{p}\}$, i.e. Q is $\mathfrak{p}$-primary. In this way, for each $\mathfrak{p} \in \mathrm{Ass}(M)$ we find a $\mathfrak{p}$-primary submodule $Q(\mathfrak{p}) \subset M$. Note
$$\mathrm{Ass}(\bigcap_{\mathfrak{p} \in \mathrm{Ass}(M)} Q(\mathfrak{p})) = \bigcap_{\mathfrak{p} \in \mathrm{Ass}(M)} \mathrm{Ass}(Q(\mathfrak{p})) = \emptyset$$
since for each $\mathfrak{p} \in \mathrm{Ass}(M)$ we have $\mathrm{Ass}(M) = \mathrm{Ass}(Q(\mathfrak{p})) \sqcup \{\mathfrak{p}\}$. Therefore
$$\bigcap_{\mathfrak{p} \in \mathrm{Ass}(M)} Q(\mathfrak{p}) = 0.$$
□

Corollary 6.2.11. *Let A be a Noetherian ring, $M \neq 0$ a finitely generated A-module. Then any submodule $N \subset M$ has a primary decomposition. If $N = Q_1 \cap \cdots \cap Q_r$ is an irredundant primary decomposition and Q_i is $\mathfrak{p}_i$-primary with $\mathfrak{p}_i \in \mathrm{Ass}(M/N)$ minimal, then Q_i is uniquely determined.*

Proof. Apply Theorem 6.2.10 to M/N. The uniqueness statement follows from Corollary 6.2.9. □

We can summarize our discussions for ideals.

Corollary 6.2.12. *Let A be a Noetherian ring, $I \subset A$ an ideal. Then we have an irredundant primary decomposition*
$$I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$$
with $\mathfrak{q}_i \subset A$ primary ideals. Let $\mathfrak{p}_i = r(\mathfrak{q}_i), 1 \leq i \leq r$. If $\mathfrak{p} \supset I$ is a minimal prime ideal containing I, then there exists an i such that $\mathfrak{p} = \mathfrak{p}_i$, and the corresponding primary ideal $\mathfrak{q}_i$ is uniquely determined.

If $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$ is an irredundant primary decomposition where $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary, then we get

$$r(I) = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r = \bigcap_{i, \mathfrak{p}_i \text{ minimal}} \mathfrak{p}_i.$$

Of course, the equality $r(I) = \bigcap_{i, \mathfrak{p}_i \text{ minimal}} \mathfrak{p}_i$ can be deduced directly by Proposition 1.1.11, Corollary 6.1.7 and Theorem 6.1.10.

If A is Artinian, then $\dim A = 0$. For an ideal $I \subset A$ and an irredundant primary decomposition $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$, any prime ideal $\mathfrak{p}_i = r(\mathfrak{q}_i)$ is minimal. Thus in this case the irredundant primary decomposition is uniquely determined. This fact was used in the proof of Theorem 5.3.16 on the structure of Artinian rings.

Exercise 6.2.13. *[2] Chapter 4, exercises 2, 4, 5.*

Chapter 7

One-dimensional integrally closed Noetherian domains

Recall that by Theorem 5.3.8,

$$\text{Noetherian rings of dimension } 0 \quad = \quad \text{Artinian rings.}$$

We may wonder what are **Noetherian rings of dimension 1**? This seems to be the next simplest class of Noetherian rings. However, to have a satisfying answer, we will have to be more restrictive. We impose the additional assumption that our rings are moreover **integral domains**.

Let A be a Noetherian (integral) domain of dimension 1. Then 0 is a prime ideal, and for any $0 \neq \mathfrak{p} \in \operatorname{Spec} A$, $\mathfrak{p}$ is maximal since $\dim A = 1$.

Proposition 7.0.1. *Let A be a Noetherian (integral) domain of dimension 1. Then any non zero ideal $\mathfrak{a} \subset A$ can be uniquely expressed as a product of primary ideals whose radicals are distinct.*

Proof. By Corollary 6.2.12, we have an irredundant primary decomposition $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ with each $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary, $\mathfrak{p}_i = r(\mathfrak{q}_i)$. As A is an integral domain of dimension 1, $\mathfrak{p}_i$ are maximal and $\mathfrak{q}_i$ are pairwise coprime, $1 \leq i \leq n$. By Lemma 5.3.14, $\bigcap_{i=1}^{n} \mathfrak{q}_i = \prod_{i=1}^{n} \mathfrak{q}_i$. The condition $\dim A = 1$ also implies that $\mathfrak{p}_i \in \operatorname{Ass}(A/\mathfrak{a})$ is minimal. Thus all the $\mathfrak{q}_i$ are uniquely determined. $\square$

Proposition 7.0.1 will be a perfect analogue of the ring of integers $\mathbb{Z}$ if all the primary ideals are powers of prime ideals. We may ask when this will happen? The answer will bring us back to chapter 4: the necessary and sufficent condition turns out to be that A should be moreover an **integrally closed integral domain**.

We will characterize **Noetherian domains of dimension 1** which are **integrally closed**. These are the algebraic counterparts of Riemann surfaces in complex analysis and algebraic curves in algebraic geometry.

7.1 Discrete valuation rings and Dedekind domains

Let K be a field. Recall that a valuation of K with values in a totally ordered abelian group Γ is a map $v : K^\times \to \Gamma$ such that for any $x, y \in K^\times$,

1. $v(xy) = v(x) + v(y)$,
2. $v(x + y) \geq \min(v(x), v(y))(x \neq -y)$.

As before we set $v(0) = \infty$ and for any $a \in \Gamma, \infty \geq a$, then we get a map $v : K \to \Gamma \cup \{\infty\}$ satisfying similar conditions.

Definition 7.1.1. *1. v is called discrete if $\Gamma \simeq \mathbb{Z}$ and v is surjective.*

2. An integral domain A is called a discrete valuation ring (DVR) if it is the valuation ring of some discrete valuation on its fraction field.

Examples 7.1.2. *1. Let p be a prime number. For any $0 \neq x \in \mathbb{Q}$, we can write $x = p^a y$ with $y = \frac{s}{r}, p \nmid s, r \in \mathbb{Z}$. Set $v_p(x) = a$. In this way we get a discrete valuation $v_p : \mathbb{Q}^\times \to \mathbb{Z}$. The associated DVR is $\mathbb{Z}_{(p)}$.*

2. Let k be a field and consider $K = k(X)$. Let $f \in k[X]$ be an irreducible polynomial with associated prime ideal $\mathfrak{p}_f = (f) \subset k[X]$. For any any $0 \neq g \in K$, we can write $g = f^a h$ with $h = \frac{s}{r}, f \nmid s, r \in k[X]$. Set $v_f(g) = a$. In this way we get a discrete valuation $v_f : K^\times \to \mathbb{Z}$. The associated DVR is $k[X]_{\mathfrak{p}_f}$.

Let A be a DVR with associated valuation $v : K^\times \to \mathbb{Z}$. We have known A is an integrally closed integral domain, and it is local with maximal ideal $\mathfrak{m} = \{x \in A \,|\, v(x) > 0\}$.

Lemma 7.1.3. *A is a Noetherian local domain of dimension 1, and any non zero ideal of A is a power of the maximal ideal $\mathfrak{m}$.*

Proof. For any non zero ideal $\mathfrak{a} \subset A$, there exists a minimal integer $k \in \mathbb{Z}_{\geq 0}$ such that $v(a) = k$ for some $a \in \mathfrak{a}$. Since $v : K^\times \to \mathbb{Z}$ is surjective, we have $\mathfrak{a} = \{y \in A \,|\, v(y) \geq k\} =: \mathfrak{m}_k$. Thus all the non zero ideals of A form a chain $\mathfrak{m} \supset \mathfrak{m}_2 \supset \mathfrak{m}_3 \supset \cdots$. In particular the a.c.c. holds and A is Noetherian.

Since $v : K^\times \to \mathbb{Z}$ is surjective, there exists an element $x \in \mathfrak{m}$ such that $v(x) = 1 \Rightarrow \mathfrak{m} = (x) \Rightarrow \mathfrak{m}_k = (x^k)$ for any $k \geq 1$. Therefore $\mathfrak{m}$ is the only non zero prime ideal, which implies than $\dim A = 1$. □

Proposition 7.1.4. *Let A be a Noetherian local domain of dimension 1, with maximal ideal $\mathfrak{m}$ and residue field $k = A/\mathfrak{m}$. TFAE:*

1. *A is a DVR.*
2. *A is integrally closed.*
3. *$\mathfrak{m}$ is a principal ideal.*
4. *$\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 1$.*
5. *Every non zero ideal is a power of $\mathfrak{m}$.*
6. *There exists an element $x \in A$ such that every non zero ideal of A is of the form (x^k) for some $k \geq 1$.*

Proof. First of all, by Proposition 5.3.11, for any $n \geq 1$, we have $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$. Next, since A is a local domain of dimension 1, we have $\operatorname{Spec} A = \{0, \mathfrak{m}\}$. Then for any ideal $0 \neq \mathfrak{a} \subsetneq A$, we have $r(\mathfrak{a}) = \mathfrak{m} \Rightarrow \mathfrak{a}$ is $\mathfrak{m}$-primary and $\mathfrak{m}^n \subset \mathfrak{a}$ for some $n > 0$.

$1 \Rightarrow 2$: see Proposition 4.2.2.

$2 \Rightarrow 3$: for any $0 \neq a \in \mathfrak{m}$, there exists an integer n such that $\mathfrak{m}^n \subset (a)$ and $\mathfrak{m}^{n-1} \not\subset (a)$. Take $b \in \mathfrak{m}^{n-1}$ such that $b \notin (a)$. Let $x = \frac{a}{b} \in K = \operatorname{Frac} A$. Then $b \notin (a) \Rightarrow x^{-1} = \frac{b}{a} \notin A$. Thus x^{-1} is not integral over A. Consider the A-module $x^{-1}\mathfrak{m}$. If $x^{-1}\mathfrak{m} \subset \mathfrak{m}$, $\mathfrak{m}$ would be a faithful $A[x^{-1}]$-module which is finitely generated over A, then x^{-1} is integral over A, a contradication. Thus $x^{-1}\mathfrak{m} \not\subset \mathfrak{m}$. Since $a \in \mathfrak{m}$ and $x^{-1} = \frac{b}{a}$, we have $x^{-1}\mathfrak{m} \subset A$. Then $x^{-1}\mathfrak{m} = A$ and thus $\mathfrak{m} = (x)$.

$3 \Rightarrow 4$: since $\dim_k(\mathfrak{m}/\mathfrak{m}^2) \leq 1$ and $\mathfrak{m} \neq \mathfrak{m}^2$.

$4 \Rightarrow 5$: $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = 1 \Rightarrow \mathfrak{m}$ is principal by Nakayama lemma. For any ideal $0 \neq \mathfrak{a} \subsetneq A$, there exists an integer n such that $\mathfrak{a} \supset \mathfrak{m}^n$. Consider the Artinian local ring $A/\mathfrak{m}^n$. By Proposition 5.3.12, $\mathfrak{a}/\mathfrak{m}^n$ is a principal ideal of $A/\mathfrak{m}^n$. Thus $\mathfrak{a}$ is a power of $\mathfrak{m}$.

$5 \Rightarrow 6$: as $\mathfrak{m} \neq \mathfrak{m}^2$, there exists an element $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Consider the ideal (x). By assumption $(x) = \mathfrak{m}^r$ for some r. Since $(x) \not\subset \mathfrak{m}^2$, we have $r = 1$ and $\mathfrak{m} = (x), \mathfrak{m}^k = (x^k)$.

$6 \Rightarrow 1$: we have $\mathfrak{m} = (x)$ and $(x^k) \neq (x^{k+1})$ for any k. For any $0 \neq a \in A$, there exists a unique $k \in \mathbb{Z}$ such that $(a) = (x^k)$. Define $v(a) = k$ and $v : K^\times \to \mathbb{Z}$ by $v(\frac{a}{b}) = v(a) - v(b)$. This is a discrete valuation and $A = R_v$.

□

Corollary 7.1.5. *Let A be a Noetherian domain of dimension 1. TFAE:*

1. *A is integrally closed.*
2. *Every primary ideal of A is prime power.*
3. *Every $A_\mathfrak{p}$ $(0 \neq \mathfrak{p} \in \operatorname{Spec} A)$ is a DVR.*

Proof. $1 \Leftrightarrow 3$ follows from Proposition 7.1.4 and Proposition 4.1.19.

$2 \Leftrightarrow 3$ follows from Proposition 7.1.4 and Corollary 6.2.9. □

Definition 7.1.6. *A Dedekind domain is a Noetherian domain of dimension 1 satisfying the condtions in Corollary 7.1.5.*

Corollary 7.1.7. *In a Dedekind domain, every non zero ideal has a unique fractorization as a product of prime ideals.*

Remark 7.1.8. *In the above, we studied Noetherian domains A under further conditions that A is integrally closed and* $\dim A = 1$. *For a general Noetherian domain A, one may first take its integral closure $\overline{A}$ in $K = \operatorname{Frac} A$ to get an integrally closed integral domain. However, it turns out that $\overline{A}$ might not be Noetherian any more. In fact, if A is a Noetherian domain of dimension 1, then $\overline{A}$ is Noetherian (see [12] Theorem 12.5 for example), thus a Dedekind domain; if A is a Noetherian domain of dimension ≥ 2, $\overline{A}$ is a possibly non Noetherian Krull ring (see [12] section 12). For an example of non Noetherian $\overline{A}$, see [13] 33. 10.*

Examples 7.1.9.

1. *Any PID (other than a field) is a Dedekind domain.*
2. *Let $K|\mathbb{Q}$ be a finite extension, O_K the integral closure of $\mathbb{Z}$ in K. Then O_K is Dedekind: By Corollary 4.1.9, O_K is integrally closed. By the theory of fields (see [2]*

Proposition 5.17) there exists a basis $v_1, \cdots, v_n$ *of* K *over* $\mathbb{Q}$ *such that* $O_K \subset \sum_{i=1}^n \mathbb{Z}v_i$. *This implies that* O_K *is a Noetherian ring. Since* O_K *is integral over* $\mathbb{Z}$, *by Theorem 4.1.14* $\operatorname{Spec} O_K \to \operatorname{Spec} \mathbb{Z}$ *is surjective. By Corollary 4.1.12 for any* $\mathfrak{p} \in \operatorname{Spec} O_K$,

$$\mathfrak{p} \text{ is maximal} \quad \Leftrightarrow \quad \mathfrak{p} \cap \mathbb{Z} \text{ is maximal} \quad \Leftrightarrow \quad \mathfrak{p} \cap \mathbb{Z} = (p)$$

for some prime number p. *Thus any non zero prime ideal of* O_K *is maximal, and* $\dim O_K = 1$.

3. *Let* S *be a compact Riemann surface,* $K = \mathbb{C}(S) = \{\text{meromorphic functions on } S\}$ *its field of meromorphic functions. Then any non constant map* $S \to \mathbb{P}^1(\mathbb{C})$ *induces an injection* $\mathbb{C}(t) \hookrightarrow K$ *which is a finite extension. We take any such map and view* $\mathbb{C}(t) \subset K$. *Consider the following set*

$$Zar(K, \mathbb{C}) = \{R \subsetneq K \text{ valuation rings of } K \mid R \supset \mathbb{C}\}.$$

Then we have a bijection

$$S \xrightarrow{\sim} Zar(K, \mathbb{C}), \quad P \mapsto O_P = \{\text{meromorphic functions which are regular at } P\}.$$

Now let A_1 *be the integral closure of* $\mathbb{C}[t]$ *in* K *and* A_2 *the integral closure of* $\mathbb{C}[t^{-1}]$ *in* K, *then one can prove similarly as above that* A_i *are Dedekind domains. By construction* $K = \operatorname{Frac} A_i$. *Let* $|\operatorname{Spec} A_i| = \operatorname{Spec} A_i \setminus \{0\}$=*set of closed points in* $\operatorname{Spec} A_i$. *We have natural inclusions*

$$|\operatorname{Spec} A_i| \hookrightarrow Zar(K, \mathbb{C}), \quad \mathfrak{p} \mapsto A_{i\mathfrak{p}},$$

and

$$Zar(K, \mathbb{C}) = |\operatorname{Spec} A_1| \cup |\operatorname{Spec} A_2|.$$

One can endow $Zar(K, \mathbb{C})$ *with some topology (this is then a Zariski-Riemann space) so that this is an open covering.*

7.2 Characterization by fractional ideals

We will give another characterization for DVRs and Dedekind domains.

Definition 7.2.1. *Let* A *be an integral domain,* $K = \operatorname{Frac} A$. *An* A*-submodule* $M \subset K$ *is called a fractional ideal of* A *if* $xM \subset A$ *for some* $0 \neq x \in A$. *In this case, denote* $(A : M) = \{x \in K \mid xM \subset A\}$, *which is an* A*-submodule of* K.

Examples 7.2.2. 1. *Any ideal* $\mathfrak{a} \subset A$ *is a fractional ideal.*

2. *For any* $x \in K, Ax \subset K$ *is fractional.*

Lemma 7.2.3. *Let* $M \subset K$ *be a finitely generated* A*-submodule, then* M *is fractional. If* A *is Noetherian, every fractional ideal is finitely generated over* A.

Proof. Exercise. □

Definition 7.2.4. *An* A*-submodule* $M \subset K$ *is called an invertible ideal if there exists an* A*-submodule* $N \subset K$ *such that* $MN = A$.

Let M be an invetible ideal, then the above N is uniquely determined:

$$N \subset (A : M) = (A : M)MN \subset AN = N, \quad \Rightarrow N = (A : M).$$

Lemma 7.2.5. *Let $M \subset K$ be an invertible ideal of A. Then M is finitely generated over A, thus it is a fractional ideal.*

Proof. SInce M is invertible, we have $M(A : M) = A$. Then there exist $x_i \in M, y_i \in (A : M), 1 \leq i \leq n$, such that $\sum_{i=1}^n x_i y_i = 1$. For any $x \in M$, we have then

$$x = \sum_{i=1}^n (y_i x) x_i, \quad y_i x \in A,$$

i.e. M is generated by $x_1, \cdots, x_n$. □

Example 7.2.6. *For any $x \in K^\times$, $Ax \subset K$ is an invertible ideal. We have $(A : Ax) = Ax^{-1}$.*

The set of invertible ideals form an abelian group with identity A with respective to the multiplicaiton.

Proposition 7.2.7. *Let M be a fractional ideal of A. TFAE:*

1. *M is invertible.*
2. *M is finitely generated and for any prime ideal $\mathfrak{p} \subset A$, $M_\mathfrak{p}$ is invertible.*
3. *M is finitely generated and for any maximal ideal $\mathfrak{m} \subset A$, $M_\mathfrak{m}$ is invertible.*

Proof. $1 \Rightarrow 2$: By Lemma 7.2.5 M is finitely generated. We have $M(A : M) = A \Rightarrow M_\mathfrak{p}(A_\mathfrak{p} : M_\mathfrak{p}) = (M(A : M))_\mathfrak{p} = A_\mathfrak{p}$ for any prime ideal $\mathfrak{p} \subset A$, thus $M_\mathfrak{p}$ is invertible. Here we used the following fact: if M is a finitely generated A-module and $S \subset A$ is a multiplicative subset, then $S^{-1}(A : M) = (S^{-1}A : S^{-1}M)$ (exercise, or see [2] Corollary 3.5).

$2 \Rightarrow 3$: trivial.

$3 \Rightarrow 1$: Let $\mathfrak{a} = M(A : M)$, then $\mathfrak{a} \subset A$ is an ideal. For any maximal ideal $\mathfrak{m}$, $\mathfrak{a}_\mathfrak{m} = M_\mathfrak{m}(A_\mathfrak{m} : M_\mathfrak{m}) = A_\mathfrak{m} \Rightarrow \mathfrak{a} = A$ and M is invertible. □

Proposition 7.2.8. *Let A be a local domain. Then A is a DVR $\Leftrightarrow$ every non zero fractional ideal of A is invertible.*

Proof. $\Rightarrow$: Write $\mathfrak{m} = (x) \subset A$ the maximal ideal. For any $0 \neq M$ fractional ideal, by definition $\exists\, y \in A$ such that $yM \subset A$. Then $yM = (x^r)$ and $M = (x^{r-s})$ where $s = v(y)$. Thus M is invertible.

$\Leftarrow$: The assumption together with Lemma 7.2.5 says that every non zero fraction ideal is finitely generated. In particular A is Noetherian. By Proposition 7.1.4, it suffices to prove every non zero ideal of A is a power of $\mathfrak{m}$ (which implies dim $A = 1$). Suppose that this is false. Let $\Sigma = \{\text{non zero ideals which are not powers of}\, \mathfrak{m}\}$, which is non empty then. As A is Noetherian, there exists a maximal element $\mathfrak{a} \in \Sigma$. Then $\mathfrak{a} \subsetneq \mathfrak{m} \Rightarrow \mathfrak{m}^{-1}\mathfrak{a} \subset \mathfrak{m}^{-1}\mathfrak{m} = A$ is an ideal and $\mathfrak{m}^{-1}\mathfrak{a} \supset \mathfrak{a}$. If $\mathfrak{m}^{-1}\mathfrak{a} = \mathfrak{a}$ then $\mathfrak{a} = \mathfrak{a}\mathfrak{m} \Rightarrow \mathfrak{a} = 0$ by Nakayama lemma. Thus $\mathfrak{m}^{-1}\mathfrak{a} \supsetneq \mathfrak{a}$ and $\mathfrak{m}^{-1}\mathfrak{a} \notin \Sigma$. Then $\mathfrak{m}^{-1}\mathfrak{a}$ is a power of $\mathfrak{m}$, which implies that $\mathfrak{a}$ is also a power of $\mathfrak{m}$, a contradication. Therefore every non zero ideal of A is a power of $\mathfrak{m}$. □

Theorem 7.2.9. *Let A be an integral domain. Then A is Dedekind $\Leftrightarrow$ every non zero fractional ideal of A is invertible.*

Proof. $\Rightarrow$: For any fractional ideal $0 \neq M$, since A is Noetherian, M is finitely generated. For any $0 \neq \mathfrak{p} \in \operatorname{Spec} A$, $M_{\mathfrak{p}}$ is a non zero fractional ideal of $A_{\mathfrak{p}}$. As $A_{\mathfrak{p}}$ is a DVR, by Proposition 7.2.8 $M_{\mathfrak{p}}$ is invertible. By Proposition 7.2.7 M is invertible.

$\Leftarrow$: The assumption together with Lemma 7.2.5 says that every non zero fraction ideal is finitely generated. In particular A is Noetherian. For any $0 \neq \mathfrak{p} \in \operatorname{Spec} A$, consider $A_{\mathfrak{p}}$. For any non zero ideal $\mathfrak{b}$ of $A_{\mathfrak{p}}$, $\mathfrak{a} := \mathfrak{b} \cap A \subset A$ is invertible by assumption. Then $\mathfrak{a}_{\mathfrak{p}} = \mathfrak{b}$ is invertible. By Proposition 7.2.8 $A_{\mathfrak{p}}$ is a DVR. Then A is a Dedekind domain.

□

Corollary 7.2.10. *Let A be a Dedekind domain. Then the set $I = \{\text{non zero fractional ideals of } A\}$ is an abelian group with respect to the multiplication.*

By Corollary 7.1.7, I is a free group over $\mathfrak{p} \in \operatorname{Spec} A \setminus \{0\}$. Consider the map

$$\phi : K^{\times} \to I, \quad u \mapsto Au.$$

Let $H = I/\mathrm{Im}\phi$, an abelian group, usually called the ideal class group of A.

Remark 7.2.11.

1. *If $A = O_K$ for $K|\mathbb{Q}$ finite extension, then H is a fintie group. This is a basic result of algebraic number theory. However, this is not ture in general: Claborn proved that every abelian group is a class group for some Dedekind domain, see [4, 9].*

2. *Let A be an arbitrary Dedekind domain, then we have*

$$H \text{ is trivial} \quad \Leftrightarrow \quad A \text{ is a PID} \quad \Leftrightarrow \quad A \text{ is a UFD}.$$

   The first $\Leftrightarrow$ is easy. In particular the group H is always trivial for DVRs.

   For the second $\Leftrightarrow$, the direction $\Rightarrow$ is well known; for the direction $\Leftarrow$: by Corollary 7.1.7, it suffices to show that any prime ideal is principal. For any $0 \neq \mathfrak{p} \in \operatorname{Spec} A$, take $0 \neq a \in \mathfrak{p}$. Since A is a UFD, we have $a = p_1 \cdots p_n$ with $p_i \in A$ irreducible elements. Then $0 \neq (p_i) \subset A$ are prime ideals. As $p_1 \cdots p_n \in \mathfrak{p}$ and $\mathfrak{p}$ is prime, there exists an i such that $p_i \in \mathfrak{p} \Rightarrow (p_i) \subset \mathfrak{p}$. Since $\dim A = 1$, (p_i) is maximal and thus $\mathfrak{p} = (p_i)$.

Exercise 7.2.12. *[2] Chapter 9, exercises 2, 4, 7, 9.*

Chapter 8

Graded rings and Completions

In Chapter 2 we learned an important technique, that is the localization of rings and modules. In this chapter we will introduce another two useful techniques: gradings and completions of rings and modules.

Let A be a ring, $M \in \mathrm{Mod}_A$. A filtration of M is a chain of submodules of M

$$M = M_0 \supset M_1 \supset \cdots \supset M_n \supset \cdots,$$

which we simply denote by (M_n).

Definition 8.0.1. *Let $I \subset A$ be an ideal. (M_n) is called an I-filtration (or I-admissible) if $IM_n \subset M_{n+1}$ for all n. It is called a stable I-filtration (or essentially I-adic) if it is an I-filtration and moreover, $IM_n = M_{n+1}$ for $n >> 0$. It is called I-adic if $M_n = I^n M$ for all n.*

By definition I-adic filtrations are stable I-filtrations.

Lemma 8.0.2. *If (M_n) and (M'_n) are stable I filtrations of M, then there exists an integer n_0 such that $M_{n+n_0} \subset M'_n$ and $M'_{n+n_0} \subset M_n$ for all $n \geq 0$.*

Proof. Since $(I^n M)$ is a stable I-filtration, it is enough to take $M'_n = I^n M$ for any n. Then since (M_n) is an I-filtration,

$$IM_n \subset M_{n+1}, \forall n \quad \Rightarrow \quad I^{n+n_0} M \subset I^n M \subset M_n, \forall n.$$

Since (M_n) is stable, there exists an integer n_0 such that

$$IM_n = M_{n+1}, \forall n \geq n_0 \quad \Rightarrow \quad M_{n+n_0} = I^n M_{n_0} \subset I^n M, \forall n.$$

□

Starting from an I-filtration (M_n), we can introduce a graded module and a topology on M, which are important tools to study M.

8.1 Graded rings and modules

Definition 8.1.1. *1. A graded ring is a ring A together with a family of subgroups $(A_n)_{n\geq 0}$ of the underlying abelian group of A, such that $A = \bigoplus_{n=0}^{\infty} A_n$ and $A_m A_n \subset A_{m+n}$ for all $m, n \geq 0$.*

2. *Let A be a graded ring. A graded A-module is an A-module M together with a family of subgroups $(M_n)_{n\geq 0}$, such that $M = \bigoplus_{n=0}^{\infty} M_n$ and $A_m M_n \subset M_{n+m}$ for all $m, n \geq 0$.*

Let A be a graded ring, M a graded A-module. Then $A_0 \subset A$ is a subring, A_n, M_n are A_0-modules for all n, and $A_+ := \bigoplus_{n>0} A_n \subset A$ is an ideal.

Definition 8.1.2. *1. An element $x \in M$ is called homogeneous of degree n if $x \in M_n$. (Thus for any element of M is a finite sum of homogeneous elements.)*

2. *Let M, N be graded A-modules. A homomorphism of graded A-modules M, N is a homomorphism of A-module $f : M \to N$ such that $f(M_n) \subset N_n$ for all $n \geq 0$.*

Proposition 8.1.3. *Let A be a graded ring. TFAE:*

1. *A is Noetherian.*

2. *A_0 is Noetherian and A is finitely generated as an A_0-algebra.*

Proof. $2 \Rightarrow 1$ follows from the Hilbert Basis theorem.

$1 \Rightarrow 2$: As $A_0 \simeq A/A_+$, A_0 is Noetherian. As A_+ is finitely generated as an ideal of A, we can take a set of generators $x_1, \cdots, x_s$ which we can assume to be homogenous of degrees $k_1, \cdots, k_s$ (all > 0) respectively. Consider $A' = A_0[x_1, \cdots, x_s] \subset A$ the subring generated by A_0 and $x_1, \cdots, x_s$. We show $A_n \subset A'$ for all $n \geq 0$ by induction. The case $n = 0$ is OK. Let $n > 0$. For any $y \in A_n \subset A_+$, we write $y = \sum_{i=1}^{s} a_i x_i$ with $a_i \in A_{n-k_i}$ (set $A_m = 0$ if $m < 0$). Since each $k_i > 0$, by induction hypothesis $a_i \in A' \Rightarrow y \in A' \Rightarrow A_n \subset A'$. Thus $A = A'$. □

8.2 Artin-Rees lemma

Let A be a ring, $I \subset A$ an ideal, set

$$A^* = \bigoplus_{n=0}^{\infty} I^n, \quad I^0 = A.$$

This is a graded ring. Let $M \in \mathrm{Mod}_A$ and (M_n) be an I-filtration of M, set

$$M^* = \bigoplus_{n=0}^{\infty} M_n.$$

This is a graded A^*-module since $I^m M_n \subset M_{m+n}$. If A is Noetherian, then I is finitely generated, and A^* is finitely generated as an A-algebra, thus A^* is also Noetherian. The following lemma is the key point to transfer the information of I-filtration to graded modules.

Lemma 8.2.1. *Let A be a Noetherian ring, M a finitely generated A-module, (M_n) an I-filtration of M. We have the associated A^* and M^* as above. TFAE:*

1. *M^* is a finitely generated A^*-module.*

2. *The filtration (M_n) is stable.*

Proof. Since A is Noetherian and M finitely generated over A, all the submodules M_n are finitely generated over A. Thus for any $n \geq 0$, the module $Q_n := \bigoplus_{r=0}^{n} M_r$ is finitely generated over A. By construction $Q_n \subset M^*$ is a subgroup for each n. Let $M_n^* \subset M^*$ be the sub A^*-module generated by Q_n by

$$M_n^* = M_0 \oplus \cdots \oplus M_n \oplus IM_n \oplus \cdots \oplus I^r M_n \oplus \cdots .$$

Then M_n^* is finitely generated over A^*. Letting n vary, M_n^* form an ascending chain and $M^* = \bigcup_n M_n^*$. Since A^* is Noetherian, we have

$$\begin{aligned} M^* \text{ is finitely generated over } A^* \quad &\Leftrightarrow \quad M^* = M_{n_0}^* \text{ for some } n_0 \\ &\Leftrightarrow \quad M_{n_0+r} = I^r M_{n_0}, \forall\, r \geq 0 \\ &\Leftrightarrow \quad \text{the filtration } (M_n) \text{ is stable.} \end{aligned}$$

□

Theorem 8.2.2 (Artin-Rees Lemma)**.** *Let A be a Noetherian ring, $I \subset A$ an ideal, M a finitely generated A-module, (M_n) a stable I-filtration of M. If $N \subset M$ is a submodule, then $(N \cap M_n)$ is a stable I-filtration of N.*

Proof. Since

$$I(N \cap M_n) \subset IN \cap IM_n \subset N \cap M_{n+1},$$

$(N \cap M_n)$ is an I-filtration. Consider the A^*-module M^*. $(N \cap M_n)$ defines a sub A^*-module $N^* \subset M^*$. As A is Notherian, A^* is Noetherian. By Lemma 8.2.1, (M_n) is stable $\Leftrightarrow$ M^* finitely generatd over A^* $\Rightarrow$ N^* finitely generate over A^* $\Leftrightarrow$ the filtration $(N \cap M_n)$ is stable. □

Corollary 8.2.3. *Let A, I, M, N be as above. Then there exists an integer k such that*

$$(I^n M) \cap N = I^{n-k}\big((I^k M) \cap N\big), \forall\, n \geq k.$$

Proof. Consider the I-adic filtration $M_n = I^n M$ of M, which is stable. The induced filtration $I^n M \cap N$ of N is stable by Theorem 8.2.2. □

Corollary 8.2.4. *Let A, I, M be as above, and $N = \bigcap_{n=0}^{\infty} I^n M$. Then $IN = N$.*

Proof. Apply Corollary 8.2.3 to $N \subset M$: for $n >> 0$, we have

$$N = I^n M \cap N = I^{n-k}\big((I^k M) \cap N\big) \subset IN \subset N, \quad \Rightarrow IN = N.$$

□

Corollary 8.2.5. *Let A, I, M be as above and assume moreover $I \subset J(A)$ (the Jacobson radical), then*

$$\bigcap_{n=0}^{\infty} I^n M = 0.$$

In particular, $\bigcap_{n=0}^{\infty} J(A)^n = 0$.

Corollary 8.2.6. *Let A be a Noetherian domain, $I \subsetneq A$ an ideal. Then*

$$\bigcap_{n=0}^{\infty} I^n = 0.$$

Proof. Let $N = \bigcap_n I^n$, then by Corollary 8.2.4 we have $IN = N$. Then there exists an element $x \in I$ such that $(1+x)N = 0$. As A is an integral domain and $1 + x \neq 0$, we have $N = 0$. □

8.3 Application: local criterion of flatness

We discuss another application of the Artin-Rees lemma. Let A be a ring, $I \subset A$ an ideal, set

$$G_I(A) = \bigoplus_{n=0}^{\infty} I^n/I^{n+1},$$

which is a graded ring. For $M \in \mathrm{Mod}_A$, set

$$G_I(M) = \bigoplus_{n=0}^{\infty} I^n M/I^{n+1}M,$$

which is a graded $G_I(A)$-module. Let $A_0 = A/I$ and $M_0 = M/IM$. If A is Noetherian, then I is finitely generated and A_0 is Noetherian, $G_I(A)$ is finitely generated as A_0-algbra thus also Noetherian. For any $n \geq 0$, we have canonical homomorphisms

$$\gamma_n : I^n/I^{n+1} \otimes_{A_0} M_0 \to I^n M/I^{n+1}M,$$

which induce a degree preserving homomorphism

$$G_I(A) \otimes_{A_0} M_0 \to G_I(M).$$

Theorem 8.3.1. *Let A be a ring, $I \subset A$ an ideal, $M \in \mathrm{Mod}_A$. Assume either (α) I is nilpotent, or (β) A is Noetherian and for any ideal $J \subset A$, we have $\bigcap_n I^n(J \otimes_A M) = 0$. Then TFAE:*

1. *M is flat over A.*
2. *$\mathrm{Tor}_1^A(M, N) = 0$ for any $N \in \mathrm{Mod}_{A_0}$.*
3. *M_0 is flat over A_0 and $I \otimes_A M \simeq IM$.*
4. *M_0 is flat over A_0 and $\mathrm{Tor}_1^A(M, A_0) = 0$.*
5. *M_0 is flat over A_0 and $\gamma_n : I^n/I^{n+1} \otimes_{A_0} M_0 \xrightarrow{\sim} I^n M/I^{n+1}M$ are isomorphisms for all $n \geq 0$, i.e. $G_I(A) \otimes_{A_0} M_0 \xrightarrow{\sim} G_I(M)$.*
6. *$M_n := M/I^{n+1}M$ are flat over $A_n := A/I^{n+1}$ for all $n \geq 0$.*

Proof. First of all, we remark that the implications $1 \Rightarrow 2 \Leftrightarrow 3 \Leftrightarrow 4 \Rightarrow 5 \Rightarrow 6$ hold without the above condition (α) nor (β).

We first show $1 \Leftrightarrow 6$.

The direction $1 \Rightarrow 6$ follows from the base change property of flatness. We show $6 \Rightarrow 1$ under the additional assumption. If (α) holds, then it is trivial to conclude. Assume now (β) holds. Then A is Noetherian. We show for any ideal $J \subset A$, the canonical map $j : J \otimes_A M \to M$ is injective. By assumption, it suffices to show

$$\mathrm{Ker}(j) \subset I^n(J \otimes_A M), \quad \forall\, n \geq 0.$$

Fix an n. By Theorem 8.2.2, there exists an integer $k > n$ such that $I^k \cap J \subset I^n J$. Consider the natural maps

$$J \otimes M \xrightarrow{f} \frac{J}{I^k \cap J} \otimes M \xrightarrow{g} \frac{J}{I^n J} \otimes M = \frac{J \otimes M}{I^n(J \otimes M)}.$$

Since $M_{k-1} = M/I^kM = M \otimes_A A_{k-1}$ is flat over $A_{k-1} = A/I^k$, the injection $J/I^k \cap J \hookrightarrow A_{k-1}$ induces an injection

$$\frac{J}{I^k \cap J} \otimes_A M = \frac{J}{I^k \cap J} \otimes_{A_{k-1}} M_{k-1} \hookrightarrow M_{k-1}.$$

Consider the following commutative diagram

$$\begin{tikzcd} J \otimes M \arrow[r, "f"] \arrow[d, "j"] & \frac{J}{I^k \cap J} \otimes M \arrow[r, "g"] \arrow[d, hook] & \frac{J \otimes M}{I^n(J \otimes M)} \\ M \arrow[r] & M_{k-1}, & \end{tikzcd}$$

we get

$$\mathrm{Ker}(j) \subset \mathrm{Ker}(f) \subset \mathrm{Ker}(g \circ f) = I^n(J \otimes M),$$

as desired. □

8.4 Topology and completion

Let G be a topological abelian group. Recall that this is both a topological space and an abelian group, such that the two structues are compatible: the operations

$$G \times G \to G, \quad (x, y) \mapsto x + y$$

and

$$G \to G, \quad x \mapsto -x$$

are continuous. If $0 \subset G$ is closed, then G is Hausdorff, since $\Delta = \phi^{-1}(0) \subset G \times G$ is closed, where $\phi : G \times G \to G, (x, y) \mapsto x - y$. For any $a \in G$, the map $T_a : G \to G, x \mapsto x + a$ is a homeomorphism. For any open neighborhood U of 0, $U + a = T_a(U)$ is an open neighborhood of a. Thus the topology on G is uniquely determined by the open neighborhoods of 0.

Remark 8.4.1. *There is another (more natural) point of view for topological abelian groups (rings/modules) due to Scholze: the associated condensed abelian groups (rings/modules)... See [14].*

Lemma 8.4.2. *Let G be a topological ableian group and $H = \bigcap_{U \text{ nbhds of } 0} U$. Then*

1. *H is a subgroup of G.*
2. *$H = \overline{\{0\}}$.*
3. *G/H is Hausdorff.*
4. *G is Hausdorff $\Leftrightarrow$ $H = 0$.*

Proof. Exercise. □

From now on we assume for simplicity $0 \in G$ has countable fundamental system of neighborhoods.

Definition 8.4.3. 1. *A Cauchy sequence in G is a sequence (x_n) of elements of G such that for any open neighborhood U of 0, there exists an integer $s(U)$ such that $x_m - x_n \in U$ for any $m, n \geq s(U)$.*

2. *Let (x_n) and (y_n) be Cauchy sequences in G. They are equivalent, written $(x_n) \sim (y_n)$, if $x_n - y_n \to 0$ in G.*

3. *Let $\widehat{G} = \{$Cauchy sequences in $G\}/\sim$, called the completion of G.*

Then one can check that $\widehat{G}$ is a topological ableian group. Moreover, we have a canonical map $\phi : G \to \widehat{G}$, sending an element of G to the class of constant Cauchy sequence, which is a homorphism of topolocial abelian groups.

Lemma 8.4.4. *We have*

$$\mathrm{Ker}\phi = \bigcap_{U \text{ nbhds of } 0} U.$$

Thus ϕ is injective $\Leftrightarrow$ G is Hausdorff.

Proof. Exercise. □

If $f : G \to H$ is a continuous homomorphism, then it induces a continuous homomorphism $\widehat{f} : \widehat{G} \to \widehat{H}$. If $G \xrightarrow{f} H \xrightarrow{g} K$ is a sequence of continuous homomorphisms, we have $\widehat{g \circ f} = \widehat{g} \circ \widehat{f}$. In other words, the completion $\widehat{}$ forms a functor from the category of topological abelian groups to itself.

From now on, we assume the topology of G is defined by a fundamental system of neighborhood of 0 consisting of subgroups

$$G = G_0 \supset G_1 \supset \cdots \supset G_n \supset \cdots$$

Then each G_n is both open and closed. The quotient groups G/G_n are then discrete. Consider $\varprojlim_n G/G_n$, which is a topological abelian group.

Proposition 8.4.5. *Under the above notation, we have an isomorphism of topological abelian groups*

$$\widehat{G} \cong \varprojlim_n G/G_n.$$

Proof. Exercise. □

Let $\{A_n\}, \{B_n\}$ and $\{C_n\}$ are inverse systems of abelian groups. Suppose we have an exact sequence of inverse systems

$$0 \to \{A_n\} \to \{B_n\} \to \{C_n\} \to 0,$$

i.e. for any n a commutative diagram of exact sequences

$$\begin{array}{ccccccccc}
0 & \longrightarrow & A_{n+1} & \longrightarrow & B_{n+1} & \longrightarrow & C_{n+1} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & A_n & \longrightarrow & B_n & \longrightarrow & C_n & \longrightarrow & 0.
\end{array}$$

Proposition 8.4.6. *Under the above notation, the sequence*

$$0 \to \varprojlim_n A_n \to \varprojlim_n B_n \to \varprojlim_n C_n$$

is always exact. If $\{A_n\}$ is a surjective system, i.e. for any n, $A_{n+1} \to A_n$ is surjective, then

$$0 \to \varprojlim_n A_n \to \varprojlim_n B_n \to \varprojlim_n C_n \to 0$$

is exact.

Proof. Exercise. □

Corollary 8.4.7. *Let*

$$0 \to G' \to G \to G'' \xrightarrow{\pi} 0$$

be an exact sequence of abelian groups. Let G have the topology defined by a sequence $\{G_n\}$ of subgroups, and G', G'' have the induced topology, i.e. the topology defined by $\{G_n \cap G'\}$ and $\{\pi(G_n)\}$ respectively. Then the sequence

$$0 \to \widehat{G'} \to \widehat{G} \to \widehat{G''} \to 0$$

is exact.

Proof. Apply Proposition 8.4.6 to $A_n := G'/(G' \cap G_n), B_n = G/G_n$ and $C_n = G''/\pi(G_n)$. □

Corollary 8.4.8. *For any n, $\widehat{G_n}$ is a subgroup of $\widehat{G}$, and $\widehat{G}/\widehat{G_n} \simeq G/G_n$. Thus $\widehat{\widehat{G}} \simeq \widehat{G}$.*

Proof. For any n, by Corollary 8.4.7 we have an exact sequence

$$0 \to \widehat{G_n} \to \widehat{G} \to \widehat{G''} \to 0,$$

where $G'' = G/G_n$, which is discrete, thus $\widehat{G''} = G''$, which gives $\widehat{G}/\widehat{G_n} \simeq G/G_n$. Taking inverse limit over n, we get $\widehat{\widehat{G}} \simeq \widehat{G}$. □

Let G be a topological abelian group as above. If the canonical map $\phi : G \to \widehat{G}$ is an isomorphism, then we say G is complete (in particular G is Hausdorff).

We apply the above constructions to rings and modules. Let A be a ring, $I \subset A$ an ideal. Then the chain

$$A \supset I \supset \cdots \supset I^n \supset \cdots$$

defines a toplogy on A, called the I-adic topology. In this case, one checks that the multiplication

$$A \times A \to A, \quad (x, y) \mapsto xy$$

is also continuous. Thus A is a topological ring. The I-adic topology is Hausdorff $\Leftrightarrow$ $\bigcap_n I^n = 0$. Moroeover, the completion

$$\widehat{A} \cong \varprojlim_n A/I^n$$

is also a topological ring, and the canonical map $\phi : A \to \widehat{A}$ is a continuous ring homomorphism with $\operatorname{Ker} \phi = \bigcap_n I^n$.

Let $M \in \mathrm{Mod}_A$, then the filtration

$$M \supset IM \supset \cdots \supset I^n M \supset \cdots$$

defines a toplogy on M, called the I-adic topology. One checks also the map

$$A \times M \to M, \quad (a, x) \mapsto ax$$

is continuous, i.e. M is a topological module over the topological ring A. The completion

$$\widehat{M} = \varprojlim_n M/I^n M$$

is then a topological module over $\widehat{A}$. If $f : M \to N$ is a homomorphism of A-modules such that $f(I^n M) \subset I^n N$, then f is continuous fo the I-adic toplogy on M and N, thus it induces $\widehat{f} : \widehat{M} \to \widehat{N}$.

Examples 8.4.9. 1. *Let k be a field and $A = k[X]$. Consider the ideal $I = (X) \subset A$. Then the I-adic completion $\widehat{A} = k[[X]]$.*

2. *Let $A = \mathbb{Z}$ and p be a prime number. Consider the ideal $I = (p) \subset \mathbb{Z}$. The I-adic completion is $\widehat{A} = \mathbb{Z}_p = \{\sum_{n=0}^{\infty} a_n p^n \mid 0 \leq a_n \leq p-1\}$. Note that $p^n \to 0$ in $\mathbb{Z}_p$ when $n \to \infty$.*

   For $A = \mathbb{Z}_{(p)}, I = p\mathbb{Z}_{(p)}$, we have also $\widehat{A} = \mathbb{Z}_p$.

Let A be a ring, $I \subset A$ an ideal and $M \in \mathrm{Mod}_A$. Recall that a stable I-filtration on M is given by

$$M = M_0 \supset \cdots \supset M_n \supset \cdots$$

such that for any n we have $IM_n \subset M_{n+1}$ and for $n >> 0$, $IM_n = M_{n+1}$. By Lemma 8.0.2, a stable I-filtration defines the same topology as the I-adic topology on M. If A is Noetherian and M is finitely generated over A, for any submodule $N \subset M$, the Artin-Rees Lemma (Theorem 8.2.2) says that the subspace topology on N induced by the I-adic topology on M is the I-adic topology of N.

8.5 Applications to Noetherian rings

We apply the above constructions to Noetherian rings. Let A be a Noetherian ring, $I \subset A$ an ideal, $\widehat{A}$ the I-adic completion of A. In this section we will show the canonical map $A \to \widehat{A}$ is flat and $\widehat{A}$ is also Noetherian.

Proposition 8.5.1. *Let A be a Noetherian ring, $I \subset A$ an ideal, $0 \to M' \to M \to M'' \to 0$ an exact sequence of finitely generated A-modules. Then the I-adic completion*

$$0 \to \widehat{M'} \to \widehat{M} \to \widehat{M''} \to 0$$

is also exact.

Proof. By the Artin-Rees Lemma (Theorem 8.2.2), the I-adic topology on M' is the topology defined by $I^n M \cap M'$. Then apply Corollary 8.4.7. $\square$

Let A be a ring, $I \subset A$ an ideal, $\widehat{A}$ the I-adic completion of A, and $A \to \widehat{A}$ the canonical map. For any $M \in \mathrm{Mod}_A$, we get an $\widehat{A}$-module $\widehat{A} \otimes_A M$. On the other hand we have also the $\widehat{A}$-module $\widehat{M}$, the I-adic completion of M, together with the canonical map $M \to \widehat{M}$. Then we get homomorphisms

$$\widehat{A} \otimes_A M \to \widehat{A} \otimes_A \widehat{M} \to \widehat{A} \otimes_{\widehat{A}} \widehat{M} = \widehat{M}.$$

Consider the composition $\widehat{A} \otimes_A M \to \widehat{M}$.

Proposition 8.5.2. *For any ring A, if M is finitely generated over A, then $\widehat{A} \otimes_A M \to \widehat{M}$ is surjective. If moreover A is Noetherian, then $\widehat{A} \otimes_A M \to \widehat{M}$ is an isomorphism.*

Proof. Firstly, one checks easily that I-adic completion commutes with finite direct sums. Thus if $F \simeq A^n$ is a free module of finite rank, we have $\widehat{F} \simeq \widehat{A}^n \simeq \widehat{A} \otimes_A F$. If M is finitely generated over A, we have an exact sequence

$$0 \to N \to F \to M \to 0$$

for some n and $F \simeq A^n$. Then we get a commutative diagram

$$\begin{tikzcd}
 & \widehat{A} \otimes_A N \arrow[r] \arrow[d, "\gamma"] & \widehat{A} \otimes_A F \arrow[r] \arrow[d, "\beta", "\simeq"'] & \widehat{A} \otimes_A M \arrow[r] \arrow[d, "\alpha"] & 0 \\
0 \arrow[r] & \widehat{N} \arrow[r] & \widehat{F} \arrow[r] & \widehat{M} \arrow[r] & 0,
\end{tikzcd}$$

where horizontal lines are exact (by the right exactness of tensor product and by Corollary 8.4.7, where we take the induced subspace topology on N). Since β is an isomorphism, a diagram chasing shows that α is surjective. If A is Noetherian, then N is finitely generated over A, thus γ is surjective by what we just proved. Then another diagram chasing shows that α is injective thus an isomorphism.

□

Corollary 8.5.3. *Let A be a Noetherian ring, $I \subset A$ an ideal, $\widehat{A}$ the I-adic completion of A, then the canonical map $A \to \widehat{A}$ is flat.*

Proof. Apply Propositions 8.5.1, 8.5.2 and Corollary 3.1.2. □

Proposition 8.5.4. *Let A be a Noetherian ring, $I \subset A$ an ideal, $\widehat{A}$ the I-adic completion of A, then we have*

1. $\widehat{I} = \widehat{A}I \cong \widehat{A} \otimes_A I$.
2. $\widehat{(I^n)} = \widehat{I}^n$ *for any* $n \geq 0$.
3. $I^n/I^{n+1} \simeq \widehat{I}^n/\widehat{I}^{n+1}$ *for any* $n \geq 0$.
4. $\widehat{I} \subset J(\widehat{A})$.

Proof. 1. Apply Proposition 8.5.2.

2. Apply 1 to I^n: $\widehat{(I^n)} = \widehat{A}I^n = (\widehat{A}I)^n = \widehat{I}^n$.

3. Since for any n we have $A/I^n \simeq \widehat{A}/\widehat{I}^n$, which are compatible when n varies, $\Rightarrow$ $I^n/I^{n+1} \simeq \widehat{I}^n/\widehat{I}^{n+1}$.

4. Since $\widehat{A}$ is I-adically complete, for any $x \in \widehat{I}$,

$$1 + x + x^2 + \cdots \in \widehat{A}$$

defines an element, which is the inverse of $1 - x$, thus $1 - x \in (\widehat{A})^\times$. The same argument implies that for any $y \in \widehat{A}$, $1 - xy \in (\widehat{A})^\times$. By Lemma 1.2.9, $x \in J(\widehat{A})$. Therefore, $\widehat{I} \subset J(\widehat{A})$. □

Corollary 8.5.5. *Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $\widehat{A}$ the $\mathfrak{m}$-adic completion. Then $\widehat{A}$ is a local ring with maximal ideal $\widehat{\mathfrak{m}}$.*

Proof. Since $A/\mathfrak{m} \simeq \widehat{A}/\widehat{\mathfrak{m}}$ is a field, $\widehat{\mathfrak{m}} \subset \widehat{A}$ is a maximal ideal. By Proposition 8.5.4 4, $\widehat{\mathfrak{m}} = J(\widehat{A})$, thus $\widehat{A}$ is a local ring with the unique maximal ideal $\widehat{\mathfrak{m}}$. □

Corollary 8.5.6. *Let A be a Noetherian ring, $I \subset A$ an ideal, M a finitely generated A-module, $\widehat{M}$ the I-adic completion of M, $\phi : M \to \widehat{M}$ the canonical map. Then*

$$\operatorname{Ker} \phi = \{x \in M \mid \exists\, y \in I, (1 + y)x = 0\}.$$

Proof. We know $\operatorname{Ker} \phi = \bigcap_n I^n M := N$. By Corollary 8.2.4, we have $IN = N$. Thus $(1 - \alpha)N = 0$ for some $\alpha \in I$. Conversely, if $(1 - \alpha)x = 0$ for some $\alpha \in I$, then

$$x = \alpha x = \alpha^2 x = \cdots \in \bigcap_n I^n M = N.$$

Thus $N = \{x \in M \mid \exists\, y \in I, (1 + y)x = 0\}$. □

Corollary 8.5.7. *Let A be a Noetherian ring, $I \subset A$ an ideal, $S = 1 + I \subset A$ the induced multiplicative subset, $\widehat{A}$ the I-adic completion of A. Then the canonical map $A \to \widehat{A}$ factors through $S^{-1}A$ and induces an injection $S^{-1}A \hookrightarrow \widehat{A}$.*

Proof. Let $f : A \to S^{-1}A, \phi : A \to \widehat{A}$ be the canonical maps. Then for any $s = 1 - \alpha \in S, \alpha \in I$,

$$1 + \phi(\alpha) + \phi(\alpha)^2 + \cdots \in \widehat{A}$$

defines an element, which is the inverse of $\phi(s) = 1 - \phi(\alpha)$. Thus ϕ factors thourh $S^{-1}A$ via f. By Corollary 8.5.6, $\operatorname{Ker} f = \operatorname{Ker} \phi$. Thus the induced map $S^{-1}A \to \widehat{A}$ is injective. □

Let A be a ring, $I \subset A$ an ideal, then we have the associated graded ring

$$G_I(A) = \bigoplus_{n=0}^{\infty} I^n / I^{n+1}.$$

Let $M \in \mathrm{Mod}_A$, (M_n) an I-filtration of M, then we have the associated graded $G_I(A)$-module

$$G(M) = \bigoplus_{n=0}^{\infty} M_n / M_{n+1}.$$

If A is Noetherian, then I is finitely generated, $A_0 := A/I$ is Noetherian, and $G_I(A)$ is a finitely generated A_0-algebra thus also Noetherian.

Proposition 8.5.8. *Let A be a Noetherian ring, $I \subset A$ an ideal, $\widehat{A}$ the I-adic completion of A. Then we have*

1. *$G_I(A) \simeq G_{\widehat{I}}(\widehat{A})$ as graded rings.*

2. *If M is a finitely generated A-module, (M_n) a stable I-filtration, then $G(M)$ is a finitely generated $G_I(A)$-module.*

Proof. 1. By Proposition 8.5.4, $I^n/I^{n+1} \simeq \widehat{I}^n/\widehat{I}^{n+1}$ for any $n \geq 0$, thus $G_I(A) \simeq G_{\widehat{I}}(\widehat{A})$.

2. Since (M_n) is stable, there exists an integer n_0 such that $M_{n_0+r} = I^r M_{n_0}$ for any $r \geq 0$. So $G(M)$ is generated bby $\bigoplus_{n \leq n_0} M_n/M_{n+1}$. As each M_n/M_{n+1} is a finitely generated A/I-module, $\bigoplus_{n \leq n_0} M_n/M_{n+1}$ is a finitely generated A/I-module. This implies $G(M)$ is finitely generated over $G_I(A)$. □

A filtered abelian group is an abelian group A with a filtration

$$A = A_0 \supset A_1 \supset \cdots \supset A_n \supset \cdots .$$

Associated to such a filtered abelian group, we can define the graded group $G(A) = \bigoplus_{n=0}^{\infty} A_n/A_{n+1}$ and the completion $\widehat{A} = \varprojlim_n A/A_n$ for the topology on A with (A_n) a fundamental system of neighborhood of 0.

Lemma 8.5.9. *Let $\phi : A \to B$ be a homomorphism of filtered abelian groups, i.e. $\phi(A_n) \subset B_n$. Let $G(\phi) : G(A) \to G(B)$ and $\widehat{\phi} : \widehat{A} \to \widehat{B}$ be the induced maps. Then*

1. *$G(\phi)$ injective $\Rightarrow$ $\widehat{\phi}$ injective.*

2. *$G(\phi)$ surjective $\Rightarrow$ $\widehat{\phi}$ surjective.*

Proof. Exercise. □

Proposition 8.5.10. *Let A be a ring, $I \subset A$ an ideal, $M \in \mathrm{Mod}_A$, and (M_n) an I-filtration of M. Suppose A is I-adically complete, $\bigcap_n M_n = 0$ and $G(M)$ finitely generated over $G_I(A)$. Then M is finitely generated over A. If $G(M)$ is moreover a Noetherian $G_I(A)$-module, then M is a Noetherian A-module.*

Proof. Let $\xi_1, \cdots, \xi_n$ be a set of homogenous generators of $G(M)$ with $\deg \xi_i = k_i$, i.e. $\xi_i \in M_{k_i}/M_{k_{i+1}}$. Let $x_i \in M_{k_i}$ be such that $\overline{x_i} = \xi_i$ for any $1 \leq i \leq n$. For each i, consider the filtration on A defined by $(I^{m+k_i})_m$. Consider $F = A^n$ with the direct sum filtration. We get a map of filtered groups

$$\phi : F \to M, \quad v_i \mapsto x_i$$

where $v_i = (0, \cdots, 0, 1, 0, \cdots, 0)$ are the standard basis of A^n. Let $G(\phi) : G(F) \to G(M)$ be the associated map on graded groups. By assumption, $G(\phi)$ is surjective. By Lemma 8.5.9, $\widehat{\phi} : \widehat{F} \to \widehat{M}$ is surjective. Consider the following commutative diagram

$$\begin{tikzcd} F \arrow[r, "\phi"] \arrow[d, "\alpha"] & M \arrow[d, "\beta"] \\ \widehat{F} \arrow[r, "\widehat{\phi}"] & \widehat{M}. \end{tikzcd}$$

Since F is free and $A = \widehat{A}$ by assumption, $F \simeq \widehat{F}$, that is, α is an isomorphism. Since $\bigcap_n M_n = 0$, we have $\mathrm{Ker}\, \beta = 0$ i.e. β is injective. As $\widehat{\phi}$ is surjective, and the above diagram is commutative, ϕ is then surjective, i.e. M is finitely generated over A.

If $N \subset M$ is any submodule, we get an I-filtration (N_n) on N with $N_n = M_n \cap N$. Since $\bigcap_n N_n = 0$, the injection $N_n/N_{n+1} \hookrightarrow M_n/M_{n+1}$ induces an injection $G(N) \hookrightarrow G(M)$. We view $G(N)$ as a sub $G_I(A)$-module. If $G(M)$ is Noetherian, $G(N)$ is then finitely generated over $G_I(A)$. By the above paragrah, N is finitely generated over A. This implies that M is Noetherian. □

Corollary 8.5.11. *Let A be a Noetherian ring, $I \subset A$ an ideal, $\widehat{A}$ the I-adic completion of A. Then $\widehat{A}$ is Noetherian.*

Proof. By Proposition 8.5.8 we have $G_I(A) = G_{\widehat{I}}(\widehat{A})$, which is Noetherian. Then apply Proposition 8.5.10 to $\widehat{A}, \widehat{I}, M = \widehat{A}$. □

Corollary 8.5.12. *If A is a Noetherian ring, then so is $A[[X_1, \cdots, X_n]]$.*

Proof. A is Noetherian, then by Hilbert Basis theorem, $A[X_1, \cdots, X_n]$ is Noetherian. Consider the ideal $I = (X_1, \cdots, X_n) \subset A[X_1, \cdots, X_n]$. $A[[X_1, \cdots, X_n]] = \widehat{A[X_1, \cdots, X_n]}$ is the I-adic completion of $A[X_1, \cdots, X_n]$, thus it is Noetherian by Corollary 8.5.11. □

Remark 8.5.13. *In the above we deduced Corollary 8.5.12 from Corollary 8.5.11. However, one can argue inversely. More precisely, we can first prove directly that power series rings over Noetherian rings are Noetherian (similar to the Hilbert Basis Theorem 5.2.8), see [12] Theorem 3.3. Then we deduce that the I-adic completion $\widehat{A}$ of a Noetherian ring A is Noetherian by establishing*

$$\widehat{A} \simeq A[[X_1, \cdots, X_n]]/(X_1 - a_1, \cdots, X_n - a_n)$$

where $I = (a_1, \cdots, a_n)$, see [12] Theorem 8.12.

Remark 8.5.14. *So far we have mainly studied topological rings which are Noetherian and the topology is I-adic for some ideal I. However, there are interesting topological rings which are neither Noetherian nor with I-adic topology. See [15, 6].*

Exercise 8.5.15. *[2] Chapter 10, exercises 3, 4, 6, 7, 9.*

Chapter 9

Dimension theory of Noetherian rings

Let A be a ring. Recall its dimension

$$\begin{aligned}\dim A &= \text{supremum of lengths of prime ideals chains}\\ &= \sup_{\mathfrak{p}\in \operatorname{Spec} A} \dim A_{\mathfrak{p}}.\end{aligned}$$

A closely related notion is the height of an ideal.

Definition 9.0.1. *1. For any $\mathfrak{p} \in \operatorname{Spec} A$, we define its height as*

$$ht\, \mathfrak{p} = \textit{supremum of lengths of } \mathfrak{p} = \mathfrak{p}_0 \supsetneqq \mathfrak{p}_1 \cdots \supsetneqq \mathfrak{p}_r, \quad \mathfrak{p}_i \in \operatorname{Spec} A.$$

Thus $ht\, \mathfrak{p} = \dim A_{\mathfrak{p}}$.

2. For any ideal $I \subset A$, we define its height as

$$ht\, I = \inf_{\mathfrak{p}\in \operatorname{Spec} A, \mathfrak{p}\supset I} ht\, \mathfrak{p}.$$

It follows from definition that for any ideal $I \subset A$, we have an unequality

$$ht\, I + \dim A/I \leq \dim A.$$

We have studied certian rings of lower dimension.

Examples 9.0.2. *1. Artinian rings are of dimension* 0.

2. Dedekind domains are of dimension 1.

We will study the general dimension theory for Noetherian rings in this chapter. It suffices to consider Noetherian local rings.

9.1 Hilbert functions

Let us come back to the setting of section 8.1. Let $A = \bigoplus_{n=0}^{\infty} A_n$ be a Noetherian graded ring. Then A_0 is Noetherian, and A is finitely generated as an A_0-algebra, say by $x_1, \cdots, x_s$, which are homogenous of degrees $k_1, \cdots, k_s$ (all > 0). Let $M = \bigoplus_{n=0}^{\infty} M_n$

be a finitely generated graded A-module, with a set of homogenous generators $m_1, \cdots, m_t$ of degrees $r_1, \cdots, r_t$. For each $n \geq 0$, $\forall\, m \in M_n$, $m = \sum_{j=1}^{t} f_j(x) m_j$, where $f_j(x) \in A = A_0[x_1, \cdots, x_s]$ is homogenous of degree $n - r_j$ for any $1 \leq j \leq t$. This implies that M_n is finitely generated over A_0.

Let λ be an additive function with values in $\mathbb{Z}$ on the category of finitely generated A_0-modules, i.e. if $0 \to N_1 \to N_2 \to N_3 \to 0$ is an exact sequence of finitely generated A_0-modules, then

$$\lambda(N_2) = \lambda(N_1) + \lambda(N_3).$$

For example, in case that A_0 is an Artinian ring, $\lambda = \ell$ is the length function. Consider the integers $\lambda(M_n) \in \mathbb{Z}$, for $n = 0, 1, 2, \cdots$. We produce the generating function of them as

$$P(M,t) = P_\lambda(M,t) = \sum_{n=0}^{\infty} \lambda(M_n) t^n \in \mathbb{Z}[[t]].$$

Theorem 9.1.1. *$P(M,t)$ is a rational function in t of the form*

$$P(M,t) = \frac{f(t)}{\prod_{i=1}^{s}(1-t^{k_i})}, \quad f(t) \in \mathbb{Z}[t].$$

Proof. See [2] Theorem 11.1. □

Definition 9.1.2. *We define an integer*

$$d(M) = d_\lambda(M) = \textit{order of the pole of } P(M,t) \textit{ at } t = 1.$$

Corollary 9.1.3. *If $k_i = 1$ for all $i = 1, \cdots, s$, then there exists a polynomial $\phi_M(t) = \phi_{M,\lambda}(t) \in \mathbb{Q}[t]$ of degree $d(M) - 1$ such that*

$$\phi_M(n) = \lambda(M_n), \quad \textit{for } n >> 0.$$

Proof. Write $d = d(M)$. We adopt the convention that the degree of the zero polynomial is -1 and

$$\binom{n}{-1} = \begin{cases} 0, & n \geq 0, \\ 1, & n = -1. \end{cases}$$

By Theorem 9.1.1, $\lambda(M_n) =$ coefficent of t^n in $f(t)(1-t)^{-s}$. Canceling powers of $(1-t)$ we may assume $s = d$ and $f(1) \neq 0$. Write $f(t) = \sum_{k=0}^{N} a_k t^k$. Since

$$(1-t)^{-d} = \sum_{k=0}^{\infty} \binom{d+k-1}{d-1} t^k,$$

we get

$$\lambda(M_n) = \sum_{k=0}^{N} a_k \binom{d+n-k-1}{d-1}, \quad \forall\, n \geq N.$$

Then there exists a polynomial $\phi_M(t) \in \mathbb{Q}[t]$ with leading term $(\sum_{k=0}^{N} a_k) \frac{t^{d-1}}{(d-1)!}$ such that $\phi_M(n) = \sum_{k=0}^{N} a_k \binom{d+n-k-1}{d-1}, \quad \forall\, n \geq N.$ □

The function $\phi_M(t)$ is called the Hilbert function or Hilbert polynomial of M (with respect to λ). From the proof of Theorem 9.1.1 we can deduce the following proposition.

Proposition 9.1.4. *For any $k \geq 0$ and $x \in A_k$ which is a non zero-divisor of M, then $d(M/xM) = d(M) - 1$.*

From now on, we assume that A_0 is Artinian and we will take $\lambda(M_n) = \ell(M_n)$ =length of the finitely generated A_0-module M_n (recall that any finitely generated module over an Artinian ring is of finite length).

Example 9.1.5. *Let k be a field, $A = k[X_1, \cdots, X_s]$ and $M = A$. Write $A = \bigoplus_{n=0}^{\infty} A_n$. Then*

$$\ell(A_n) = \dim_k(A_n) = \binom{s+n-1}{s-1}, \quad \Rightarrow P(M,t) = (1-t)^{-s}.$$

One can also compute

$$\phi_A(t) = \frac{1}{(s-1)!}(t+s-1)\cdots(t+1).$$

Proposition 9.1.6. *Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $\mathfrak{q}$ a $\mathfrak{m}$-primary ideal, M a finitely generated A-module, (M_n) a stable $\mathfrak{q}$-filtration of M. Then*

1. *M/M_n has finite length, for any $n \geq 0$.*

2. *For $n >> 0$, $\ell(M/M_n) = g(n)$ for a polynomial $g(t) \in \mathbb{Q}[t]$ of degree $\leq s$ =least number of generators of $\mathfrak{q}$.*

3. *The degree and leading coefficient of $g(t)$ depend only on M and $\mathfrak{q}$, not on the filtration.*

Proof. 1. Let

$$G(A) = G_{\mathfrak{q}}(A) = \bigoplus_{n=0} \mathfrak{q}^n/\mathfrak{q}^{n+1}, \quad G(M) = \bigoplus_{n} M_n/M_{n+1}$$

be the associated graded ring and module. Then $A_0 = A/\mathfrak{q}$ is an Artinian local ring as $\mathfrak{m}^k \subset \mathfrak{q} \subset \mathfrak{m}$ for some k. Moreover, $G(A)$ is Noetherian and $G(M)$ is a finitely generated graded $G(A)$-module. Since each $G_n(M) = M_n/M_{n+1}$ is finitely generated over A_0, thus of finite length. Then M/M_n has finite length, and

$$\ell_n := \ell(M/M_n) = \sum_{i=1}^{n} \ell(M_i/M_{i+1})$$

as ℓ is additive.

2. If $x_1, \cdots, x_s$ is a set of generators of $\mathfrak{q}$, the images $\overline{x_i}$ of x_i in $\mathfrak{q}/\mathfrak{q}^2$ generate $G(A)$ as an A_0-algebra. We have deg $\overline{x_i} = 1$ for all $1 \leq i \leq s$. By Corollary 9.1.3,

$$\ell(M_n/M_{n+1}) = \phi_M(n)$$

for some polynomial $\phi_M(t) \in \mathbb{Q}[t]$ of degree $\leq s-1$ (note $d(G(M)) \leq s$ since all deg $\overline{x_i} = 1$) and $n >> 0$. Since

$$\ell_{n+1} - \ell_n = \phi_M(n),$$

there exists a polynomial $g(t) \in \mathbb{Q}[t]$ of degree $\leq s$ such that

$$\ell_n = g(n) \quad \text{for } n >> 0.$$

In fact, deg $g(t) = $ deg $\phi_M(t) + 1 = d(G(M))$.

3. If $(\widetilde{M_n})$ is another stable $\mathfrak{q}$-filtration on M, let $\widetilde{g}(n) = \ell(M/\widetilde{M_n})$ with $\widetilde{g}(t) \in \mathbb{Q}[t]$ the associated polynomial. By Lemma 8.0.2, there exists an integer n_0 such that $M_{n+n_0} \subset \widetilde{M_n}, \widetilde{M_{n+n_0}} \subset M_n$ for all $n \geq 0$. Thus

$$g(n+n_0) \geq \widetilde{g}(n), \quad \widetilde{g}(n+n_0) \geq g(n), \quad \Rightarrow \lim_{n\to\infty} \frac{g(n)}{\widetilde{g}(n)} = 1.$$

This implies that $g(t), \widetilde{g}(t)$ have the same degree and leading coeffient.

□

Consider the $\mathfrak{q}$-adic filtration $(\mathfrak{q}^n M)$. Let $\chi_{\mathfrak{q}}^M(t) \in \mathbb{Q}[t]$ be the associated polynomial such that

$$\chi_{\mathfrak{q}}^M(n) = \ell(M/\mathfrak{q}^n M), \quad \text{for } n >> 0.$$

If $M = A$, we write $\chi_{\mathfrak{q}} = \chi_{\mathfrak{q}}^A$. Then deg $\chi_{\mathfrak{q}} \leq s =$ least number of generators of $\mathfrak{q}$.

Proposition 9.1.7. *Let $A, \mathfrak{m}, \mathfrak{q}$ be as above. Then*

$$\text{deg } \chi_{\mathfrak{q}} = \text{deg } \chi_{\mathfrak{m}}.$$

Proof. Since $\mathfrak{m}^r \subset \mathfrak{q} \subset \mathfrak{m}$ for some r, we get $\mathfrak{m}^{rn} \subset \mathfrak{q}^n \subset \mathfrak{m}^n$ and thus $\chi_{\mathfrak{m}}(n) \leq \chi_{\mathfrak{q}}(n) \leq \chi_{\mathfrak{m}}(rn)$ for $n >> 0$. This implies deg $\chi_{\mathfrak{q}} = $ deg $\chi_{\mathfrak{m}}$. □

By Corollary 9.1.3, we have

$$\text{deg } \chi_{\mathfrak{q}} = d(G_{\mathfrak{q}}(A)), \quad \text{deg } \chi_{\mathfrak{m}} = d(G_{\mathfrak{m}}(A)), \quad \Rightarrow \quad d(G_{\mathfrak{q}}(A)) = d(G_{\mathfrak{m}}(A)).$$

9.2 Dimension theory of Noetherian local rings

Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal.

Notation 9.2.1. *Consider the following two integers*

$$d(A) = d(G_{\mathfrak{m}}(A)), \quad \delta(A) = \textit{least number of generators of an } \mathfrak{m}\textit{-primary ideal of } A.$$

One sees easily that

$$\delta(A) = \min\{n \mid x_1, \cdots, x_n \in \mathfrak{m}, \quad \ell(A/(x_1, \cdots, x_n)) < \infty\}.$$

Our goal is to show

$$\dim A = d(A) = \delta(A).$$

By Proposition 9.1.7 and Proposition 9.1.6 2, we have

$$d(A) \leq \delta(A).$$

In the following we will prove

$$\delta(A) \leq \dim(A) \leq d(A).$$

Proposition 9.2.2. *Let $A, \mathfrak{m}, \mathfrak{q}$ be as above, M a finitely generated A-module, $x \in A$ a non zero divisor in M. Let $M' = M/xM$, then* deg $\chi_{\mathfrak{q}}^{M'} \leq$ deg $\chi_{\mathfrak{q}}^M - 1$.

Proof. Let $N = xM \subset M$, then $N \simeq M$ as A-modules, since by assumption $0 \to M \xrightarrow{\cdot x} M \to M' \to 0$ is exact. Let $N_n = N \cap \mathfrak{q}^n M (\simeq x^{-1}(\mathfrak{q}^n M))$, then we have exact sequences

$$0 \to N/N_n \to M/\mathfrak{q}^n M \to M'/\mathfrak{q}^n M' \to 0.$$

Let $g(t)$ be the polynomial attached to (N_n) as in Proposition 9.1.6 2, then we have

$$g(n) - \chi_{\mathfrak{q}}^{M}(n) + \chi_{\mathfrak{q}}^{M'}(n) = 0, \quad n >> 0.$$

By Artin-Rees Lemma, (N_n) is a stable $\mathfrak{q}$-filtration on N, thus $g(t)$ and $\chi_{\mathfrak{q}}^{M}(t)$ have the same leading term by Proposition 9.1.6 3. So $\deg \chi_{\mathfrak{q}}^{M'} \leq \deg \chi_{\mathfrak{q}}^{M} - 1$. □

Corollary 9.2.3. *Let A be a Noetheiran local ring, x a non zero divisor in A. Then $d(A/(x)) \leq d(A) - 1$.*

Proposition 9.2.4. *Let A be as above. Then*

$$\dim A \leq d(A).$$

In particular, $\dim A < +\infty$.

Proof. We proceed by induction on $d = d(A)$. If $d = 0$, then $\ell(A/\mathfrak{m}^n)$ is constant for $n >> 0$. This means $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for $n >> 0$. Then $\mathfrak{m}^n = 0$ by Nakayama Lemma. By Proposition 5.3.11 we know A is Artinian and thus $\dim A = 0$. Suppose $d > 0$. If $\dim A = 0$, we are done. Assume $\dim A > 0$. Let

$$\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_r$$

be a strict chain of prime ideals of A with $r \geq 1$. Take $x \in \mathfrak{p}_1 \setminus \mathfrak{p}_0$ and set $A' = A/\mathfrak{p}_0, x' =$ the image of x in A'. Then $x' \neq 0$ and A' is an integral domain. By Corollary 9.2.3, we have

$$d(A'/(x')) \leq d(A') - 1.$$

Let $\mathfrak{m}' \subset A'$ be the maximal ideal. Since $A'/(\mathfrak{m}')^n$ is a homomorphism image of $A/\mathfrak{m}^n$, we get $\ell(A/\mathfrak{m}^n) \geq \ell(A'/(\mathfrak{m}')^n)$ and thus

$$d(A) \geq d(A').$$

Then $d(A'/(x')) \leq d(A) - 1 = d - 1$. By inductive hypothesis,

$$\dim A'/(x') \leq d(A'/(x')) \leq d - 1.$$

Since the images of $\mathfrak{p}_1, \cdots, \mathfrak{p}_r$ form a chain $\mathfrak{p}_1' \subset \cdots \subset \mathfrak{p}_r'$ of length $r-1$, we get

$$r - 1 \leq \dim A'/(x') \leq d - 1, \quad \Rightarrow \quad r \leq d.$$

As r is arbitrary, we get $\dim A \leq d = d(A) < \infty$. □

Corollary 9.2.5. *Let A be any Noetherian ring, $\mathfrak{p} \in \operatorname{Spec} A$, then $ht\, \mathfrak{p} < \infty$. More generally, for any ideal $I \subsetneq A$, we have $ht\, I < \infty$.*

Proposition 9.2.6. *Let A be a Noetherian local ring of dimension d. Then there exists an $\mathfrak{m}$-primary ideal generated by d elements $x_1, \cdots, x_d$. In particular*

$$\delta(A) \leq \dim A = d.$$

Proof. We will construct elements $x_1, \cdots, x_d$ inductively such that for any prime ideal $\mathfrak{p} \supset (x_1, \cdots, x_i)$, then $ht\,\mathfrak{p} \geq i$ for each i.

Suppose $i > 0$ and $x_1, \cdots, x_{i-1}$ have been constructed. Consider the ideal $(x_1, \cdots, x_{i-1})$ and let $\mathfrak{p}_1, \cdots, \mathfrak{p}_s$ be the minimal prime ideals $\supset (x_1, \cdots, x_{i-1})$ and (if any) have height $i-1$. SInce $i-1 < d = \dim\, A = ht\,\mathfrak{m}$, we have $\mathfrak{m} \neq \mathfrak{p}_j$ for any $1 \leq j \leq s$. Then $\mathfrak{m} \neq \bigcup_{j=1}^{s} \mathfrak{p}_j$. Choose $x_i \in \mathfrak{m}$ such that $x \notin \bigcup_{j=1}^{s} \mathfrak{p}_j$. Consider $(x_1, \cdots, x_i)$. Let $\mathfrak{q} \supset (x_1, \cdots, x_i)$ be any prime ideal. Then $\mathfrak{q}$ contains some minimal prime ideal $\mathfrak{p} \supset (x_1, \cdots, x_{i-1})$. If $\mathfrak{p} = \mathfrak{p}_j$ for some j, then $x_i \in \mathfrak{q}$ and $x_i \notin \mathfrak{p}$. So $\mathfrak{q} \supsetneq \mathfrak{p}$ and $ht\,\mathfrak{q} \geq i$. If $\mathfrak{p} \neq \mathfrak{p}_j$ for any $1 \leq j \leq s$, then $ht\,\mathfrak{p} \geq i \Rightarrow ht\,\mathfrak{q} \geq i$. Thus every prime ideal $\supset (x_1, \cdots, x_i)$ has height $\geq i$.

Consider $(x_1, \cdots, x_d)$. Let $\mathfrak{p} \supset (x_1, \cdots, x_d)$ be a prime ideal, then $ht\,\mathfrak{p} \geq d = \dim\, A, \Rightarrow \mathfrak{p} = \mathfrak{m}$. Thus $(x_1, \cdots, x_d)$ is $\mathfrak{m}$-primary.

□

Here is one of the most important theorems in commutative algebra.

Theorem 9.2.7 (Krull)**.** *Let A be a Noetherian local ring. Then*

$$\dim\, A = d(A) = \delta(A).$$

Proof. By Propositions 9.1.7, 9.1.6, 9.2.4 and 9.2.6, we have

$$\delta(A) \leq \dim\, A \leq d(A) \leq \delta(A).$$

□

Example 9.2.8. *Let k be a field and $A = k[X_1, \cdots, X_n], \mathfrak{m} = (X_1, \cdots, X_n) \subset A$. Then* $\dim\, A_{\mathfrak{m}} = n$*, since $P(G_{\mathfrak{m}}(A_{\mathfrak{m}}), t) = (1-t)^{-n}$.*

Corollary 9.2.9. *Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $k = A/\mathfrak{m}$. Then*

$$\dim\, A \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2).$$

Proof. By Nakayama Lemma, $\dim_k(\mathfrak{m}/\mathfrak{m}^2)$ = least number of generators of $\mathfrak{m}$. Thus $\dim\, A = \delta(A) \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$. □

Corollary 9.2.10. *Let A be a Noetherian ring, $x_1, \cdots, x_r \in A$, then for any minimal prime ideal $\mathfrak{p} \supset (x_1, \cdots, x_r)$, we have $ht\,\mathfrak{p} \leq r$. In particular, $ht\,((x_1, \cdots, x_r)) \leq r$.*

Proof. Consider $A_{\mathfrak{p}}$, then $IA_{\mathfrak{p}}$ is $\mathfrak{p}A_{\mathfrak{p}}$-primary where $I = (x_1, \cdots, x_r)$. Then $ht\,\mathfrak{p} = \dim\, A_{\mathfrak{p}} \leq r$. □

Corollary 9.2.11. *Let A be a Noetherian ring, $x \in A$ which is neither a zero divisor nor a unit, then for any prime ideal $\mathfrak{p} \supset (x)$, we have $ht\,\mathfrak{p} = 1$.*

Proof. By Corollary 9.2.10 we have $ht\,\mathfrak{p} \leq 1$. If $ht\,\mathfrak{p} = 0$, then $\mathfrak{p}$ is a prime ideal belonging to 0, and thus $\mathfrak{p} \subset \{\text{zero divisors of } A\}$, a contradication. Thus $ht\,\mathfrak{p} = 1$. □

Corollary 9.2.12. *Let A be a Noetherian local ring, $x \in \mathfrak{m}$ a non zero divisor, then* $\dim, A/(x) = \dim\, A - 1$.

Proof. By Corollary 9.2.3 we have $\dim\, A/(x) = d(A/(x)) \leq d(A) - 1 = \dim\, A - 1$. Let $d = \dim\, A/(x)$ and $x_1, \cdots, x_d \in \mathfrak{m}$ be such that their images in $A/(x)$ generates a $\mathfrak{m}/(x)$-primary ideal. Then $(x, x_1, \cdots, x_d)$ is $\mathfrak{m}$-primary, and thus $d + 1 \geq \dim\, A, \Rightarrow d = \dim\, A - 1$. □

Corollary 9.2.13. *Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $\widehat{A}$ the $\mathfrak{m}$-adic completion of A. Then* $\dim A = \dim \widehat{A}$.

Proof. As $G_{\mathfrak{m}}(A) \simeq G_{\widehat{\mathfrak{m}}}(\widehat{A})$, we get $\chi^A_{\mathfrak{m}} = \chi^{\widehat{A}}_{\widehat{\mathfrak{m}}}$. In particular $d(A) = d(\widehat{A})$. □

Definition 9.2.14. *Let A be a Noetherian local ring of dimension d. $x_1, \cdots, x_d \in A$ is called a system of parameters for A if $(x_1, \cdots, x_d)$ is an $\mathfrak{m}$-primary ideal of A.*

Proposition 9.2.15. *Let $x_1, \cdots, x_d \in A$ be a system of parameters, $\mathfrak{q} = (x_1, \cdots, x_d)$, $f(t_1, \cdots, t_d) \in A[t_1, \cdots, t_d]$ be a homogeneous polynomial of degree s. Assume $f(x_1, \cdots, x_d) \in \mathfrak{q}^{s+1}$. Then all coefficients of f lie in $\mathfrak{m}$.*

Proof. Consider the homomorphism of graded rings

$$\alpha : A/q[t_1, \cdots, t_d] \to G_{\mathfrak{q}}(A), \quad t_i \mapsto \overline{x_i},$$

where $\overline{x_i} \in \mathfrak{q}/\mathfrak{q}^2$ are the images of $x_i, 1 \leq i \leq d$. By construction α is a surjection. Let $\overline{f} \in A/\mathfrak{q}[t_1, \cdots, t_d]$ be the reduction of f modulo $\mathfrak{q}$. The assumptions $\deg f = s$ and $f(x_1, \cdots, x_d) \in \mathfrak{q}^{s+1}$ imply $\overline{f}(\overline{x_1}, \cdots, \overline{x_d}) = \alpha(\overline{f}) = 0$. If some coefficent of f is a unit, then $\overline{f}$ is a non zero divisor in $A/q[t_1, \cdots, t_d]$ (exercise), and thus by Proposition 9.1.4

$$d = d(G_{\mathfrak{q}}(A)) \leq d(A/\mathfrak{q}[t_1, \cdots, t_d]/(\overline{f})) = d(A/\mathfrak{q}[t_1, \cdots, t_d]) - 1 = d - 1,$$

a contradication. So all coefficients of f are non units, i.e. $\in \mathfrak{m}$.

□

Corollary 9.2.16. *If $k \subset A$ is a field mapping isomorphically onto $A/\mathfrak{m}$, and if $x_1, \cdots, x_d \in A$ is a system of parameters of A, then $x_1, \cdots, x_d$ are algebracially independent over k.*

Proof. Suppose $f \in k[t_1. \cdots, t_d]$ such that $f(x_1, \cdots, x_d) = 0$. If $f \neq 0$, we can write $f = f_s +$ higher terms, where $f_s \neq 0$ is homogeneous of degree s. Then $f(x_1, \cdots, x_d) = 0 \Rightarrow f_s(x_1, \cdots, x_d) \in \mathfrak{q}^{s+1}$. By Proposition 9.2.15 all coefficients of f_s lie in $\mathfrak{m}$. But f_s has coefficents in $k \Rightarrow f_s = 0$, a contradication. Thus $f = 0$ and $x_1, \cdots, x_d$ are algebracially independent over k. □

9.3 Transcendental dimension

Let k be a field, which we assume to be algebraially closed for simplicity, A a finitely generated k-algebra which is an integral domain. Let $K = \operatorname{Frac} A$, then the transcendental degree $tr.\deg(K|k) < +\infty$ for the fields extension $K|k$. For any maximal ideal $\mathfrak{m} \subset A$, consider the Noetherian local ring $A_{\mathfrak{m}}$. By Corollary 9.2.16, we have

$$\dim A_{\mathfrak{m}} \leq tr.\deg(K|k).$$

Lemma 9.3.1. *Let $B \subset A$ be integral domains such that B is integrally closed and A is integral over B. Let $\mathfrak{m} \subset A$ be a maximal ideal and $\mathfrak{n} = \mathfrak{m} \cap B$. Then* $\dim A = \dim B_{\mathfrak{n}}$.

Proof. Since A is integral over B, by Corollary 4.1.12 $\mathfrak{n}$ is a maximal ideal of B. If

$$\mathfrak{m} \supset \mathfrak{q}_1 \supset \cdots \supset \mathfrak{q}_d$$

is a strict chain of primes in A, then $\mathfrak{n} \supset \mathfrak{q}_1 \cap B \supset \cdots \supset \mathfrak{q}_d \cap B$ is a strict chain of primes in B, by Corollary 4.1.13. Thus dim $B_{\mathfrak{n}} \geq$ dim $A_{\mathfrak{m}}$. Conversely, if

$$\mathfrak{n} \supset \mathfrak{p}_1 \supset \cdots \mathfrak{p}_d$$

is a strict chain of primes in B, by the going-down Theorem 4.1.24, we can lift it to a strict chain of primes in A, thus dim $A_{\mathfrak{m}} \geq$ dim $B_{\mathfrak{n}}$. □

Theorem 9.3.2. *Let k be an algebraically closed field, A a finitely generated k-algebra which is an integral domain, and $K = \mathrm{Frac}\, A$. Then for any maximal ideal $\mathfrak{m} \subset A$, we have*

$$\dim A = \dim A_{\mathfrak{m}} = tr.\deg(K|k).$$

Proof. We have already dim $A_{\mathfrak{m}} \leq tr.\deg(K|k)$. By Noether's normalization lemma (Theorem 4.3.1), there exist elements $t_1, \cdots, t_d \in A$ which are algebraially independent over k such that A is integral over $B := k[t_1, \cdots, t_d]$. Let $\mathfrak{n} = \mathfrak{m} \cap B$. By Lemma 9.3.1,

$$\dim A_{\mathfrak{m}} = \dim B_{\mathfrak{n}} = d = tr.\deg(\mathrm{Frac}\, B|k) = tr.\deg(K|k).$$

By definition,

$$\dim A = \sup_{\mathfrak{p} \in \mathrm{Spec}\, A} \dim A_{\mathfrak{p}} = \sup_{\mathfrak{m} \subset A\, max.} \dim A_{\mathfrak{m}} = \dim A_{\mathfrak{m}}, \quad \forall\, \mathfrak{m}.$$

□

Remark 9.3.3. *The above theorem also holds for non algebraically closed fields k.*

Chapter 10

Cohen-Macaulay rings, Normal rings, and Regular rings

We study some classes of interesting rings, which appear everywhere in arithmetic algebraic geometry. We will be quite sketchy in this chapter. For more details, see [11, 12].

10.1 Regular local rings

Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $k = A/\mathfrak{m}$. By Corollary 9.2.9 we have

$$\dim A \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2).$$

Definition 10.1.1. *A is called a regular local ring if* $\dim A = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$.

Theorem 10.1.2. *Let A be a Noetherian local ring. Then A is regular $\Leftrightarrow G_{\mathfrak{m}}(A) \simeq k[t_1, \cdots, t_d]$ for some integer d.*

Proof. $\Leftarrow$: Since $G_{\mathfrak{m}}(A) = A/m \oplus \mathfrak{m}/\mathfrak{m}^2 \oplus \cdots \simeq k[t_1, \cdots, t_d]$ as graded rings, we get $\dim A = d$ and $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = d$, thus A is regular.

$\Rightarrow$: Since A is regular, then $\mathfrak{m} = (x_1, \cdots, x_d)$ for a system of parameters $x_1, \cdots, x_d$ with $d = \dim A$. Consider the homomorphism of graded rings

$$\alpha : k[t_1, \cdots, t_d] \to G_{\mathfrak{m}}(A), \quad t_i \mapsto \overline{x_i}$$

as in the proof of Proposition 9.2.15, which is surjective. By Proposition 9.2.15, α is injective. Thus α is an isomorphism of graded rings. □

Corollary 10.1.3. *Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $\widehat{A}$ the $\mathfrak{m}$-adic completion. Then A is regular $\Leftrightarrow$ $\widehat{A}$ is regular.*

Proof. By Corollaries 8.5.5 8.5.11 $\widehat{A}$ is a Noetherian local ring. By Corollary 9.2.13 $\dim A = \dim \widehat{A}$. By Proposition 8.5.8, $G_{\mathfrak{m}}(A) \simeq G_{\widehat{\mathfrak{m}}}(\widehat{A})$. Then the corollary follows from Theorem 10.1.2. □

Examples 10.1.4.

1. *Let k be a field, $d \geq 0$, then $k[[X_1, \cdots, X_d]]$ is a regular local ring of dimension d.*

2. *Any DVR is a regualr local ring of dimension 1, by Proposition 7.1.4.*

Lemma 10.1.5. *Let A be a ring, $I \subset A$ an ideal such that $\bigcap_n I^n = 0$. Suppose $G_I(A)$ is an integral domain, then A is an integral domain.*

Proof. Exercise. □

Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, then $\bigcap_n \mathfrak{m}^n = 0$ by Corollary 8.2.5. Then by Lemma 10.1.5 and Theorem 10.1.2, regular local rings are integral domains.

Remark 10.1.6. *1. By similar ideas, one can show regular local rings are integrally closed, see [11] Theorem 34.*

2. More strongly, we have (see [11] Theorem 48, which uses homological methods)

Theorem *(Auslander-Buchsbaum) Regular local rings are UFDs.*

10.2 Regular sequences, Koszul complex, and depth

Let A be a ring, $M \in \mathrm{Mod}_A$, $a_1, \cdots, a_r \in A$. Write $(\underline{a}) = (a_1, \cdots, a_r) \subset A$ and $\underline{a}M = (\underline{a})M = \sum_{i=1}^r a_i M \subset M$.

Definition 10.2.1. *We say $a_1, \cdots, a_r$ is an M-regular sequence if*

1. *for any $1 \le i \le r$, a_i is not a zero-divisor on $M/(a_1, \cdots, a_{i-1})M$,*
2. *$M \ne \underline{a}M$.*

Note that a priori this notion depends on the order of the elements in the sequence. But one can show the following result.

Theorem 10.2.2. *Let A be a ring, $M \in \mathrm{Mod}_A$, $a_1, \cdots, a_n \in A$ and $I = (\underline{a}) \subset A$. If $a_1, \cdots, a_n$ is M-reqular, and*

$$M, \quad M/a_1M, \quad M/(a_1, a_2)M, \quad \cdots, \quad M/(a_1, \cdots, a_{n-1})M$$

are separated for the I-adic topology, then any permutation of $a_1, \cdots, a_n$ is M-regular.

The condition that $M, \quad M/a_1M, \quad M/(a_1, a_2)M, \quad \cdots, \quad M/(a_1, \cdots, a_{n-1})M$ are separated for the I-adic topology is satisfied if

(α) either A is Noetherian, M is finitely generated over A, and $I \subset J(A)$, e.g. A is a Noetherian local ring and $I = \mathfrak{m}$,

(β) or A is a graded ring, M is a graded A-module, and each a_i is homogeneous of degree > 0.

From now on, we assume A is Noetherian. If $a_1, a_2, \cdots \in A$ is an M-regular sequence, then we have a strict chain $(a_1) \subsetneq (a_1, a_2) \subsetneq \cdots$, which must be saturated. Fix an ideal $I \subset A$. Then each M-regular sequence in I can be extended to a maximal M-regular sequence in I.

Theorem 10.2.3. *Let A be a Noetherian ring, M a finitely generated A-module, $I \subset A$ an ideal such that $IM \ne M$. Let $n > 0$ be an integer. TFAE:*

1. $\mathrm{Ext}^i_A(N, M) = 0 \quad (i < n)$, *for any finitely generated A-module N with* $\mathrm{Supp}(N) \subset V(I)$.

2. $\mathrm{Ext}^i_A(A/I, M) = 0 \quad (i < n)$.

3. *There exists a finitely generated A-module N with $\mathrm{Supp}(N) = V(I)$ such that $\mathrm{Ext}^i_A(N, M) = 0 \quad (i < n)$.*

4. *There exists an M-regular sequence $a_1, \cdots, a_n$ of length n in I.*

Proof. See [11] Theorem 28. □

This theorem implies that if M is finitely generated, then any two maximal M-regular sequence have the same length.

Definition 10.2.4. *Let A, M, I be as above. The I-depth of M is the length of a maximal M-reqular sequence in I. We have*

$$\mathrm{depth}_I M = \min\{i \,|\, \mathrm{Ext}^i_A(A/I, M) \neq 0\}.$$

If $IM = M$, we define $\mathrm{depth}_I M = \infty$.

If A is a Noetherian local ring, $I = \mathfrak{m}$ the maximal ideal, we will also write

$$\mathrm{depth}\, M = \mathrm{depth}_A M := \mathrm{depth}_{\mathfrak{m}} M.$$

By definition, we have

$$\mathrm{depth}\, M = 0 \quad \Leftrightarrow \quad \mathrm{Hom}(A/\mathfrak{m}, B) \neq 0 \quad \Leftrightarrow \quad \mathfrak{m} \in \mathrm{Ass}(M).$$

For a general Noetherian ring A, $\forall\, \mathfrak{p} \in \mathrm{Spec}\, A$, we have

$$\mathrm{depth}_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \geq \mathrm{depth}_{\mathfrak{p}} M$$

since $\Big(\mathrm{Ext}^i_A(A/\mathfrak{p}, M)\Big)_{\mathfrak{p}} = \mathrm{Ext}^i_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}, M_{\mathfrak{p}})$.

We can compute $\mathrm{depth}_I M$ by an explicit complex. Let us begin with a general setup. Let A be a ring, $x_1, \cdots, x_n \in A$, we define a complex $K_\bullet$ as follows. Let

$$K_p = \begin{cases} 0, & p < 0 \text{ or } p > n, \\ A, & p = 0, \\ \bigoplus Ae_{i_1, \cdots, i_p}, & 1 \leq p \leq n, \end{cases}$$

where for $1 \leq p \leq n$, $\bigoplus Ae_{i_1, \cdots, i_p}$ is the free A-module of rank $\binom{n}{p}$ with basis

$$\{e_{i_1, \cdots, i_p} \,|\, 1 \leq i_1 < \cdots < i_p \leq n\}.$$

We have isomorphisms

$$K_p \simeq \bigwedge^p A^n, \quad 1 \leq p \leq n.$$

For the differentials, we define homomorphisms of A-modules

$$d : K_p \to K_{p-1}, \quad e_{i_1, \cdots, i_p} \mapsto \sum_{i=1}^{p} (-1)^{r-1} x_{i_r} e_{i_1, \cdots, \widehat{i_r}, \cdots, i_p}, \quad 2 \leq p \leq n.$$

For $p = 1$, set $d(e_i) = x_i$. For $p \leq 0$ or $p \geq n+1$, set $d = 0$. Then one can check that

$$d^2 = 0,$$

i.e. we get a complex.

Definition 10.2.5. *The above complex $K_\bullet = K_\bullet(\underline{x})$ is called the Koszul complex associated to $x_1, \cdots, x_n$. For any $M \in \mathrm{Mod}_A$, set*

$$K_\bullet(\underline{x}, M) = K_\bullet(\underline{x}) \otimes_A M.$$

For $n = 1$, the complex $K_\bullet(x)$ is simply

$$\cdots \to 0 \to 0 \to A \xrightarrow{x} A \to 0 \to 0 \to \cdots.$$

The general case is built from the simplest case in the sense that we have an isomorphism of complexes

$$K_\bullet(x_1, \cdots, x_n) \cong K_\bullet(x_1) \otimes \cdots \otimes K_\bullet(x_n),$$

where the right hand is the product of complexes $K_\bullet(x_i), 1 \le i \le n$.

Example 10.2.6. *The case $n = 2$ is*

$$0 \to A \xrightarrow{(-x_2, x_1)} A^2 \xrightarrow{\binom{x_1}{x_2}} A \to 0.$$

Write $H_p(\underline{x}, M) = H_p(K_\bullet(\underline{x}, M))$, the p-th homology group, then one can compute

$$H_0(\underline{x}, M) \simeq M/\underline{x}M, \quad H_n(\underline{x}, M) \simeq \{\xi \in M \,|\, x_1\xi = \cdots = x_n\xi = 0\}.$$

Remark 10.2.7. *In the above we only consider Koszul complex in the form of chain complex. One can also define Koszul complex of cochains and then consider the cohomology.*

Theorem 10.2.8. *If $x_1, \cdots, x_n$ is an M-reqular sequence, then we have*

$$H_p(\underline{x}, M) = 0, \quad \forall\, p > 0, \quad \text{and} \quad H_0(\underline{x}, M) = M/\underline{x}M.$$

Conversely, if A is Noetherian, M is finitely generated over A, $x_1, \cdots, x_n \in J(A)$ with $H_p(\underline{x}, M) = 0, \quad \forall\, p > 0, \quad$ and $\quad H_0(\underline{x}, M) = M/\underline{x}M$, then $x_1, \cdots, x_n$ is an M-reqular sequence.

Let us come back to Noetherian rings and depths.

Theorem 10.2.9. *Let A be a Noetherian ring, M a finitely generated A-module, $I = (y_1, \cdots, y_n) \subset J(A)$ such that $IM \neq M$. Let*

$$q = \sup\{i \,|\, H_i(\underline{y}, M) \neq 0\}.$$

Then

$$\mathrm{depth}_I M = n - q.$$

Proof. See [12] Theorem 16.8. □

10.3 Cohen-Macaulay rings

Theorem 10.3.1. *Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $M \neq 0$ a finitely generated A-module. Then*

$$\mathrm{depth}\, M \le \dim A/\mathfrak{p}, \quad \forall\, \mathfrak{p} \in \mathrm{Ass}(M).$$

Proof. See [11] Theorem 29. □

For M as above, let
$$\dim\, M = \dim(A/Ann(M)).$$
Since M is finitely generated, we have
$$\mathrm{Ass}(M) \subset \mathrm{Supp}(M) = V(Ann(M)).$$
Then $\forall\, \mathfrak{p} \in \mathrm{Ass}(M)$, we have
$$\dim\, A/\mathfrak{p} \leq \dim(A/Ann(M)) = \dim\, M.$$
In particular by Theorem 10.3.1
$$\operatorname{depth} M \leq \dim\, M.$$

Definition 10.3.2. *M is called Cohen-Macaulay if $M = 0$ or $\operatorname{depth} M = \dim\, M$. If A is Cohen-Macaulay as an A-module, we call A a Cohen-Macaulay ring.*

Theorem 10.3.3. *Let A be a Noetherian local ring, $\mathfrak{m} \subset A$ the maximal ideal, $M \neq 0$ a finitely generated A-module. Then*

1. *If M is Cohen-Macaulay, then for any $\mathfrak{p} \in \mathrm{Ass}(M)$,*
$$\operatorname{depth} M = \dim\, A/\mathfrak{p}.$$

2. *If $a_1, \cdots, a_r$ is an M-reqular sequence in $\mathfrak{m}$, $M' = M/\underline{a}M$, then M is Cohen-Macaulay $\Leftrightarrow$ M' is Cohen-Macaulay.*

3. *If M is Cohen-Macaulay, then for any $\mathfrak{p} \in \operatorname{Spec} A$, the $A_\mathfrak{p}$-module $M_\mathfrak{p}$ is Cohen-Macaulay, and if $M_\mathfrak{p} \neq 0$ we have*
$$\operatorname{depth}_\mathfrak{p} M = \operatorname{depth}_{A_\mathfrak{p}} M_\mathfrak{p}.$$

Proof. 1 is immediate from definition.

2. By Nakayama Lemma, $M \neq 0 \Leftrightarrow M' \neq 0$. Then the result follows from the equalities
$$\operatorname{depth} M' = \operatorname{depth} M - r \quad \text{(easy)}$$
and
$$\dim\, M' = \dim\, M - r \quad \text{(exercise)}.$$

3. We may assume $M_\mathfrak{p} \neq 0$. Then $\mathfrak{p} \in \mathrm{Supp}(M) = V(AnnM)$ and
$$\operatorname{depth}_\mathfrak{p} M \leq \operatorname{depth}_{A_\mathfrak{p}} M_\mathfrak{p} \leq \dim\, M_\mathfrak{p}.$$

We prove $\operatorname{depth}_\mathfrak{p} M = \dim\, M_\mathfrak{p}$ by induction on $\operatorname{depth}_\mathfrak{p} M$. If $\operatorname{depth}_\mathfrak{p} M = 0$, then since M is Cohen-Macaulay, by 1, for any $\mathfrak{p} \subset \mathfrak{p}' \in \mathrm{Ass}(M)$ we have then $\mathfrak{p} = \mathfrak{p}'$. This means $\dim\, M_\mathfrak{p} = 0$. Suppose $\operatorname{depth}_\mathfrak{p} M > 0$. Take an M-reqular element $a \in \mathfrak{p}$ and put $M_1 = M/aM$. Then a is also $M_\mathfrak{p}$-regular. We have
$$\dim\, M_{1\mathfrak{p}} = \dim(M_\mathfrak{p}/aM_\mathfrak{p}) = \dim\, M_\mathfrak{p} - 1, \quad \operatorname{depth}_\mathfrak{p} M_1 = \operatorname{depth}_\mathfrak{p} M - 1.$$
By 2, M' is Cohen-Macaulay, and by inductive hypothesis $dim\, M_{1\mathfrak{p}} = \operatorname{depth}_\mathfrak{p} M_1$. Then by the above equalities we get $\operatorname{depth}_\mathfrak{p} M = \dim\, M_\mathfrak{p}$

□

If $M = A$ in the above theorem, by 3, if A is Cohen-Macaulay, then $A_{\mathfrak{p}}$ is also Cohen-Macaulay, for any $\mathfrak{p} \in \operatorname{Spec} A$.

Definition 10.3.4. *Let A be a Noetherian ring. A is called Cohen-Macaulay if $A_{\mathfrak{p}}$ is a Cohen-Macaulay local ring for any $\mathfrak{p} \in \operatorname{Spec} A$ ($\Leftrightarrow A_{\mathfrak{m}}$ is Cohen-Macaulay for any maximal ideal $\mathfrak{m} \subset A$).*

Theorem 10.3.5. *Let A be a Cohen-Macaulay local ring, $\mathfrak{m}$ the maximal ideal. Then*

1. *For any ideal $I \subsetneq A$,*

$$ht\, I = \operatorname{depth}_I A, \quad ht\, I + \dim\, A/I = \dim\, A.$$

2. *A is catenary, i.e. for any prime ideals $\mathfrak{p} \supset \mathfrak{q} \in \operatorname{Spec} A$, any maximal prime ideals chain between $\mathfrak{p}$ and $\mathfrak{q}$ have the same length, which is $ht(\mathfrak{p}/\mathfrak{q})$.*

3. *For any sequence $a_1, \cdots, a_r$ in $\mathfrak{m}$, TFAE:*
   - (a) *the sequence $a_1, \cdots, a_r$ is A-reqular,*
   - (b) *$ht(a_1, \cdots, a_i) = i, (1 \le i \le r)$,*
   - (c) *$ht(a_1, \cdots, a_r) = r$.*
   - (d) *there exist $a_{r+1}, \cdots, a_n$ in $\mathfrak{m}$ such that $a_1, \cdots, a_n$ form a system of parameters of A (where $n = \dim\, A$).*

Proof. See [11] Theorem 31. □

Let A be a Noetherian local ring, if one system of parameters of A is an A-reqular sequence, then $\operatorname{depth} A = \dim\, A$, i.e. A is Cohen-Macaulay, then by Theorem 10.3.5 3, any system of parameters of A is A-regular.

Proposition 10.3.6. *Let A be a regular local ring. Then A is Cohen-Macaulay.*

Proof. Let $x_1, \cdots, x_d$ be a system of parameters such that $\mathfrak{m} = (x_1, \cdots, x_d)$. Then $x_1, \cdots, x_d$ is A-regular. □

Let A be a Noetherian ring, $I \subset A$ an ideal. Let

$$\operatorname{Ass}_A(A/I) = \{\mathfrak{p}_1, \cdots, \mathfrak{p}_s\} \subset \operatorname{Supp}_A(A/I) = V(I).$$

We know the minimal prime ideals containing I lie in $\operatorname{Ass}_A(A/I)$. Recall

$$ht\, I = \inf_{\mathfrak{p} \in V(I)} ht\, \mathfrak{p} = \inf_{\mathfrak{p} \in \operatorname{Ass}_A(A/I) \text{ min.}} ht\, \mathfrak{p}$$

and if $I = (a_1, \cdots, a_r)$ then $ht\, I \le r$.

Definition 10.3.7.

1. *We say I is unmixed if $ht\, I = ht\, \mathfrak{p}_i$, $\forall\, \mathfrak{p}_i \in \operatorname{Ass}_A(A/I)$.*
2. *We say the unmixed theorem holds in A if $\forall\, r \ge 0$ and $\forall\, I = (a_1, \cdots, a_r)$ and $ht\, I = r$, then I is unmixed.*

Theorem 10.3.8. *Let A be a Noetherian ring. Then A is Cohen-Macaulay $\Leftrightarrow$ the unmixed theorem holds in A.*

Proof. $\Rightarrow$: First of all, note that the unmixed theorem holds in $A \Leftrightarrow$ for any maximal ideal $\mathfrak{m} \subset A$, the unmixed theorem holds in $A_{\mathfrak{m}}$. So we may assume that A is a Cohen-Macaulay local ring. Then by Theorem 10.3.5 1, 0 is unmixed. Let $(a_1, \cdots, a_r)$ be an ideal of height $r > 0$. By Theorem 10.3.5 3, $a_1, \cdots, a_r$ is A-regular. By Theorem 10.3.3, $A/\underline{a}$ is Cohen-Macaulay, which implies then $(\underline{a})$ is unmixed.

$\Leftarrow$: Suppose the unmixed theorem holds in A. Let $\mathfrak{p} \in \operatorname{Spec} A$ and $ht\,\mathfrak{p} = r$. Then we may find $a_1, \cdots, a_r \in \mathfrak{p}$ such that $ht(a_1, \cdots, a_i) = i, 1 \leq i \leq r$. By assumption $(a_1, \cdots, a_i)$ is unmixed. This implies that a_{i+1} lies in no $\mathfrak{p} \in \operatorname{Ass}_A(A/(a_1, \cdots, a_i))$. Thus $a_1, \cdots, a_r$ is an A-regular sequence in $\mathfrak{p}$, and

$$r \leq \operatorname{depth}_{\mathfrak{p}} A \leq \operatorname{depth} A_{\mathfrak{p}} \leq \dim A_{\mathfrak{p}} = r,$$

so $A_{\mathfrak{p}}$ is a Cohen-Macaulay local ring. Since $\mathfrak{p}$ is arbitrary, A is Cohen-Macaulay by definition. $\square$

10.4 Normal rings

Let A be an integral domain. Recall that A is integrally closed $\Leftrightarrow$ $A_{\mathfrak{p}}$ is integrally closed, for any $\mathfrak{p} \in \operatorname{Spec} A \Leftrightarrow A_{\mathfrak{m}}$ is integrally closed, for any $\mathfrak{m} \subset A$ maximal ideal. Examples of integrally closed integral domains include valuation rings of a field, regular local rings, etc.

Definition 10.4.1. *A ring A is called normal if $A_{\mathfrak{p}}$ is an integrally closed integral domain, for any $\mathfrak{p} \in \operatorname{Spec} A$.*

Thus a normal (integral) domain is just another terminology for an integrally closed integral domain.

Lemma 10.4.2. *A Noetherian normal ring is a finite product of Noetherian normal domains.*

Proof. Let A be a Noetherian normal ring. Since $A_{\mathfrak{p}}$ is an integral domain, we get $nilrad(A_{\mathfrak{p}}) = 0$ for any $\mathfrak{p} \in \operatorname{Spec} A$. As $nilrad(A)_{\mathfrak{p}} = nilrad(A_{\mathfrak{p}}) = 0, \forall\, \mathfrak{p} \in \operatorname{Spec} A$, $nilrad(A) = 0$. Let $\mathfrak{p}_1, \cdots, \mathfrak{p}_r$ be all the minimal prime ideals of A. Then $0 = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r$. For any $i \neq j$, $\mathfrak{p}_i + \mathfrak{p}_j = A$, thus $A \simeq A/\mathfrak{p}_1 \times \cdots \times A/\mathfrak{p}_r$, with each $A/\mathfrak{p}_i$ Noetherian normal domain. $\square$

Theorem 10.4.3. *Let A be a Noetherian local ring of dimension 1. Then*

$$A \text{ is regular} \quad \Leftrightarrow \quad A \text{ is normal} \quad (\text{in which case } A \text{ is a DVR.}).$$

Proof. $\Rightarrow$: see Remark 10.1.6 1.

$\Leftarrow$: A is a one-dimensional integrally closed Noetherian local domain, thus a DVR by Proposition 7.1.4, which is regular. $\square$

Remark 10.4.4. *A normal ring is not necessary Noetherian. For example, let A be a valuation ring of field K with $A \subsetneq K$. Then A is normal. One can show (exercise) that A is Noetherian $\Leftrightarrow$ A is a DVR. There exist too many valuation rings which are not DVRs.*

Let A be an integral domain with fraction field K. Then one checks easily

$$A = \bigcap_{\mathfrak{m} \subset A\, max.} A_{\mathfrak{m}} = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} A_{\mathfrak{p}} \quad \text{in} \quad K.$$

Theorem 10.4.5. *Let A be a Noetherian normal domain. Then*

1. *Any non zero principal ideal is unmixed.*
2. *We have*
$$A = \bigcap_{ht\,\mathfrak{p}=1} A_{\mathfrak{p}}, \quad in \quad K.$$
(Note the local rings $A_{\mathfrak{p}}$ on the right hand side are DVRs, thus A is a Kull ring in the sense of [12] section 12.)
3. *If* $\dim A \leq 2$*, then A is Cohen-Macaulay.*

Proof. See [11] Theorem 38. □

Let A be a Noetherian ring. For any integer $k \geq 0$, consider the following conditions

$$(S_k): \quad \text{depth}\, A_{\mathfrak{p}} \geq \inf(k, ht\,\mathfrak{p}), \quad \forall\, \mathfrak{p} \in \text{Spec}\, A.$$
$$(R_k): \quad \text{If } \mathfrak{p} \in \text{Spec}\, A \text{ with } ht\,\mathfrak{p} \leq k, \quad \text{then } A_{\mathfrak{p}} \text{ is regular.}$$

One checks that

- A is Cohen-Macaulay $\Leftrightarrow$ (S_k) holds, for any $k \geq 0$.
- A is reduced, i.e. $nilrad(A) = 0$, $\Leftrightarrow$ (S_1) and (R_0) hold.

Here is the Serre criterion for a Noetherian ring to be normal.

Theorem 10.4.6 (Serre)**.** *Let A be a Noetherian ring. Then A is normal $\Leftrightarrow$ (S_2) and (R_1) hold.*

Proof. See [11] Theorem 39. □

10.5 Homological dimension and Regular rings

We come back to regular local rings introduced in section 10.1. We will study them by methods of homological algebra.

Let A be a ring, $M \in \text{Mod}_A$. Recall

$$proj.\dim M = \text{length of a shortest projective resolution of } M,$$
$$inj.\dim M = \text{length of a shortest injective resolution of } M.$$

We have
$$0 \leq proj.\dim M \leq +\infty, \quad 0 \leq inj.\dim M \leq +\infty.$$

Lemma 10.5.1. *Let $n \geq 0$ be an integer. TFAE:*

1. $proj.\dim M \leq n$, $\forall\, M \in \text{Mod}_A$.
2. $proj.\dim M \leq n$, $\forall\, M \in \text{Mod}_A$ *finitely generated.*
3. $inj.\dim M \leq n$, $\forall\, M \in \text{Mod}_A$.
4. $\text{Ext}_A^{n+1}(M, N) = 0$, $\forall\, M, N \in \text{Mod}_A$.

Proof. Exercise. □

The above lemma implies that

$$\sup_{M\in \mathrm{Mod}_A} proj.\dim M = \sup_{M\in \mathrm{Mod}_A} inj.\dim M.$$

Definition 10.5.2. *The homological dimension of A is*

$$gl.\dim A := \sup_{M\in \mathrm{Mod}_A} proj.\dim M = \sup_{M\in \mathrm{Mod}_A} inj.\dim M.$$

By definition

$$0 \leq gl.\dim A \leq +\infty.$$

Lemma 10.5.3. *Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $k = A/\mathfrak{m}$, M a finitely generated A-module, $n \geq 0$. Then*

$$proj.\dim M \leq n \quad \Leftrightarrow \quad \mathrm{Tor}^A_{n+1}(M, k) = 0.$$

Proof. Exercise. □

Lemma 10.5.4. *Let A be a Noetherian ring.*

1. *Let M be a finitely generated A-module. Then*

   (a)
   $$proj.\dim M = \sup_{\mathfrak{m}\subset A\, max.} proj.\dim M_{\mathfrak{m}}.$$

   (b) Let $n \geq 0$ be an integer.
   $$proj.\dim M \leq n \quad \Leftrightarrow \quad \mathrm{Tor}^A_{n+1}(M, A/\mathfrak{m}) = 0, \quad \forall\, \mathfrak{m} \subset A \text{ maximal ideal.}$$

2. *Let $n \geq 0$ be an integer. TFAE:*

   (a) $gl.\dim A \leq n$.
   (b) $proj.\dim M \leq n$, $\forall\, M \in \mathrm{Mod}_A$ *finitely generated.*
   (c) $inj.\dim M \leq n$, $\forall\, M \in \mathrm{Mod}_A$ *finitely generated.*
   (d) $\mathrm{Ext}^{n+1}_A(M, N) = 0$, $\forall\, M, N \in \mathrm{Mod}_A$ *finitely generated.*
   (e) $\mathrm{Tor}^A_{n+1}(M, N) = 0$, $\forall\, M, N \in \mathrm{Mod}_A$ *finitely generated.*

3. *We have*
   $$gl.\dim A = \sup_{\mathfrak{m}\subset A\, max.} gl.\dim A_{\mathfrak{m}}.$$

Proof. See [11] p. 130. □

Theorem 10.5.5. *Let A be a Noetherian local ring, $\mathfrak{m}$ the maximal ideal, $k = A/\mathfrak{m}$. Then*

$$gl.\dim A \leq n \quad \Leftrightarrow \quad \mathrm{Tor}^A_{n+1}(k, k) = 0.$$

Consequently, for the A-module $k = A/\mathfrak{m}$, we have

$$gl.\dim A = proj.\dim k.$$

Proof. $\Rightarrow$: By Lemma 10.5.4.

$\Leftarrow$: By Lemma 10.5.3, $\mathrm{Tor}_{n+1}^A(k,k) = 0 \Leftrightarrow proj.\dim k \leq n$. Then $\mathrm{Tor}_{n+1}^A(M,k) = 0, \forall M \in \mathrm{Mod}_A, \quad \Rightarrow proj.\dim M \leq n, \forall M \in \mathrm{Mod}_A, \quad \Rightarrow gl.\dim A \leq n$. □

Corollary 10.5.6. *Let A be a regular local ring, then*

$$gl.\dim A = \dim A.$$

Proof. Let $\mathfrak{m}, k$ be as above. By Theorem 10.5.5, $gl.\dim A = proj.\dim k$. Let $\dim A = n$ and $x_1, \cdots, x_n$ be a system of parameters, then this is A-regular. By Theorem 10.2.8, the associated Koszul complex $K_\bullet(\underline{x})$ is a free resolution of $A/(x_1, \cdots, x_n) = k$. Moreover, this is minimal in the sense of [11] 18.E. As $K_n \neq 0, K_{n+1} = 0$, we get $proj.\dim k = n$. □

Corollary 10.5.7 (Hilbert's Syzygy Theorem)**.** *Let k be a field, $A = k[X_1, \cdots, X_n]$, then $gl.\dim A = n$.*

Proof. By Lemma 10.5.4 3,

$$gl.\dim A = \sup_{\mathfrak{m} \subset A\, max.} gl.\dim A_{\mathfrak{m}}.$$

By Corollary 10.5.6,

$$gl.\dim A_{\mathfrak{m}} = \dim A_{\mathfrak{m}} = n.$$

□

Theorem 10.5.8 (Auslander-Buchsbaum)**.** *Let A be a Noetherian local ring, $M \neq 0$ a finitely generated A-module. Suppose $proj.\dim M < +\infty$. Then*

$$proj.\dim M + \operatorname{depth} M = \operatorname{depth} A.$$

Proof. See [12] Theorem 19.1. □

Now we arrive at the following striking result in Algebra.

Theorem 10.5.9 (Serre)**.** *Let A be a Noetherian local ring. Then*

$$A \text{ is regular} \quad \Leftrightarrow \quad gl.\dim A = \dim A \quad \Leftrightarrow \quad gl.\dim A < +\infty.$$

Proof. The first $\Rightarrow$ follows from Corollary 10.5.6. The second $\Rightarrow$ follows from Proposition 9.2.4. We prove that if $gl.\dim A < +\infty$ then A is regular.

Let $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = s$. Then by Corollary 9.2.9 $\dim A \leq s$. We need to show that this is in fact an equality. The theory of minimal free resolution (see [11] 18.E) implies that

$$\mathrm{Tor}_s^A(k,k) \neq 0.$$

By Theorem 10.5.5,

$$gl.\dim A = proj.\dim k \geq s.$$

By Theorem 10.5.8 (take $M = k$), we get

$$proj.\dim k = \operatorname{depth} A$$

(as $\operatorname{depth} k = 0$). On the other hand by Theorem 10.3.1 $\operatorname{depth} A \leq \dim A$. To summarize, we get

$$\dim A \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2) \leq gl.\dim A = proj.\dim k = \operatorname{depth} A \leq \dim A,$$

so all the above $\leq$ are in fact equalities. We are done.

□

Corollary 10.5.10. *If A is a regular local ring, then so is $A_{\mathfrak{p}}$, for any $\mathfrak{p} \in \operatorname{Spec} A$.*

Proof. For any $M \in \mathrm{Mod}_{A_{\mathfrak{p}}}$, consider it as an A-module, then there exists a finite length projective resolution

$$0 \to P_n \to \cdots \to P_0 = M \to 0,$$

since $gl.\dim A < +\infty$ by Theorem 10.5.9. We have then

$$n \leq gl.\dim A.$$

Apply the functor $- \otimes_A A_{\mathfrak{p}}$, we get an exact sequence

$$0 \to (P_n)_{\mathfrak{p}} \to \cdots \to (P_0)_{\mathfrak{p}} \to M_{\mathfrak{p}} = M \to 0.$$

This is a projective resolution of M in $\mathrm{Mod}_{A_{\mathfrak{p}}}$ of length n. Since M is arbitrary, we get

$$gl.\dim A_{\mathfrak{p}} \leq gl.\dim A < +\infty,$$

which implies $A_{\mathfrak{p}}$ is regular by Theorem 10.5.9.

□

Thanks to Corollary 10.5.10, we can now define general regular rings.

Definition 10.5.11. *Let A be a Noetherian ring. A is called regular if $A_{\mathfrak{p}}$ is a regular ring for any $\mathfrak{p} \in \operatorname{Spec} A$ ($\Leftrightarrow$ $A_{\mathfrak{m}}$ is regular for any maximal ideal $\mathfrak{m} \subset A$).*

Thus, we have the following inclusions of some basic classes of rings:

- Regular rings $\subset$ Cohen-Macaulay rings $\subset$ Noetherian rings.
- Regular rings $\subset$ Noetherian normal rings $\subset$ Normal rings.

The reader is encouraged to find further relations among the above rings.

So far, we have mainly focused on the theory of Noetherian rings. To conclude, let us mention the following recent breakthrough in commutative algebra:

Theorem 10.5.12 (André, Bhatt)**.** *Let R be a regular ring, $R \subset S$ ring extensions such that S is finitely generated as a R-module. Consider the exact sequence of R-modules*

$$0 \to R \to S \to S/R \to 0.$$

Then this exact sequence is split.

The above theorem was called the "direct summand conjecture", which is equivalent to other versions of several deep conjectures, namely the homological conjectures on Noetherian local rings in classical commutative algebra. The proofs given by André and Bhatt (respectively) use crucially the theory of perfectoid rings [15, 6], which are very interesting non Noetherian rings! For an overview, see André's ICM 2018 report [1].

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